



香港科技大學  
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

# ELEC 2100 Signals and Systems

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# Outline Chapter 1 Basic concepts of signals and systems



THE HONG KONG  
UNIVERSITY OF SCIENCE  
AND TECHNOLOGY

**1 Basic concepts**

**2 Description and classification of signals**

**3 Continuous-time signals**

**4 Unit step signal and unit impulse signal**

**5 Decomposition of signals**

**6 Discrete-time signals**

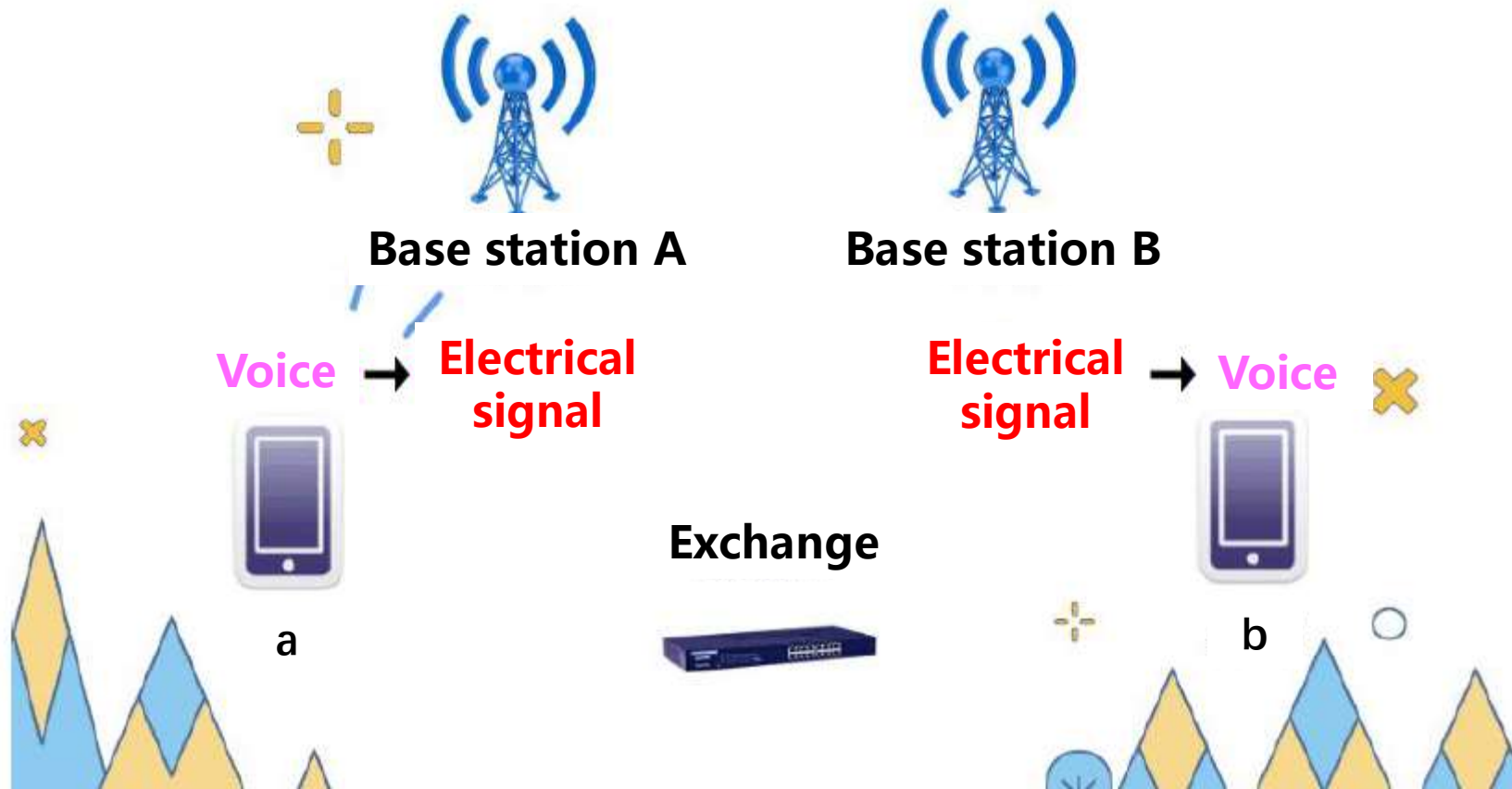
**7 System**

## **Main Content**

- 1. Basic concepts of signals and systems**
- 2. Theory of Signals and Systems**
- 3. Learning method**
- 4. Reference book**

# 1. Basic concepts of signals and systems

## 1. Basic concepts of signals and systems



# 1. Basic concepts of signals and systems

## (1) Signal

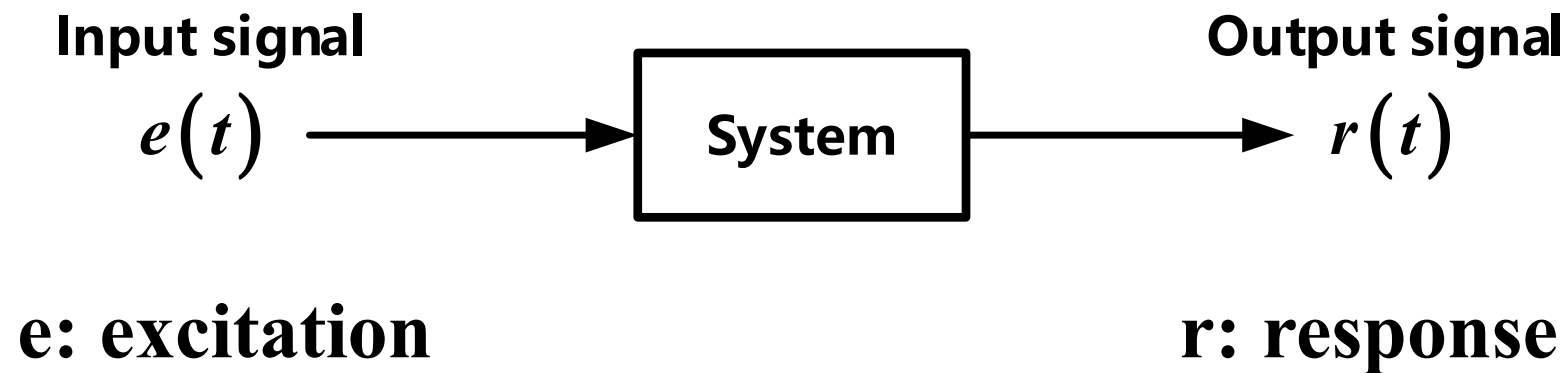
- **Message:** In communication systems, voice, text, images, and data are collectively referred to as messages
- **Signal: Refers to the representation and transmission medium of a message**
- A signal serves as both the representation and transmission medium of a message, while the message constitutes the actual content being conveyed by the signal

# 1. Basic concepts of signals and systems

## (2) System

### System

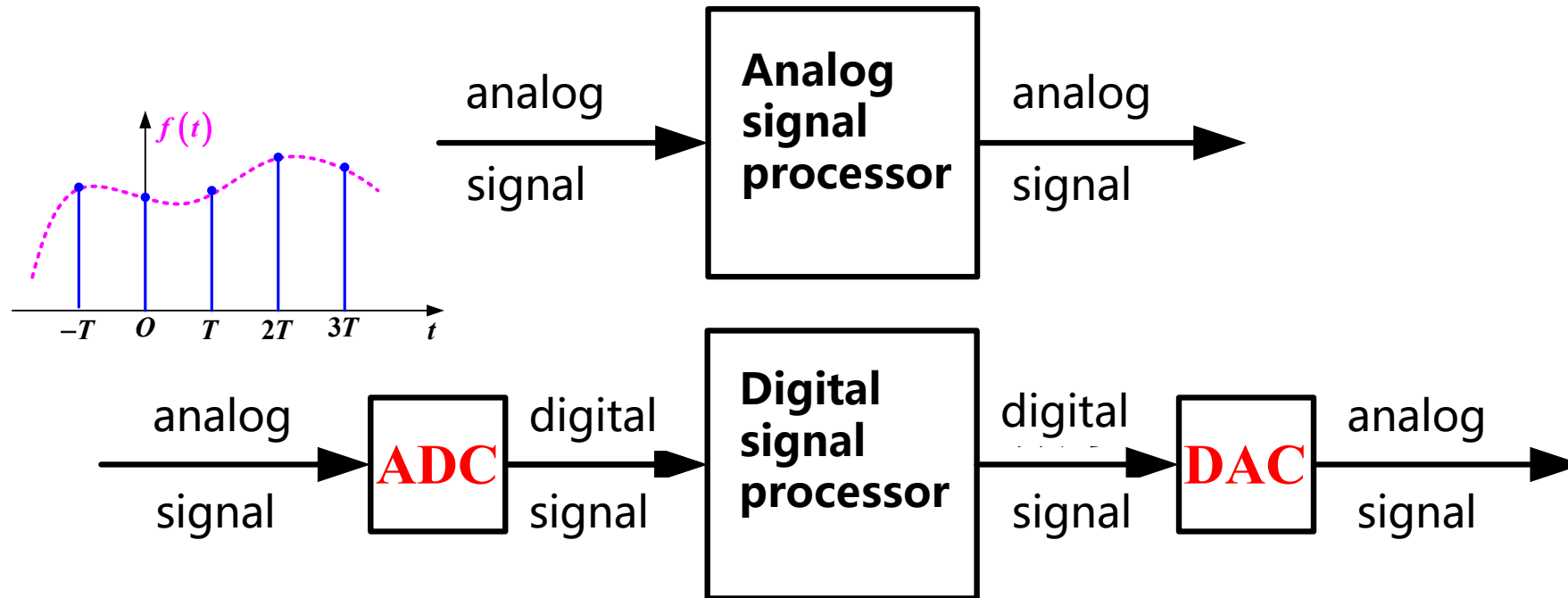
An integrated entity composed of multiple interacting and interdependent components, possessing stable functionality



# 1. Basic concepts of signals and systems

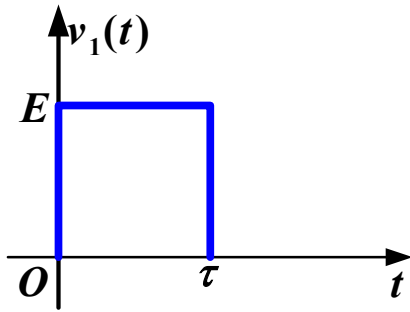
## (2) System

**The basic function of a system:** To process and transform input signals into the desired output signals

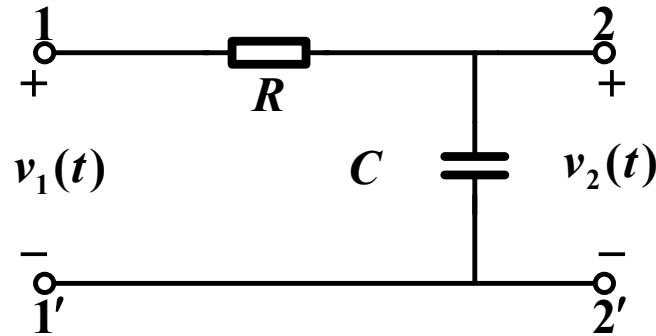


# 1. Basic concepts of signals and systems

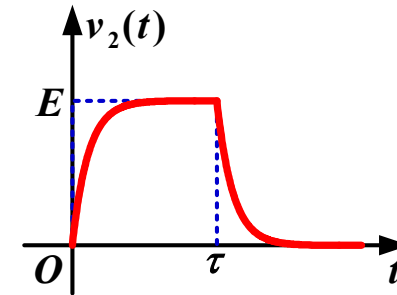
## (2) System



Input signal



System



Output signal

- Will the input signal be distorted after transmission through the system?
- How does the system affect the transmission of the input signal?
- Does the system process the signal to achieve the desired effect?

# 1. Basic concepts of signals and systems

## 2. Theory of Signals and Systems

### Signal Theory

**Signal Analysis:**  
Studies the  
description and  
properties of  
signals

Signal Transmission

Signal Processing

### System Theory

**System Analysis:**  
For a given system,  
examines the  
output generated  
in response to  
input signals

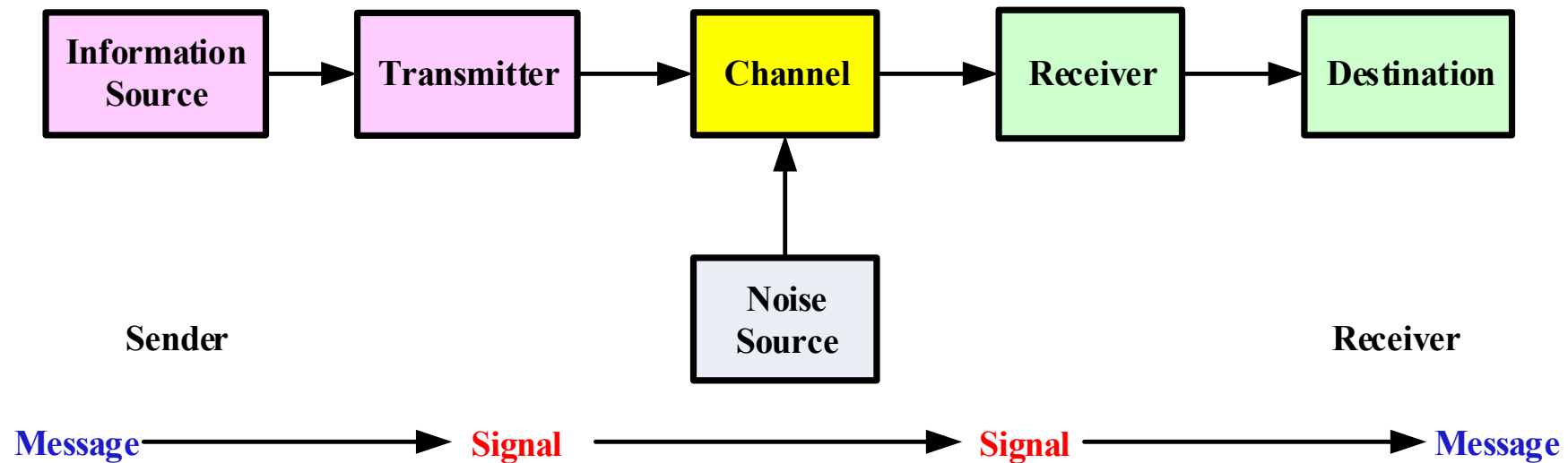
**System Synthesis:**  
Designs  
(synthesizes)  
systems according  
to specified  
requirements

# 1. Basic concepts of signals and systems

## (1) Communication system

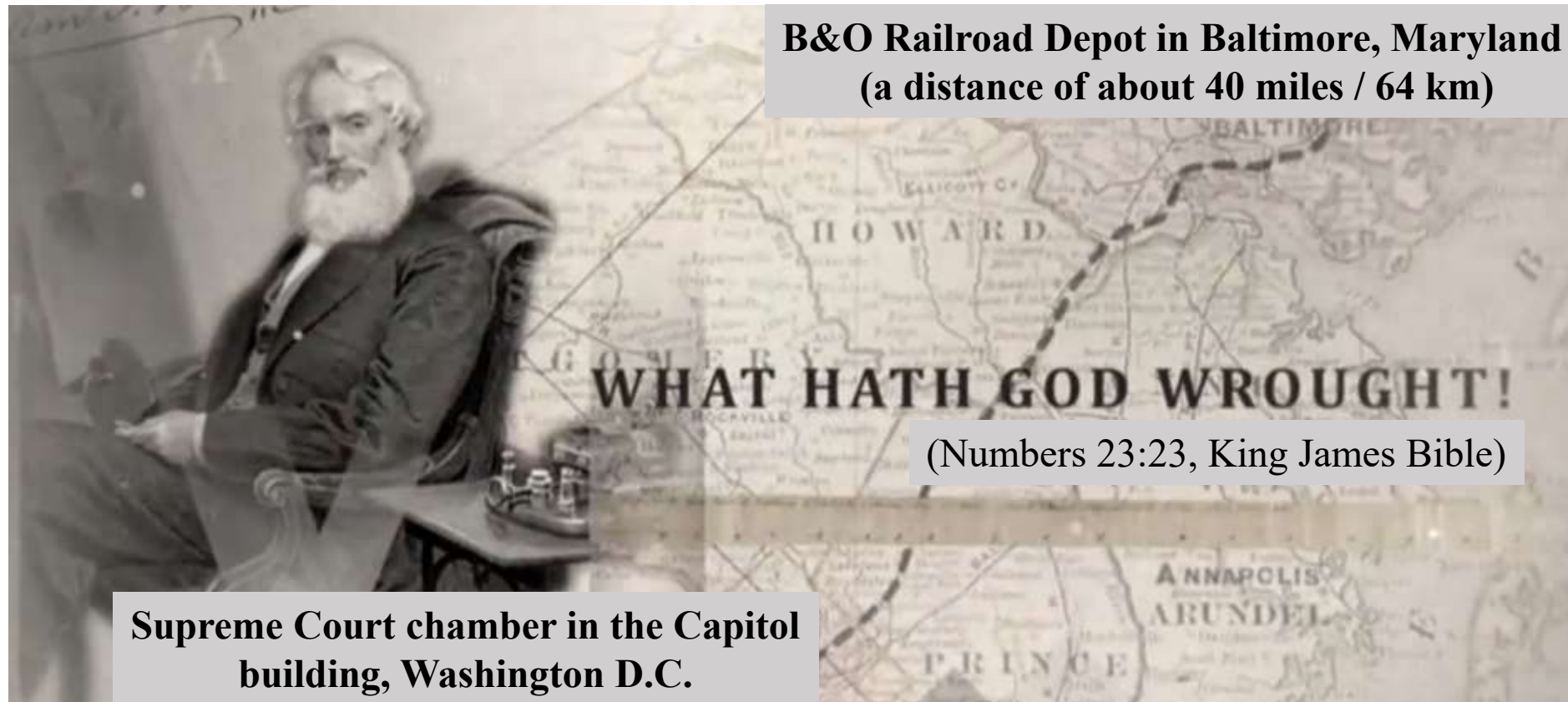
The purpose of communication is to achieve message transmission

**Communication System:** A complete set of technical equipment (including transmission channels) installed for the purpose of message delivery



# 1. Basic concepts of signals and systems

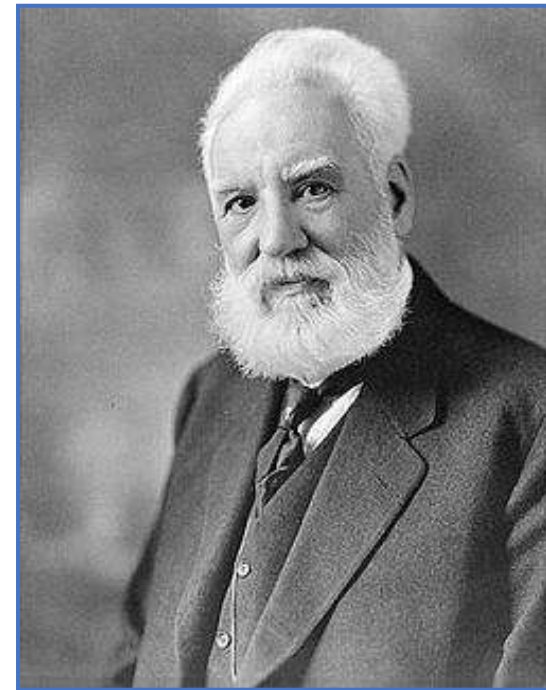
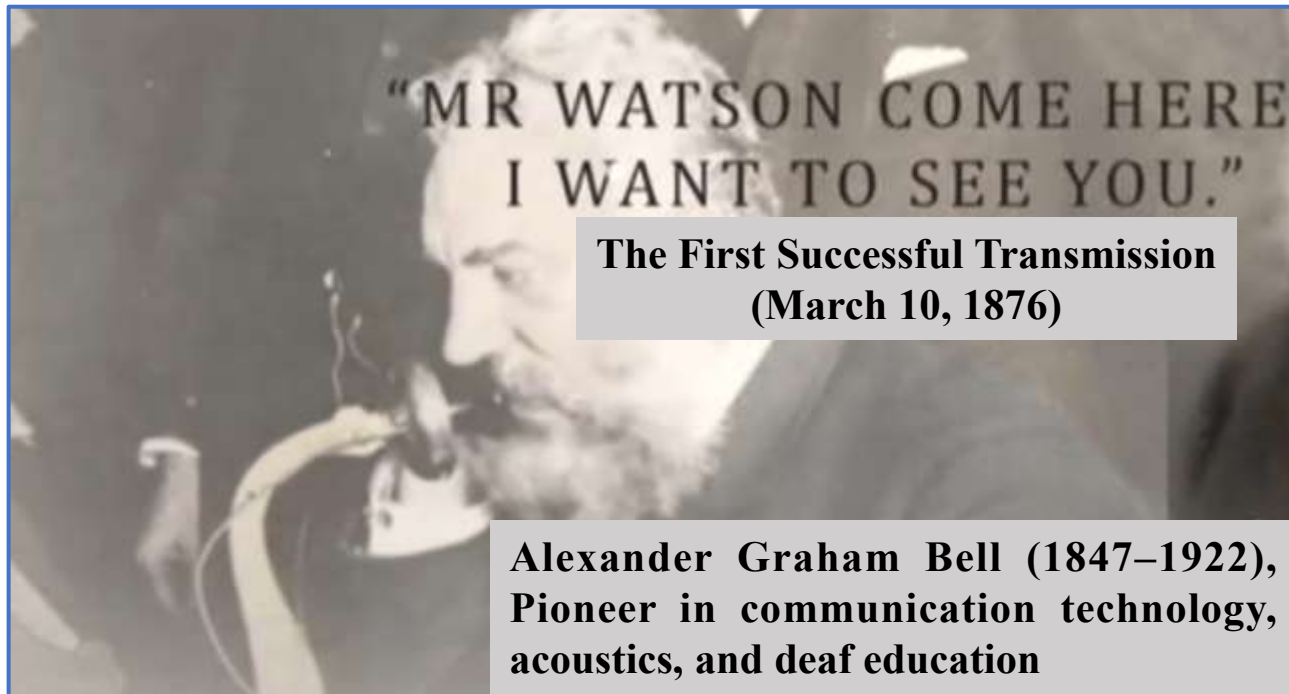
**(1) Communication system: uses electrical signals to transmit messages**



**In 1844, Morse successfully transmitted the first telegraph message in history**

# 1. Basic concepts of signals and systems

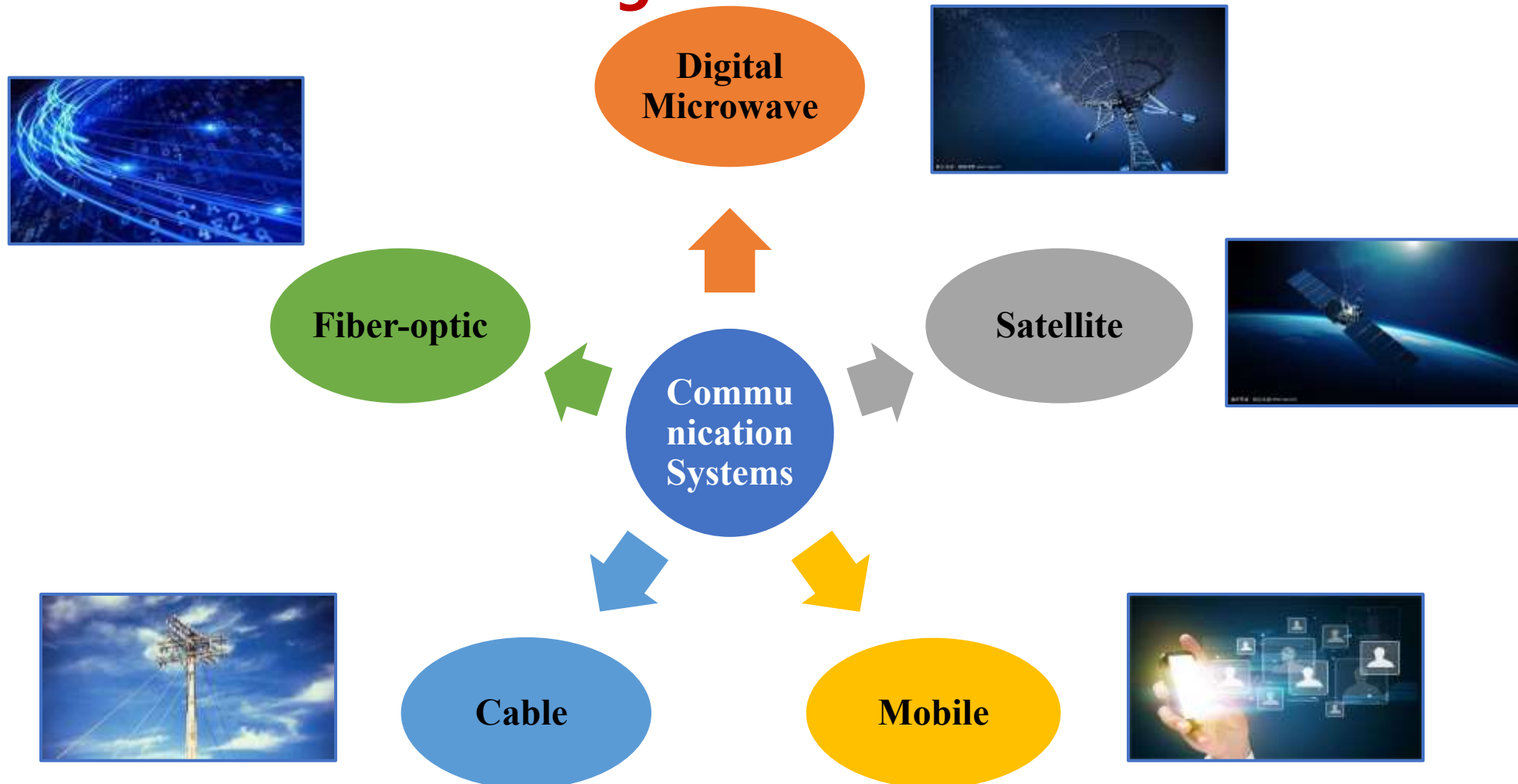
## (1) Communication system: Telephone, transmitting electrical signals through wires



In 1876, Alexander Graham Bell invented the telephone, marking a new era in communication

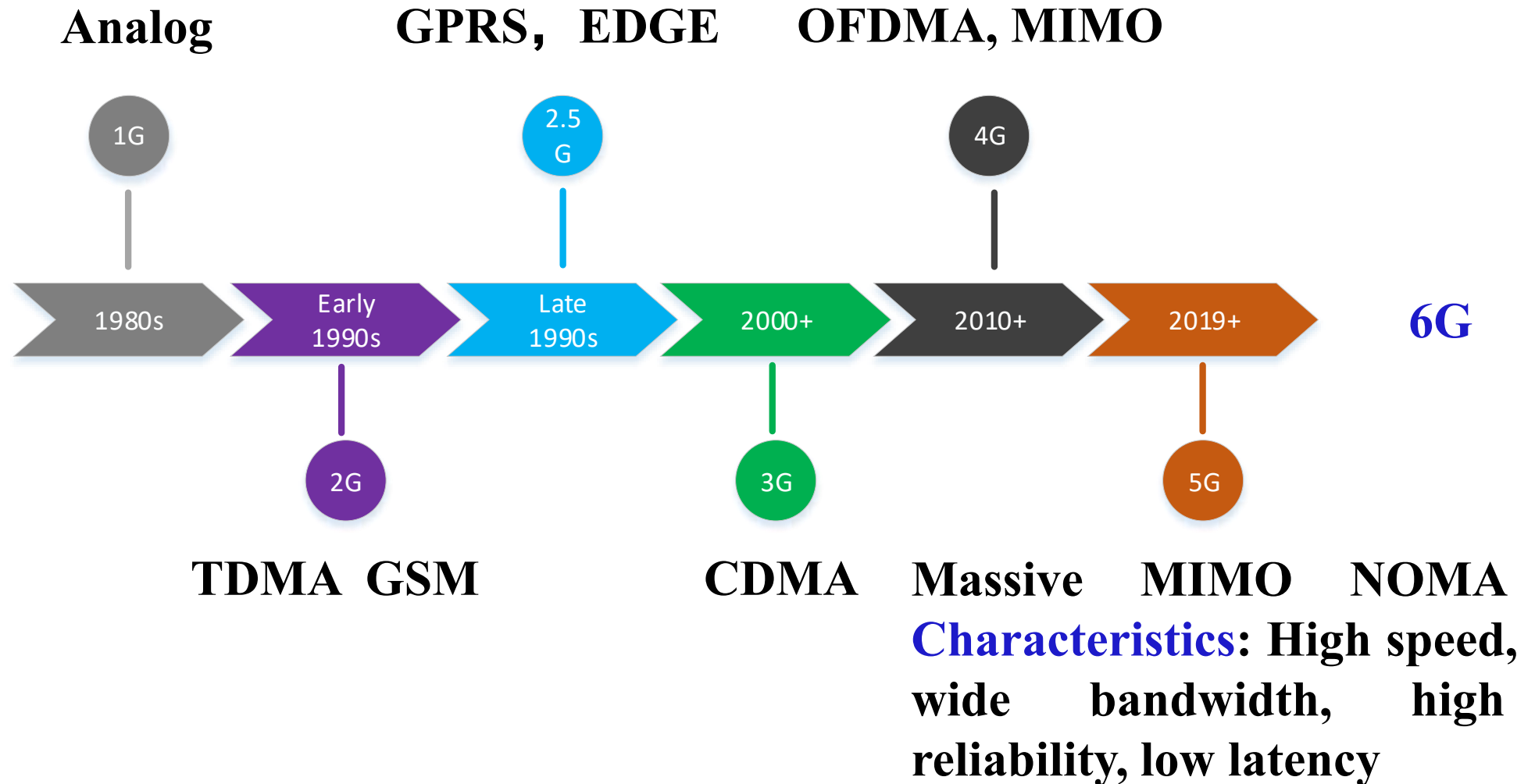
# 1. Basic concepts of signals and systems

## (1) Communication system: Evolution of Modern Communication Technologies



# 1. Basic concepts of signals and systems

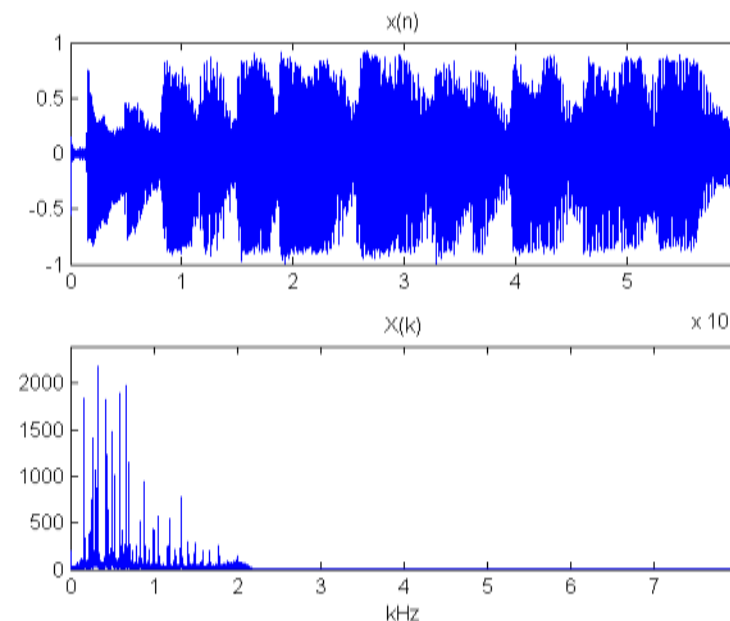
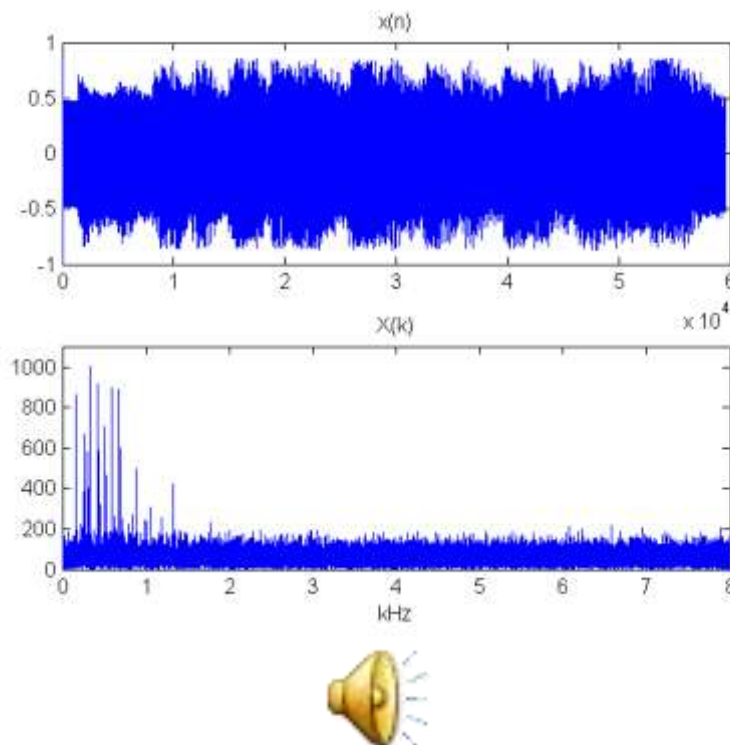
## (1) Communication system: Mobile Communication



# 1. Basic concepts of signals and systems

## (2) Signal Processing: Noise Reduction

- A segment of music mixed with noise, severely affecting listening quality
- Through **filtering processing**, the purpose of noise reduction is achieved



# 1. Basic concepts of signals and systems

## (2) Signal Processing: Noise Reduction



**Objective:** To transform signals into forms that facilitate analysis and identification, enabling efficient estimation and selection of their characteristic parameters

# 1. Basic concepts of signals and systems

## 3. Learning method

Focus on the correspondence between **physical concepts** and **mathematical analysis**, do not calculate blindly

Pay attention to the **physical interpretation** of analysis results, the physical meaning when various parameters change, and the consequences they produce

The same problem may have **multiple solutions**; one should seek the simplest and most reasonable solution, and compare the advantages and disadvantages of each method

Even a considerable time after completing this course, it is still necessary to **repeatedly study** the basic concepts of this course

# 1. Basic concepts of signals and systems

## 4. Reference Book



## Main Content

### 1. Signal Representation Methods

### 2. Description and Classification of Signals (by **Temporal Characteristics**)

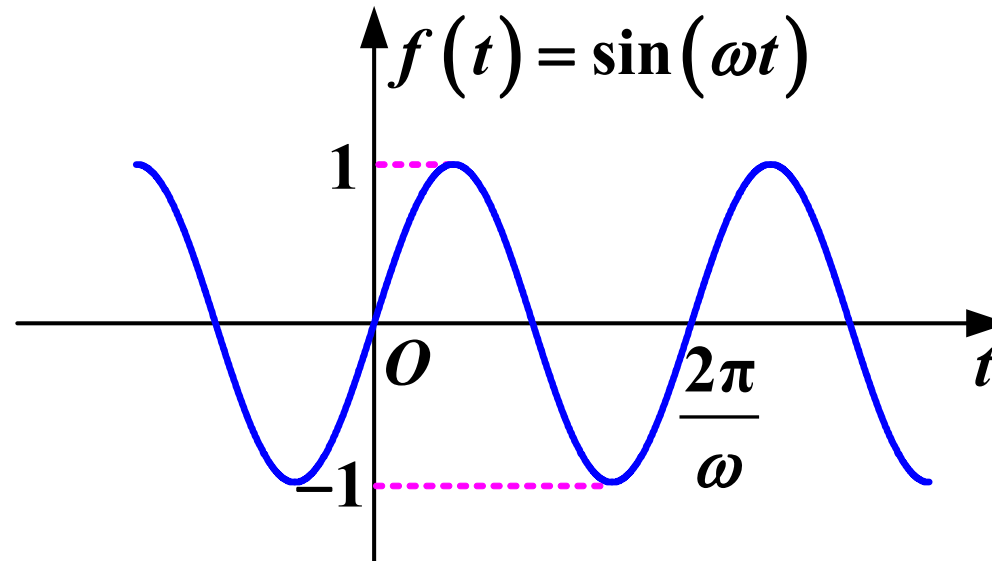
- (1) Deterministic Signals vs. Random Signals
- (2) Continuous Signals vs. Discrete Signals
- (3) Periodic Signals vs. Aperiodic Signals
- (4) Energy Signals vs. Power Signals
- (5) One-Dimensional Signals vs. Multidimensional Signals

## 2. Description and classification of signals

### 1. Methods for Signal Description

Signals can be represented as functions of one or more independent variables

- Graphical representation (waveform plotting)
- Tabular representation



## 2. Classification of Signals

By **Temporal Characteristics:**

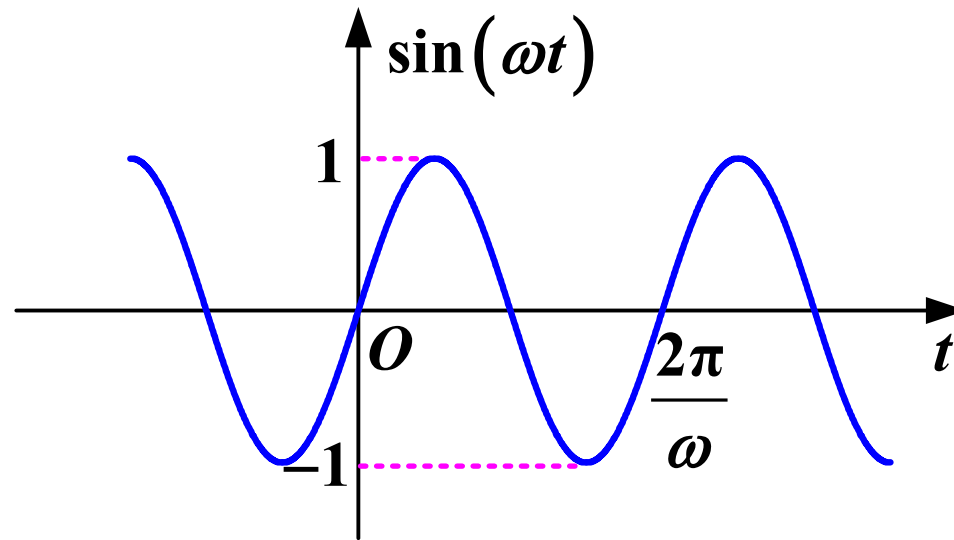
- (1) Deterministic Signals vs. Random Signals
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## 2. Description and classification of signals

### (1) Deterministic Signals vs. Random Signals

#### Deterministic Signals

For any given time instant  $t$ , the signal value  $f(t)$  is precisely specified

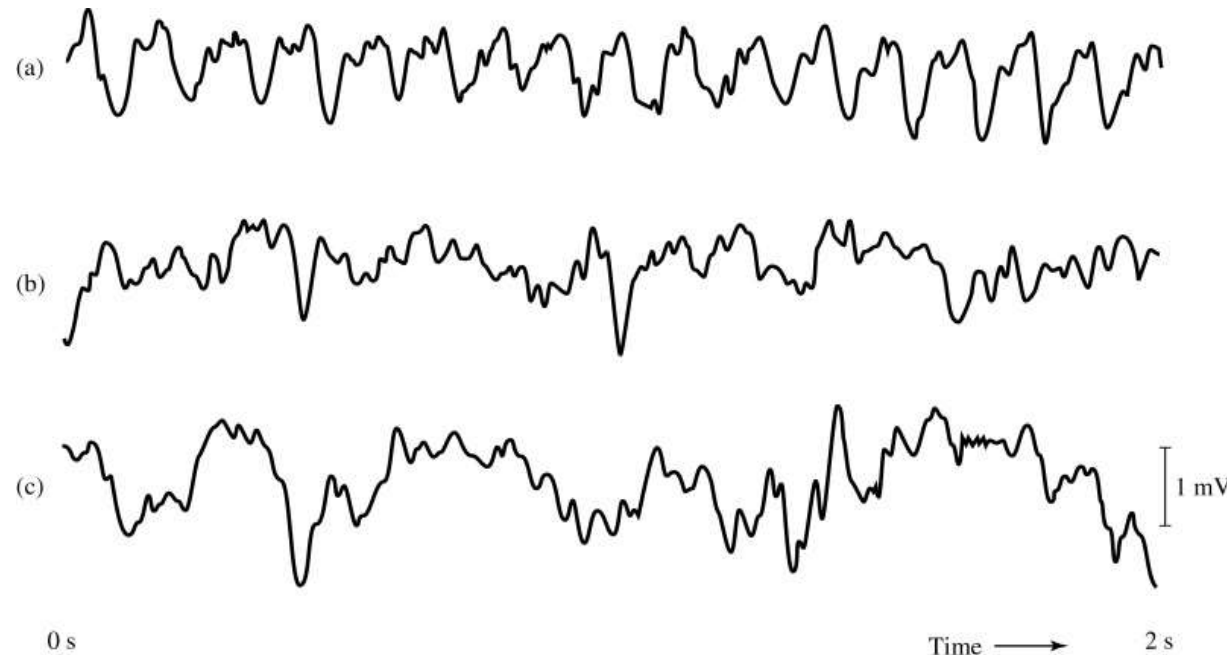


## 2. Description and classification of signals

### (1) Deterministic Signals vs. Random Signals

#### Random Signals

Signals exhibiting unpredictable uncertainty, where instantaneous values cannot be precisely determined beforehand



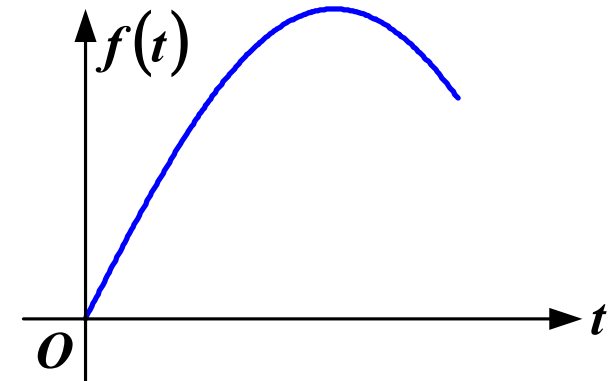
**Rodent Electroencephalogram Signals**

## 2. Description and classification of signals

### (2) Continuous Signals vs. Discrete Signals

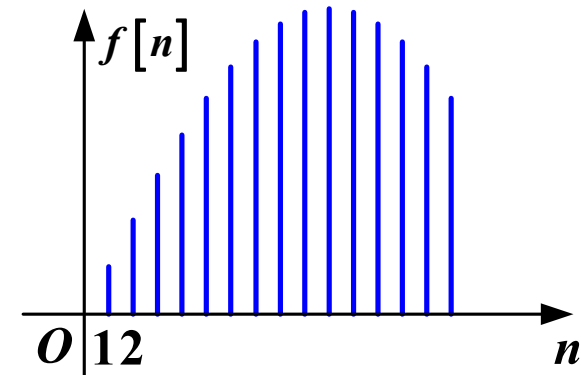
#### Continuous-Time Signals

Within the time domain of the signal's existence, the signal is defined at every instant (except for a finite number of discontinuity points), meaning a deterministic function value can be assigned at each point  
Denoted by the continuous-time variable  $t$



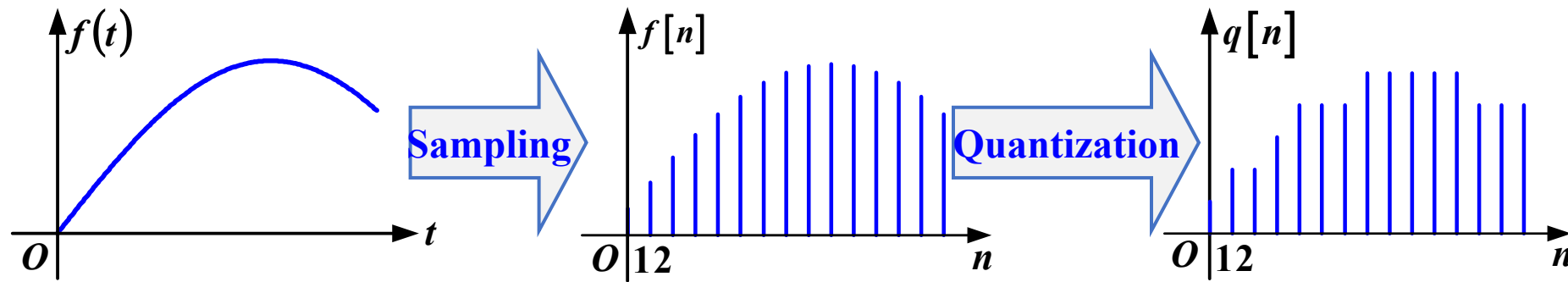
#### Discrete-Time Signals

Temporally discrete, these signals are defined only at specific, non-consecutive instants in time and are undefined at all other times  
Denoted by the discrete-time variable  $n$



## 2. Description and classification of signals

### (2) Continuous Signals vs. Discrete Signals



**Continuous Signal:**

**Time** continuous

**Amplitude** continuous

**Sampled Signal:**

**Time** discrete

**Amplitude** continuous

**Digital Signal:**

**Time** discrete

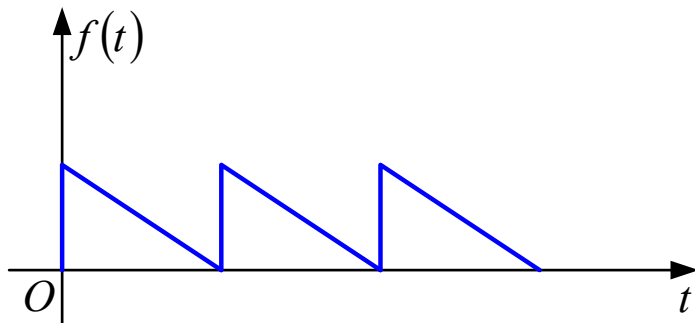
**Amplitude** discrete

## 2. Description and classification of signals

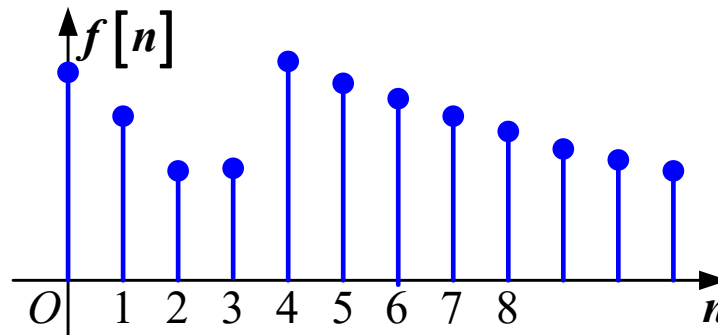
### (2) Continuous Signals vs. Discrete Signals

#### Example 1:

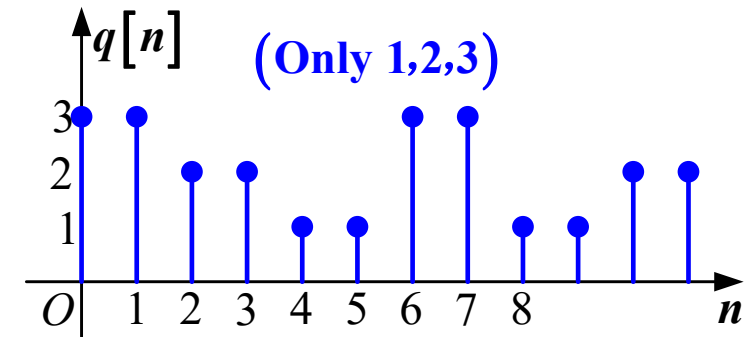
Determine whether the given waveform is a continuous-time signal or discrete-time signal, and if it is a discrete-time signal, identify whether it is a digital signal



Continuous Signal



Discrete Signal



Discrete Signal  
Digital Signal

## 2. Description and classification of signals

### (3) Periodic Signals vs. Aperiodic Signals

#### Periodic Signals

A signal that repeats its pattern indefinitely, with no beginning or end in its idealized form

$$f(t) = f(t + nT) \quad n = 0, \pm 1, \pm 2, \dots (\text{Any integer})$$

**Fundamental Period ( $T$ ):** The **smallest positive value**  $T$

**Units:** Seconds (s)

**Fundamental Frequency ( $f_0$ )**

$$f_0 = \frac{1}{T}$$

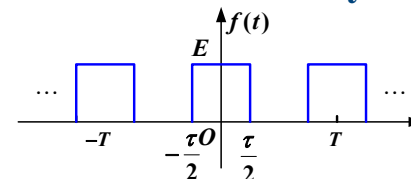
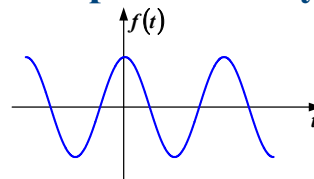
**Units:** Hertz (Hz)

**Fundamental Angular Frequency ( $\omega_0$ )**

$$\omega_0 = 2\pi f_0 = \frac{2\pi}{T}$$

**Units:** Radians per second (rad/s)

Periodic Signals {  
    Sinusoidal Periodic Signal (Harmonic Signal)  
        Single frequency component (pure tone)  
    Complex Periodic Signals (Any periodic signal not purely sinusoidal (e.g., square, triangle waves))  
        Represented by Fourier series: Sum of harmonically related sinusoids



## 2. Description and classification of signals

### (3) Periodic Signals vs. Aperiodic Signals

**Example 2:** Is the signal  $f(t) = \cos 10t + \cos 30t$  periodic?

If yes, determine its fundamental period  $T$  and fundamental angular frequency  $\omega_0$

$$\text{For } \cos 10t: \quad T_1 = \frac{2\pi}{10}$$

$$\text{For } \cos 30t: \quad T_2 = \frac{2\pi}{30}$$

**Check Periodicity Condition:** A sum of periodic signals is periodic if and only if the ratio of their periods is a rational number

$$T = k_1 T_1 = k_2 T_2, \text{ where } k_1 \text{ and } k_2 \text{ are positive integers}$$

$$\frac{T_1}{T_2} = \frac{k_2}{k_1} = 3 \quad \text{is a rational number, therefore it is a periodic signal}$$

## 2. Description and classification of signals

### (3) Periodic Signals vs. Aperiodic Signals

**Example 2:** Is the signal  $f(t) = \cos 10t + \cos 30t$  periodic?

If yes, determine its fundamental period  $T$  and fundamental angular frequency  $\omega_0$

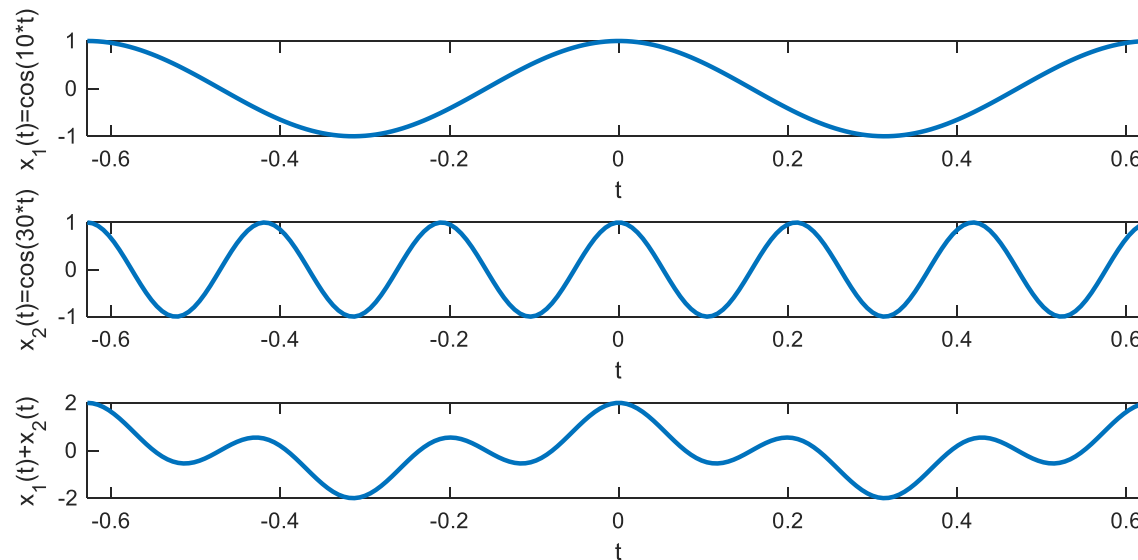
The minimum value of  $k_1$  is 1

Fundamental Frequency

$$T = k_1 T_1 = \frac{2\pi}{10}$$

Fundamental Angular Frequency

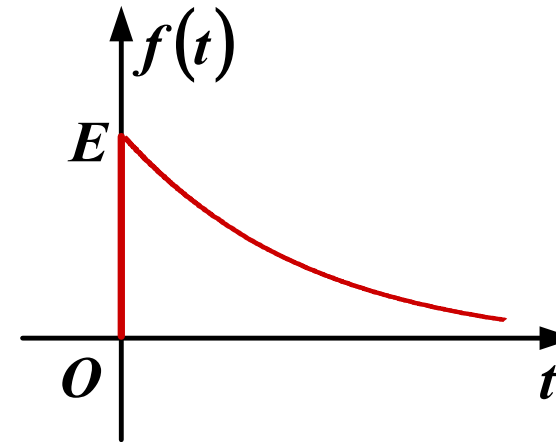
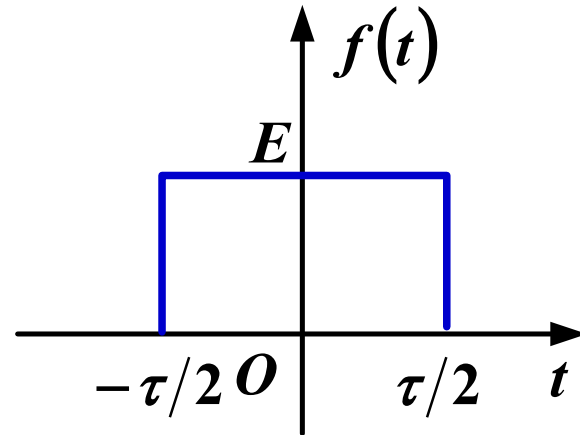
$$\omega = \frac{2\pi}{T} = 10$$



## 2. Description and classification of signals

### (3) Periodic Signals vs. Aperiodic Signals

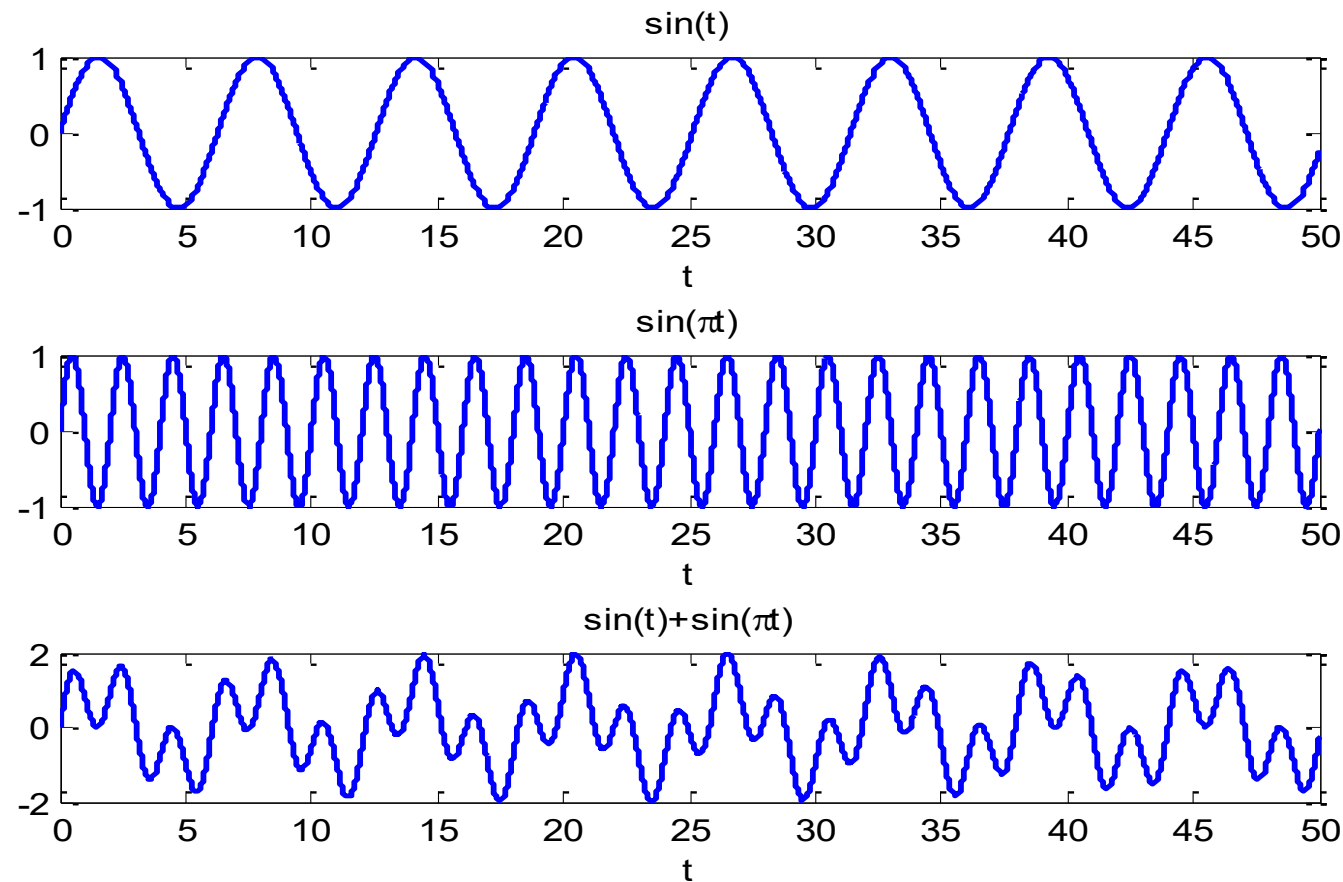
Aperiodic signals { Transient signals (Pulses, Decaying functions)  
Quasi-periodic signals



## 2. Description and classification of signals

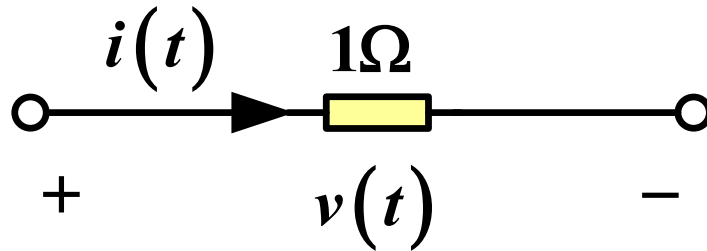
### (3) Periodic Signals vs. Aperiodic Signals

Quasi-periodic signal:  $\sin(\pi t) + \sin(t)$



## 2. Description and classification of signals

### (4) Energy Signals vs. Power Signals



Normalized **instantaneous power**

$$p(t) = |f(t)|^2$$

**Normalized energy** of signal  $f(t)$  (or simply signal **energy**)

The energy dissipated when signal voltage (or current) is applied to a  $1\Omega$  resistor

$$E = \int_{-\infty}^{\infty} |f(t)|^2 dt$$

#### **Average Power of Signal:**

The power dissipated by signal voltage (or current) across a  $1\Omega$  resistor. The average power over the entire time domain is:

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt$$

## 2. Description and classification of signals

### (4) Energy Signals vs. Power Signals

**Energy signal:** A signal with finite total energy over the time interval  $(-\infty, \infty)$  but zero average power

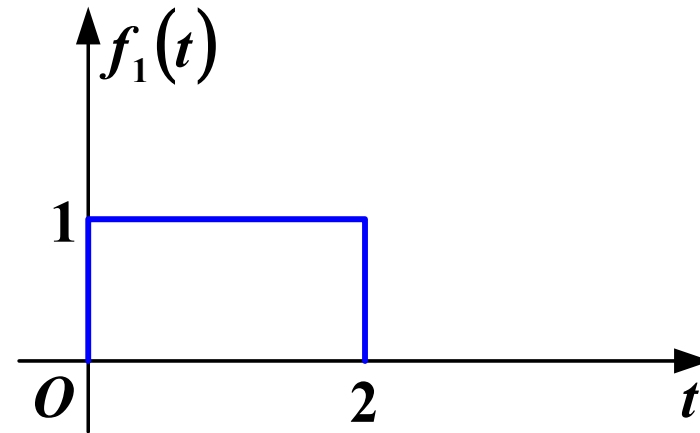
Most time-limited signals are energy signals

**Example 3:**

$$E_1 = \int_0^2 1^2 dt = 2$$

$$0 < E_1 < \infty$$

$$P_1 = 0$$



$f_1(t)$  is the energy signal

## 2. Description and classification of signals

### (4) Energy Signals vs. Power Signals

**Power signal:** A signal with infinite total energy over the time interval  $(-\infty, \infty)$  but finite average power

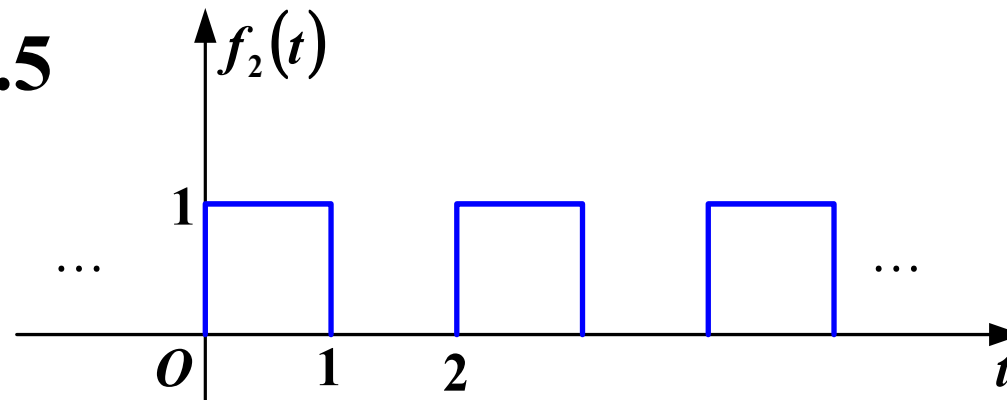
Examples include periodic signals, step functions, and signum functions

**Example 4:**

$$P_2 = \frac{1}{2} \int_0^1 1^2 dt = 0.5$$

$$0 < P_2 < \infty$$

$$E_2 = \infty$$

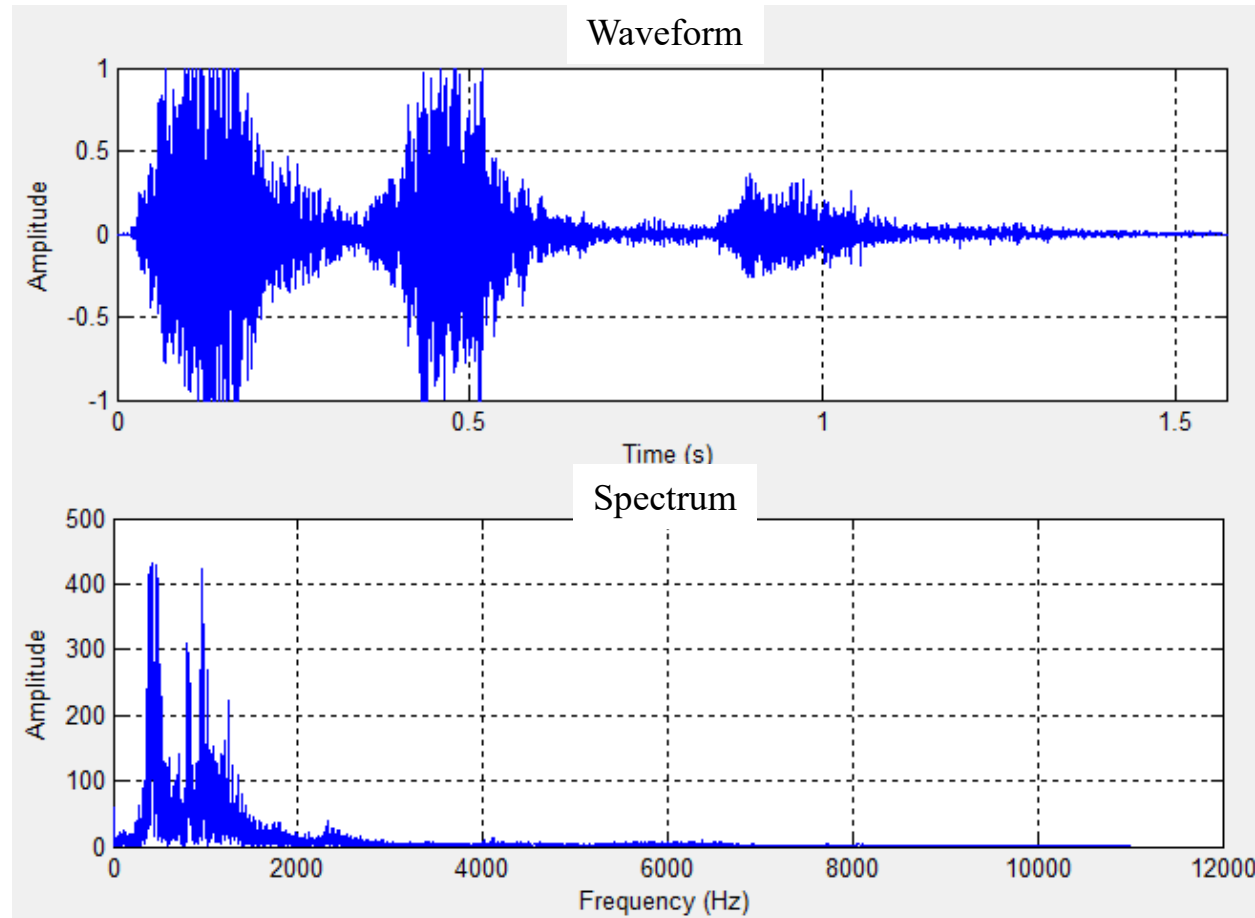


$f_2(t)$  is the power signal

## 2. Description and classification of signals

### (5) One-Dimensional Signals vs. Multidimensional Signals

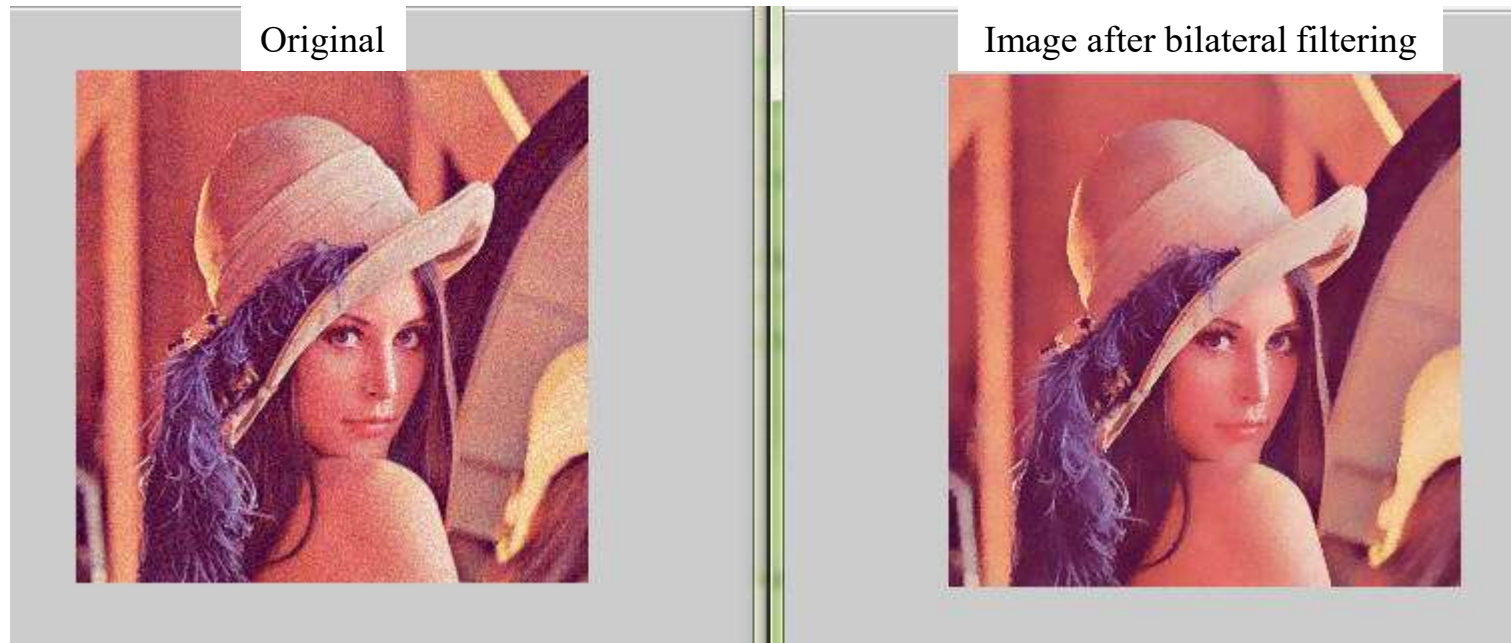
**One-dimensional signal:** A signal represented by a single independent variable (typically time or space), such as an audio signal



## 2. Description and classification of signals

### (5) One-Dimensional Signals vs. Multidimensional Signals

**Multidimensional signal:** A signal requiring multiple independent variables for full characterization, such as image signals, electromagnetic field distributions, etc.



**Lenna image, a widely used test image in image processing research**

## 2. Description and classification of signals

### (5) One-Dimensional Signals vs. Multidimensional Signals

**Multidimensional Signals:** Such as Wi-Fi signals. Luis Hernan, a Ph.D. student at Newcastle University in the UK, designed a device capable of capturing and visualizing Wi-Fi signals. The system continuously scans wireless networks and translates signal strength into dynamic displays of colored LEDs, with the intensity of each light band representing the corresponding signal strength



## Summary

### Signal Representation Methods

### Description and Classification of Signals (by **Temporal Characteristics**)

- (1) Deterministic Signals vs. Random Signals
- (2) Continuous Signals vs. Discrete Signals
- (3) Periodic Signals vs. Aperiodic Signals
- (4) Energy Signals vs. Power Signals
- (5) One-Dimensional Signals vs. Multidimensional Signals

## Main Content

1. Exponential signal
2. Sinusoidal signal
3. Complex exponential signal
4. Sampling signal
5. Gaussian pulse signal

# 3.1 Typical continuous signals

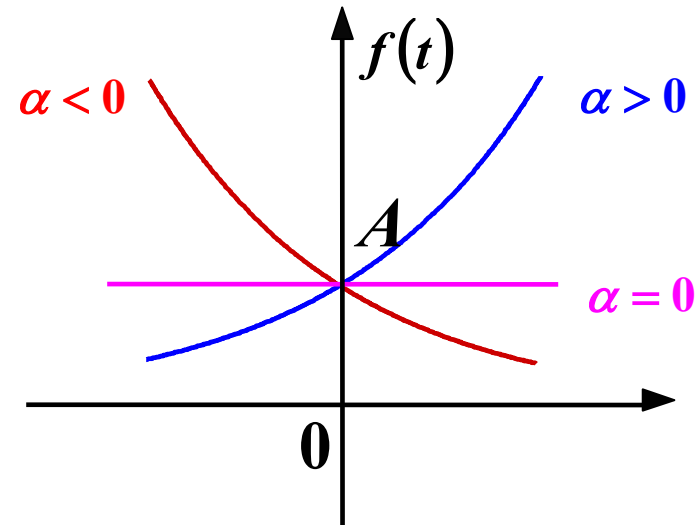
## 1. Exponential signal

$$f(t) = Ae^{\alpha t}$$

$\alpha = 0$  Constants, DC

$\alpha < 0$  ↘

$\alpha > 0$  ↗



**Key property:** Their differentiation and integration with respect to time still yield **exponential** forms

# 3.1 Typical continuous signals

## 1. Exponential signal

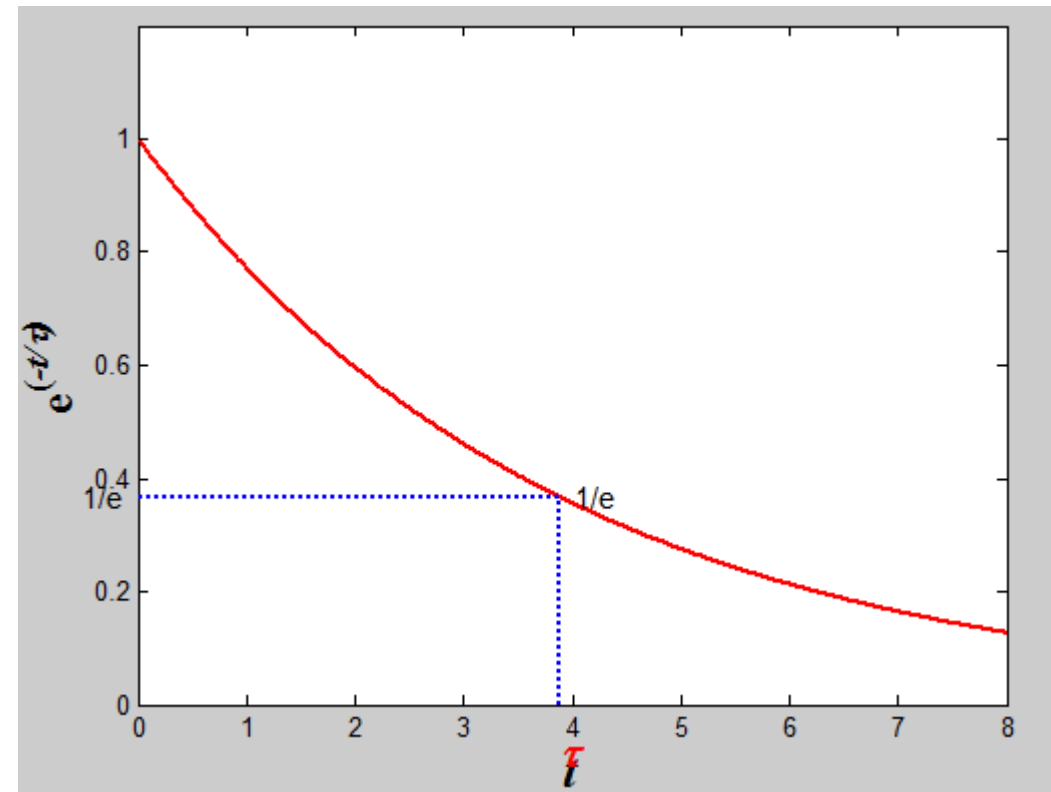
- **Single-sided exponential signal**

$$f(t) = e^{-\frac{t}{\tau}} \quad (t > 0)$$

- $\tau$ , time constant
- Has the dimension of time
- Represents the signal decay rate

When  $t = \tau$

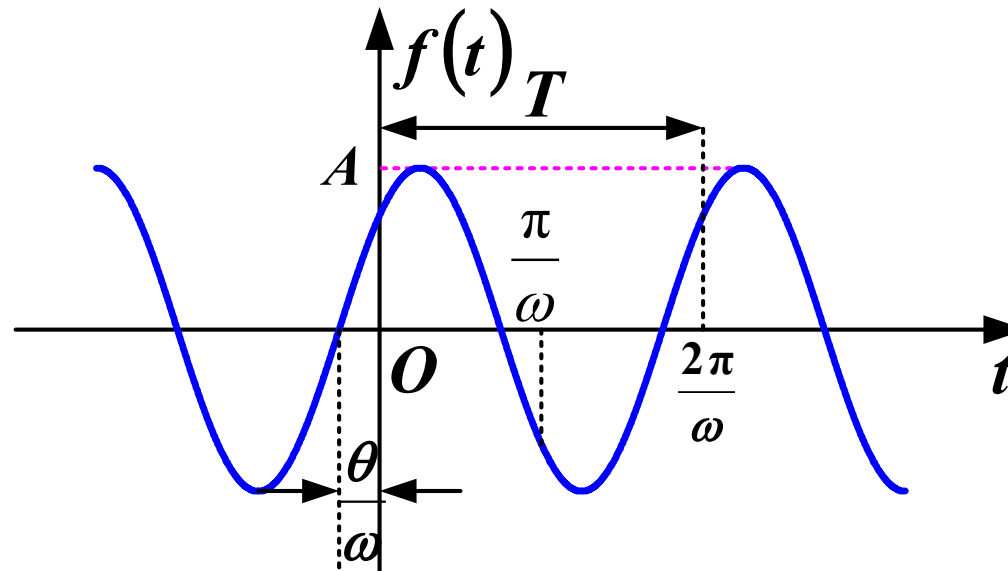
$$f(t) = \frac{1}{e}$$



# 3.1 Typical continuous signals

## 2. Sinusoidal signal

$$f(t) = A \sin(\omega t + \theta)$$



**Amplitude:**  $A$

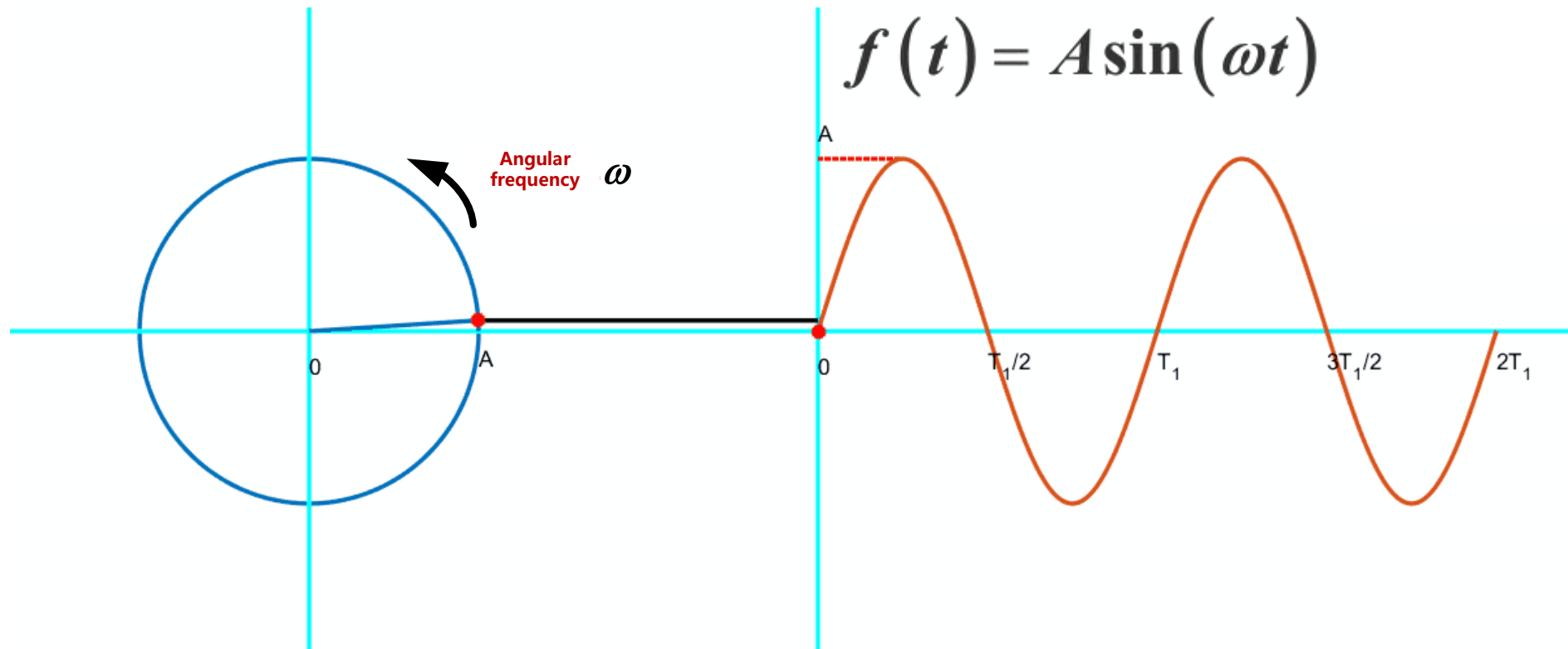
**Angular frequency:**  $\omega$

**Initial phase:**  $\theta$

# 3.1 Typical continuous signals

## 2. Sinusoidal signal

**Angular Frequency:**  $\omega$     **Frequency:**  $f = \frac{\omega}{2\pi}$     **Period:**  $T = \frac{1}{f}$



# 3.1 Typical continuous signals

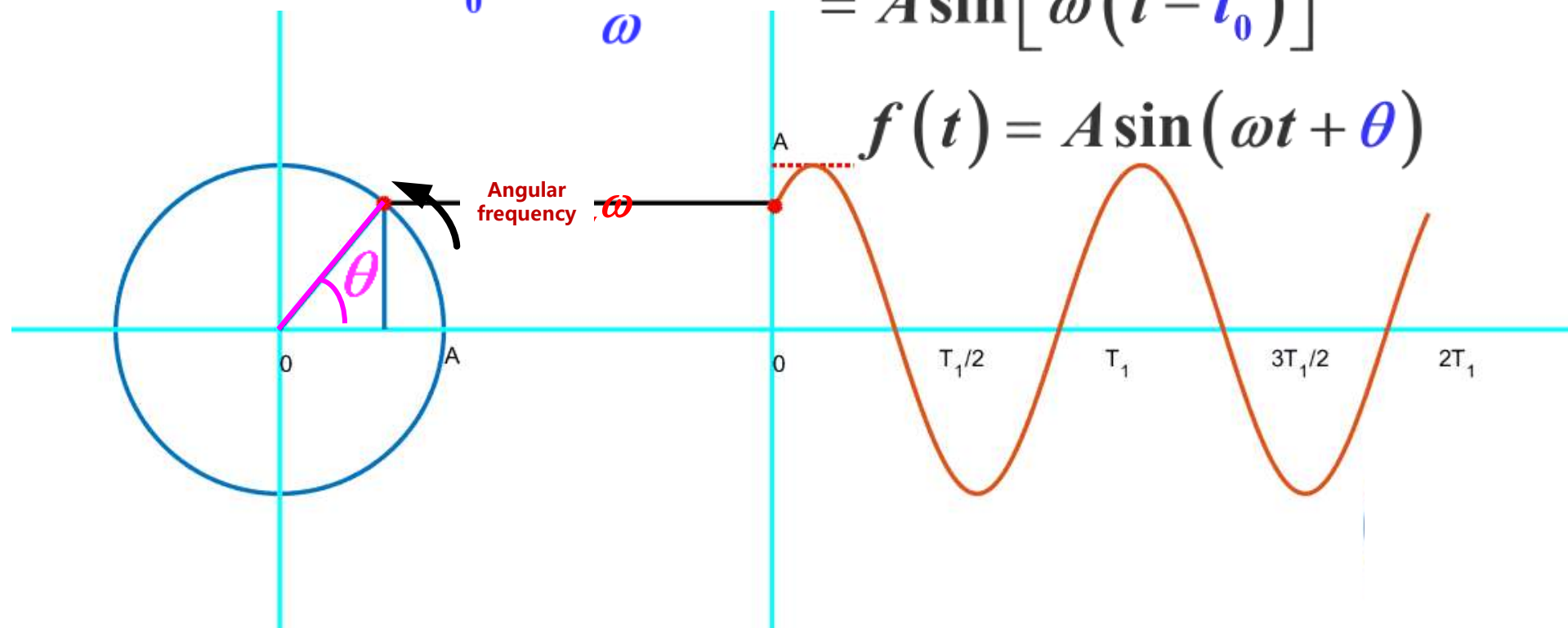
## 2. Sinusoidal signal

- Initial phase

$$f(t) = A \sin(\omega t + \theta) = A \sin \left[ \omega \left( t + \frac{\theta}{\omega} \right) \right]$$

Let  $t_0 = -\frac{\theta}{\omega}$

$$= A \sin \left[ \omega (t - t_0) \right]$$



## 2. Sinusoidal signal

- Euler's Formula



Leonhard Euler (1707–1783), Swiss mathematician and natural scientist. Widely regarded as one of the four greatest mathematicians in history, alongside Archimedes, Newton, and Gauss.

He introduced numerous mathematical notations, including:

- $\pi$  (1736)
- $i$  (imaginary unit, 1777)
- $e$  (base of natural logarithm, 1748)
- $\sin$  and  $\cos$  (1748)
- $\tan$  (1753)
- $\Delta x$  (increment notation, 1755)
- $\Sigma$  (summation symbol, 1755)
- $f(x)$  (function notation, 1734)

(Note: Euler's formula,  $e^{i\theta} = \cos\theta + i\sin\theta$ , is hailed as the "bridge of mathematics" for its profound connection between exponential and trigonometric functions, with far-reaching impacts in signal analysis, quantum mechanics, and beyond.)

# 3.1 Typical continuous signals

## 2. Sinusoidal signal

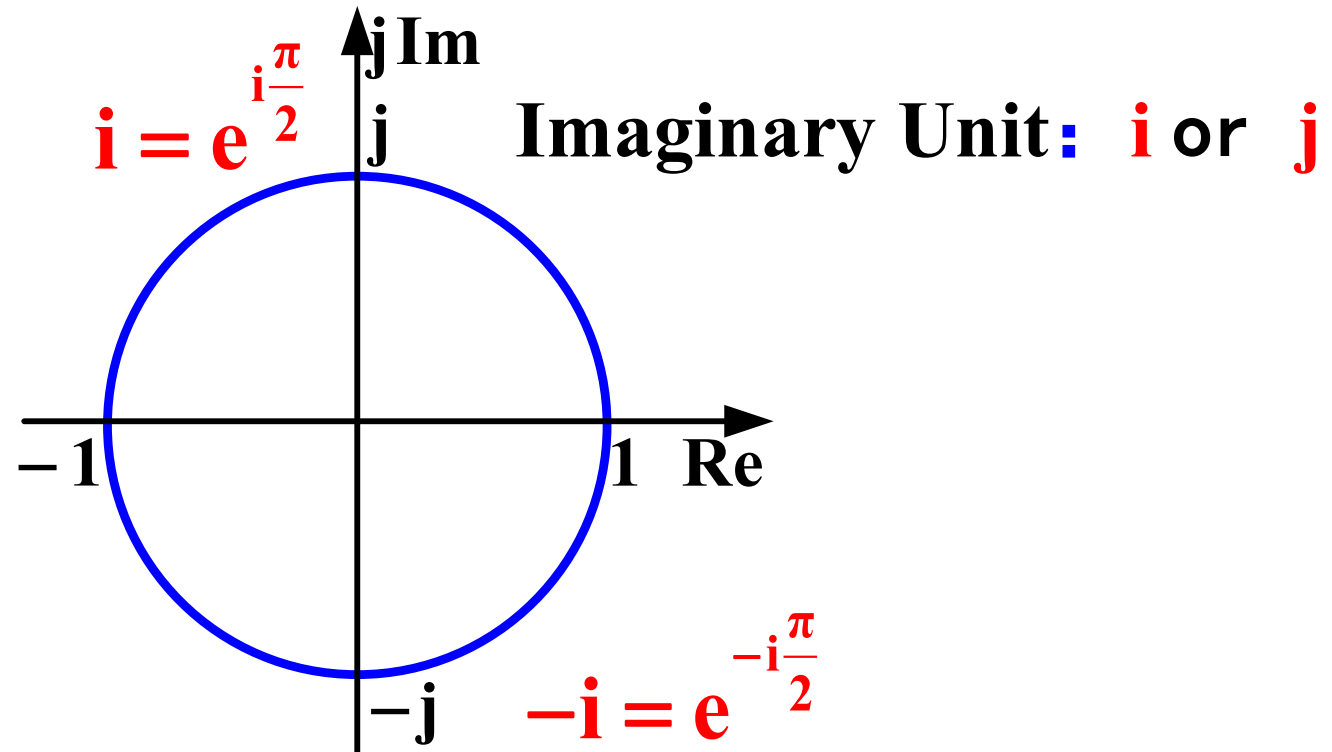
- Euler's Formula

$$-1 = e^{i\pi} = e^{i(-\pi)}$$

Euler's identity

$$e^{i\pi} + 1 = 0$$

**God's Formula**



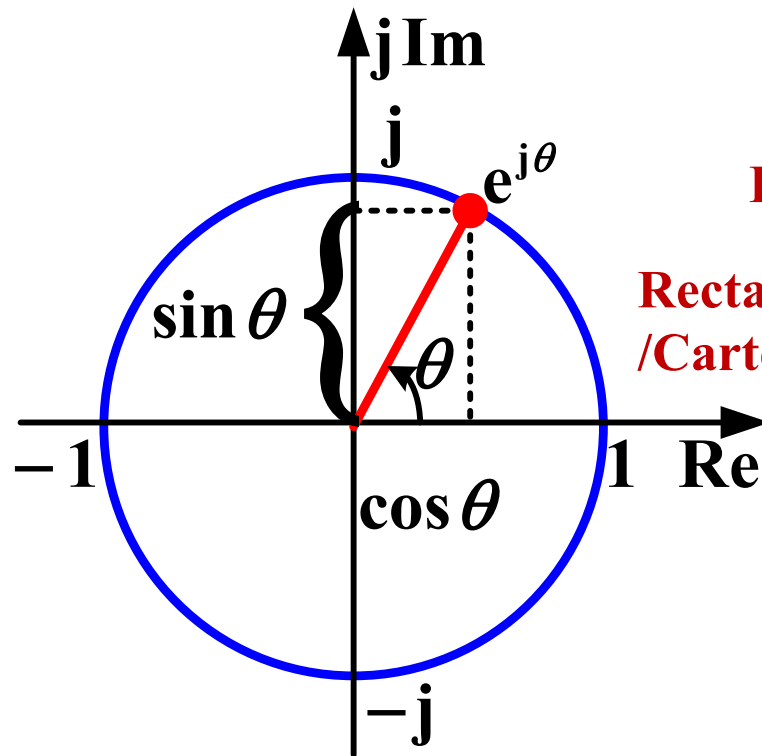
**Complex Plane**

# 3.1 Typical continuous signals

## 2. Sinusoidal signal

- **Euler's Formula**

On the complex plane, a ray from the origin to a point on the unit circle, forming an angle  $\theta$  with the real axis, can be expressed as



**Polar Form:**  $1 \angle \theta = e^{j\theta}$

**Rectangular / Cartesian Form:**  $\cos \theta + j \sin \theta$

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$e^{-j\theta} = \cos \theta - j \sin \theta$$

# 3.1 Typical continuous signals

## 2. Sinusoidal signal

- Euler's Formula

Let  $\theta = \omega t$

$$\begin{cases} e^{j\omega t} = \cos(\omega t) + j\sin(\omega t) \\ e^{-j\omega t} = \cos(\omega t) - j\sin(\omega t) \end{cases}$$

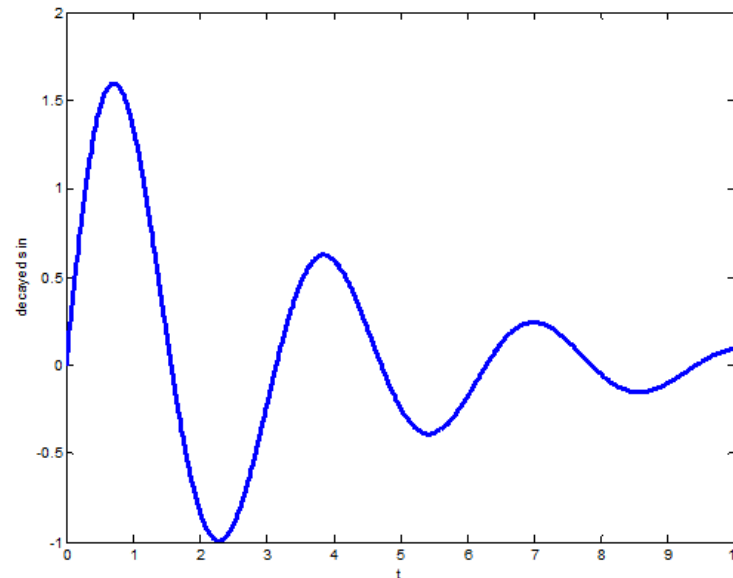
$$\begin{cases} \cos(\omega t) = \frac{1}{2}(e^{j\omega t} + e^{-j\omega t}) \\ \sin(\omega t) = \frac{1}{2j}(e^{j\omega t} - e^{-j\omega t}) \end{cases}$$

The Bridge Between Complex Exponential Functions and Trigonometric Functions

## 2. Sinusoidal signal

- **Single-Sided Damped Sinusoidal Signal**

$$f(t) = \begin{cases} K e^{-\alpha t} \sin(\omega t) & t \geq 0 \\ 0 & t < 0 \end{cases} \quad (\alpha > 0)$$



# 3.1 Typical continuous signals

## 3. Complex exponential signal

$$f(t) = Ke^{st} \quad (-\infty < t < \infty)$$

- $s = \sigma + j\omega$  is a complex number, known as complex frequency
- $\sigma, \omega$  are all constants
- $\sigma$ , the dimension is  $1/s$ ,  $\omega$ , the dimension is  $\text{rad}/s$

$$\omega = 0$$

$$f(t) = Ke^{\sigma t}$$

Exponential signal

# 3.1 Typical continuous signals

## 3. Complex exponential signal

$$\omega \neq 0$$

$$f(t) = Ke^{st}$$

$$= Ke^{(\sigma + j\omega)t}$$

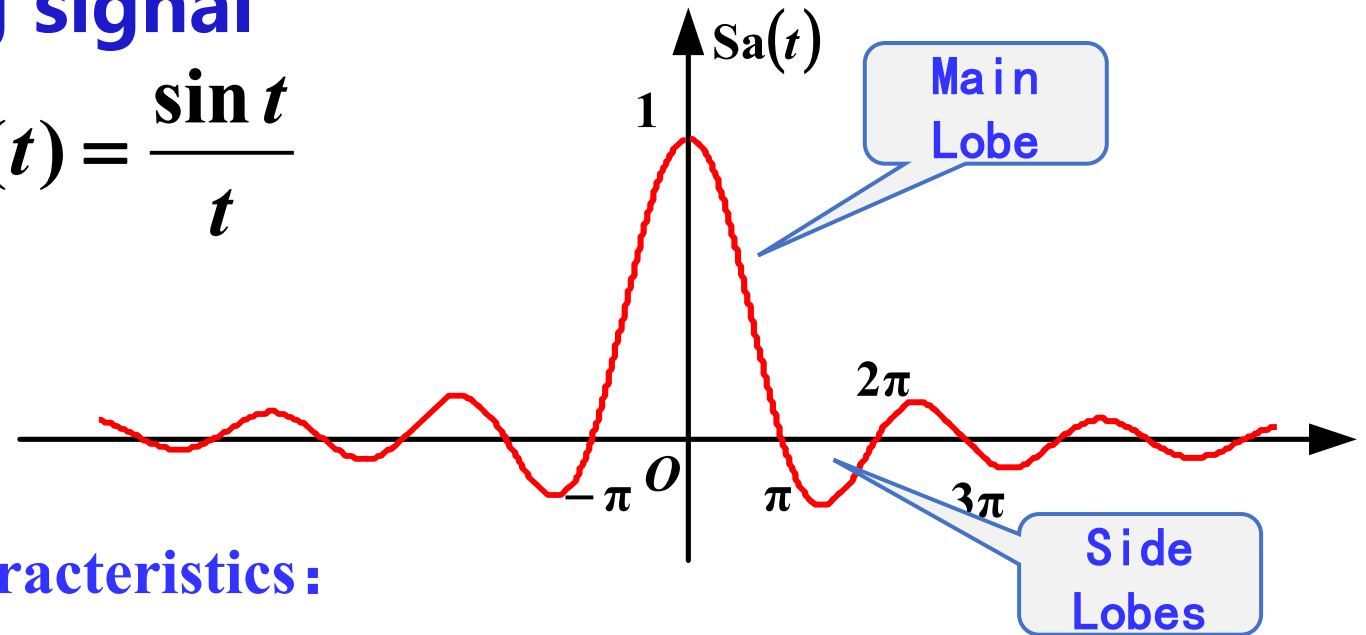
$$= Ke^{\sigma t} \cos(\omega t) + jKe^{\sigma t} \sin(\omega t)$$

$$\text{Real and Imaginary Parts} \left\{ \begin{array}{ll} \sigma = 0, & \text{Constant-Amplitude} \\ \sigma > 0, & \text{Amplifying} \\ \sigma < 0, & \text{Decaying} \end{array} \right\} \text{Oscillation}$$

# 3.1 Typical continuous signals

## 4. Sampling signal

$$Sa(t) = \frac{\sin t}{t}$$



**Characteristics:**

$$Sa(-t) = Sa(t), \text{ Even Function}$$

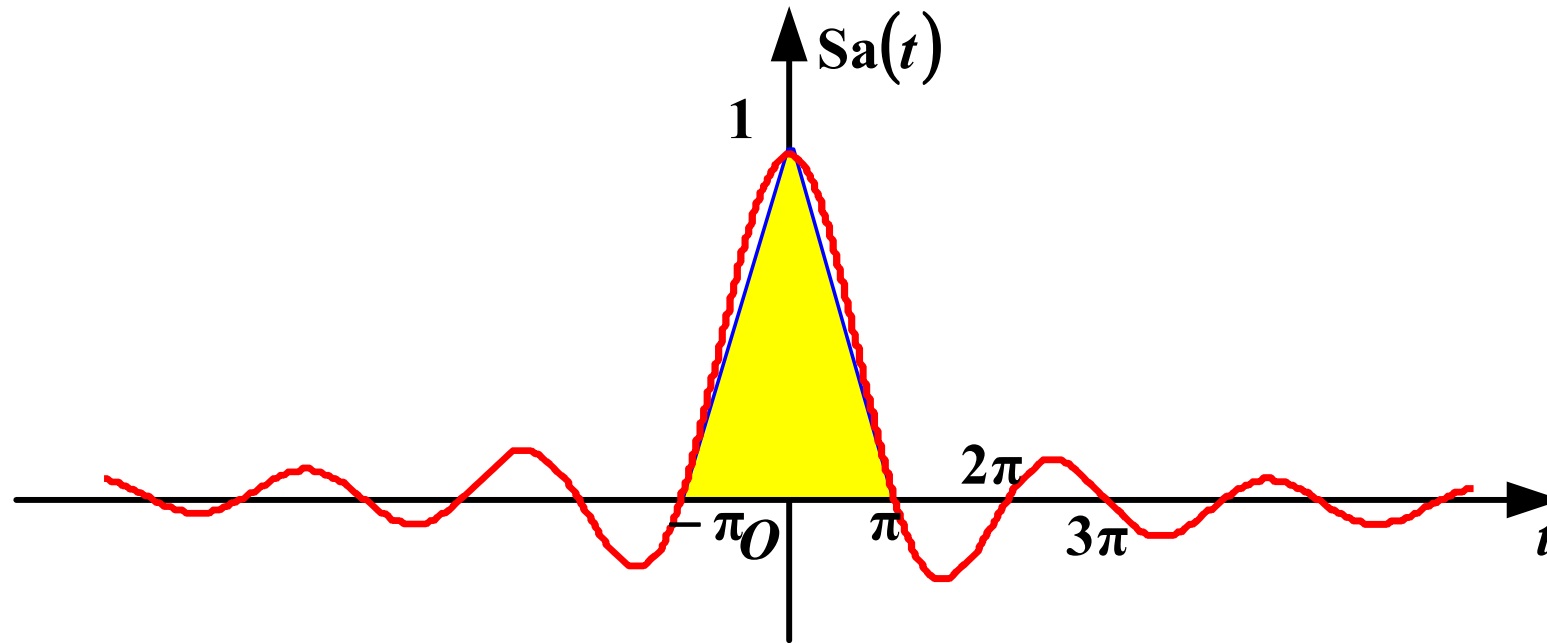
$$t = 0, Sa(t) = 1, \text{ thus } \lim_{t \rightarrow 0} Sa(t) = 1$$

$$Sa(t) = 0, \quad t = \pm n\pi, \quad n = 1, 2, 3 \dots$$

$$\lim_{t \rightarrow \pm\infty} Sa(t) = 0$$

# 3.1 Typical continuous signals

## 4. Sampling signal



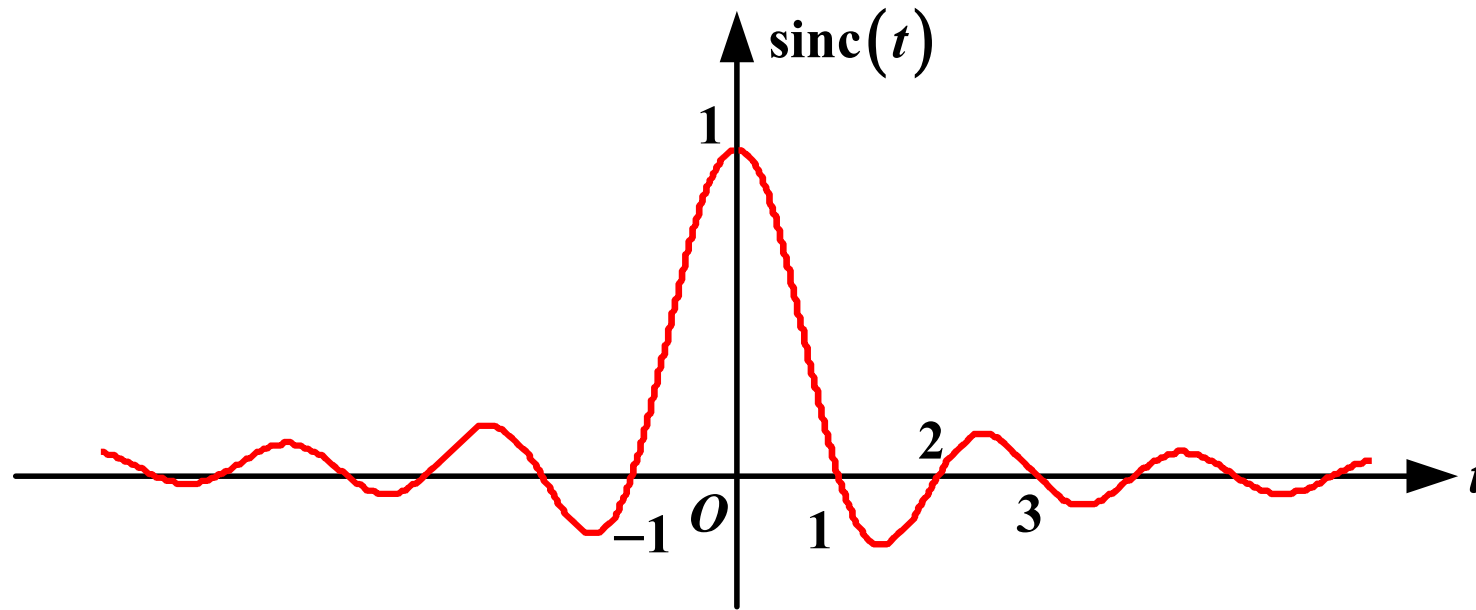
$$\int_0^{\infty} \text{Sa}(t) dt = \frac{\pi}{2}$$

$$\int_{-\infty}^{\infty} \text{Sa}(t) dt = \pi$$

Precisely equals  
the area of the  
inscribed triangle

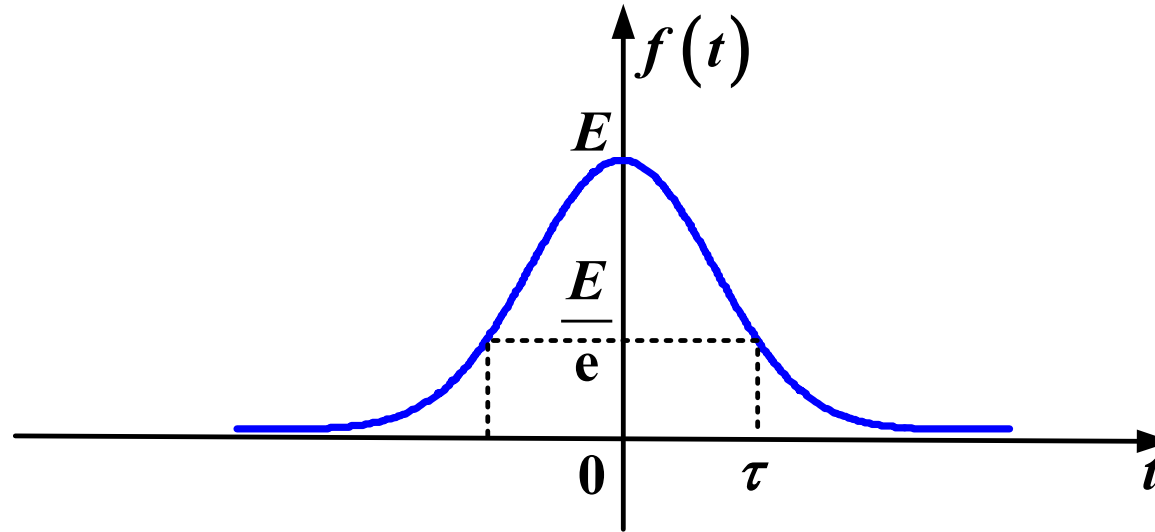
# 3.1 Typical continuous signals

## 4. Sampling signal



$$\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$

## 5. Gaussian pulse signal



$$f(t) = E e^{-\left(\frac{t}{\tau}\right)^2}$$

## Summary

1. Exponential signal
2. Sinusoidal signal
3. Complex exponential signal
4. Sampling signal
5. Gaussian pulse signal

## Main Content

### 1. Dependent Variable Operations

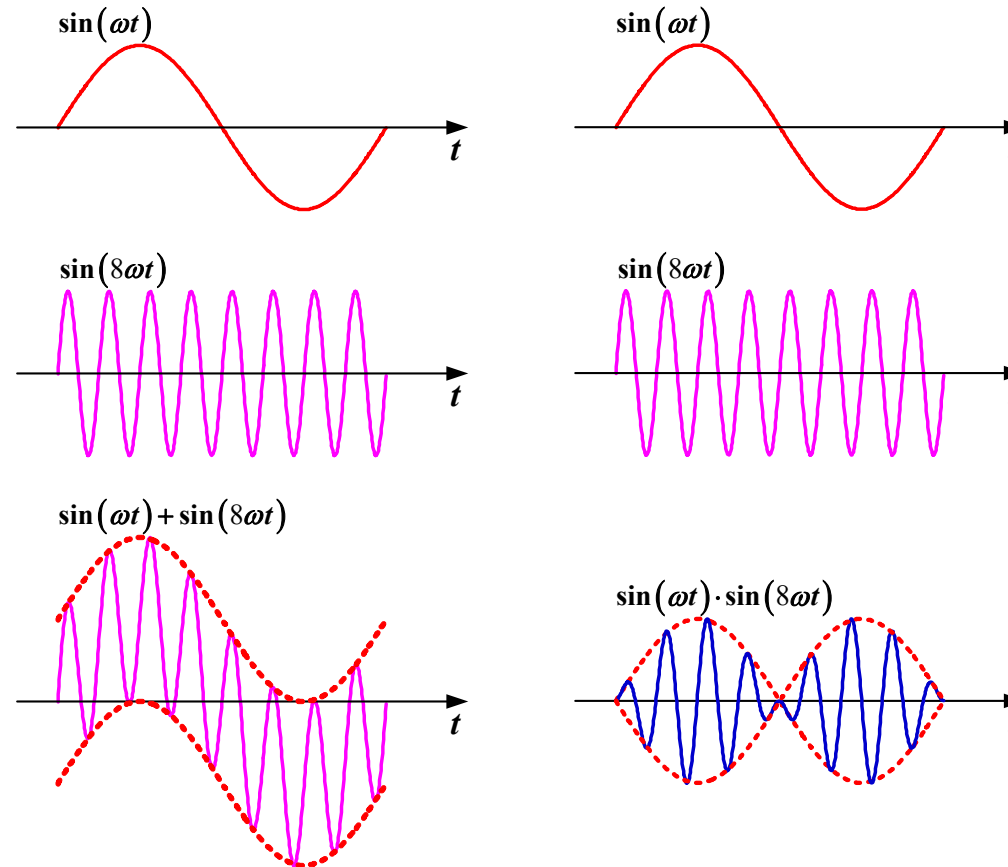
- Addition and Multiplication
- Differentiation and Integration

### 2. Independent Variable Transformations

- Time Shifting (Translation)
- Time Reversal
- Time Scaling

## 1. Addition and Multiplication

The operations where two signals are combined by adding or multiplying their corresponding values at each identical time instant



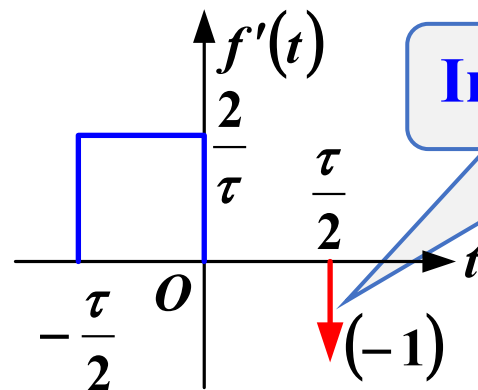
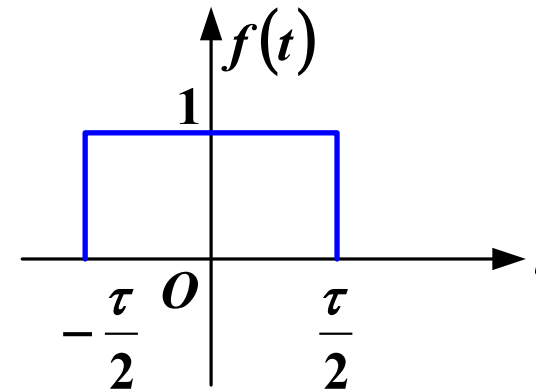
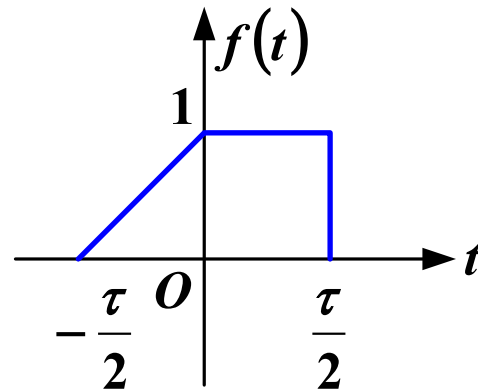
# 3.2 Operations on continuous-time signals

## 2. Differentiation and Integration

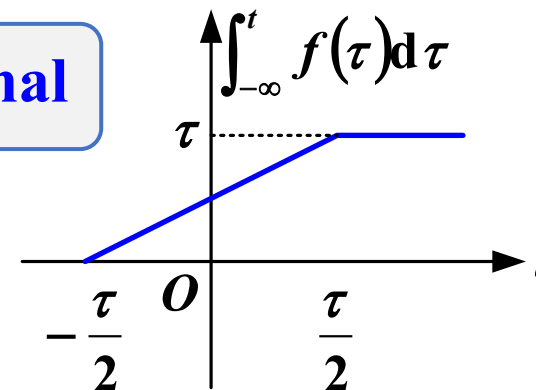
Differentiation:  $f'(t) = \frac{d f(t)}{d t}$

Integration:  $\int_{-\infty}^t f(\tau) d\tau$

Running  
integral



Impulse Signal



### 3. Independent Variable Transformations

Including shifting, reversal, and scaling transformations

$$f(t) \rightarrow f(at + b)$$

Where

$f$  : represents a function, a mapping

$at + b$  : argument

$t$  : independent variable

After transformation, the horizontal coordinate  $t$  of the waveform remains **unchanged**

### (1) Shifting

$$f(t) \rightarrow f(t - \tau)$$

Translate the signal  $f(t)$  along the  $t$ -axis to obtain the shifted signal  $f(t - \tau)$ , where  $\tau$  is a constant

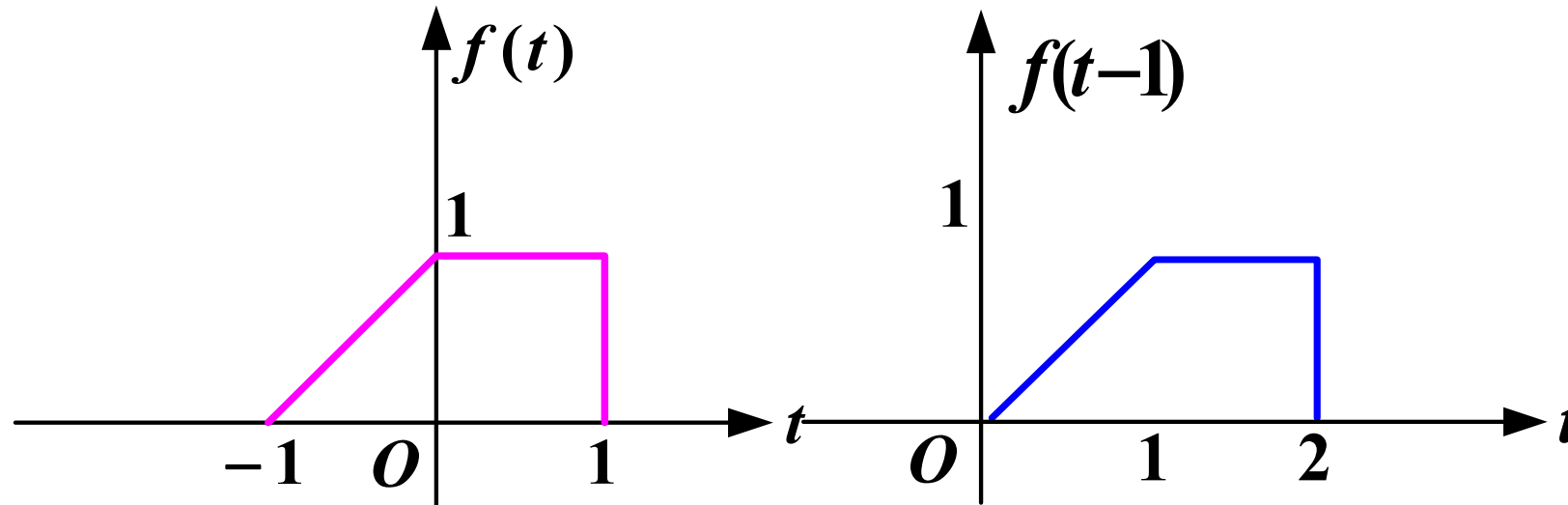
$\tau > 0$ , Right shift (delay)

$\tau < 0$ , Left shift (advance)

## 3.2 Operations on continuous-time signals

### Example 1

Given the waveform of  $f(t)$ , plot the waveform for  $f(t-1)$



**Preserve amplitudes, Identical outputs, Shifting time coordinates**

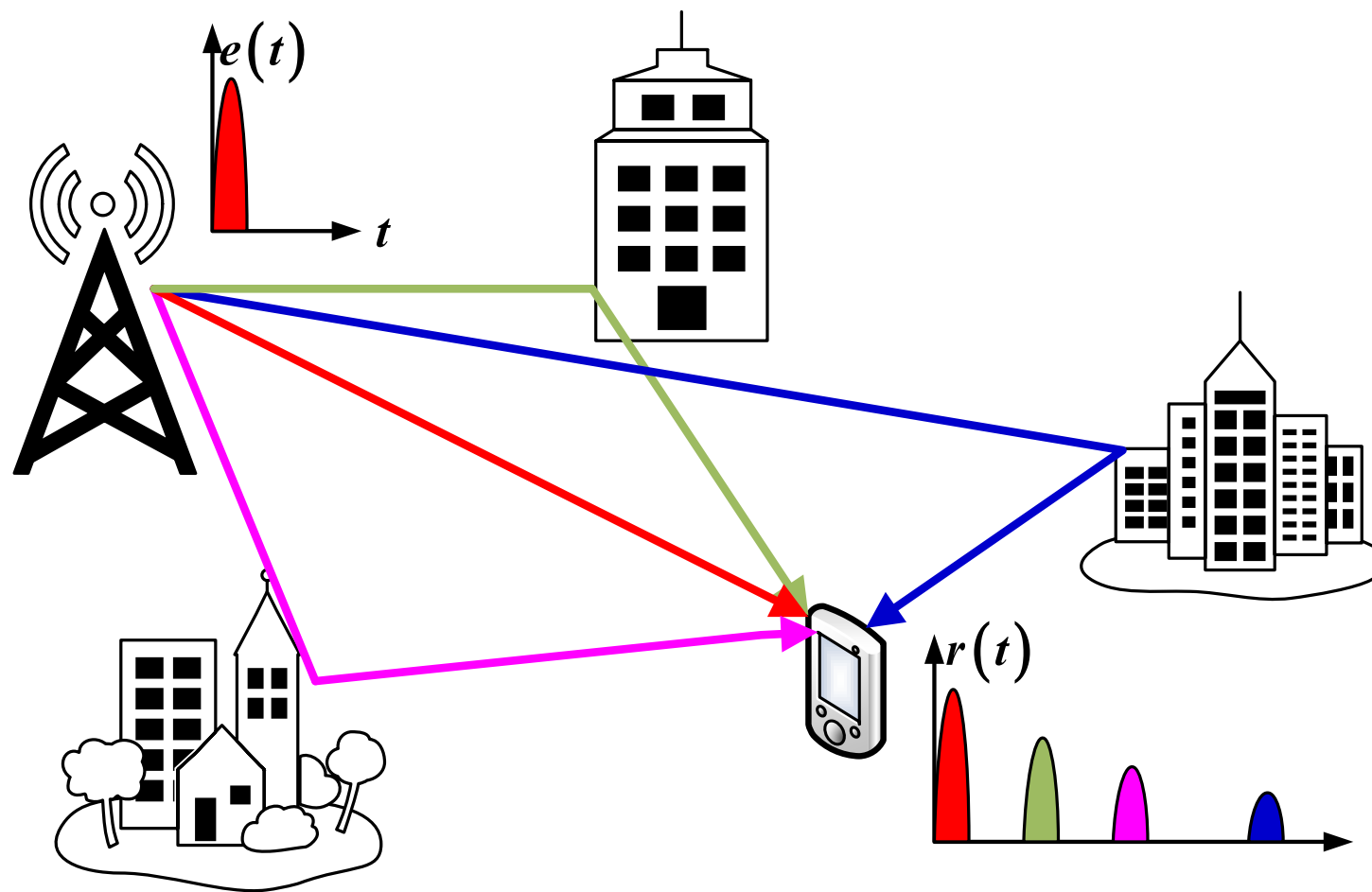
$$\begin{cases} t = 0 \\ f(t) = 1 \end{cases}$$

$$\begin{cases} t - 1 = 0 \\ f(t - 1) = 1 \end{cases}$$

$$\begin{cases} t = 1 \\ f(t - 1) = 1 \end{cases}$$

## 3.2 Operations on continuous-time signals

### Signal Delay Example – Multipath Transmission



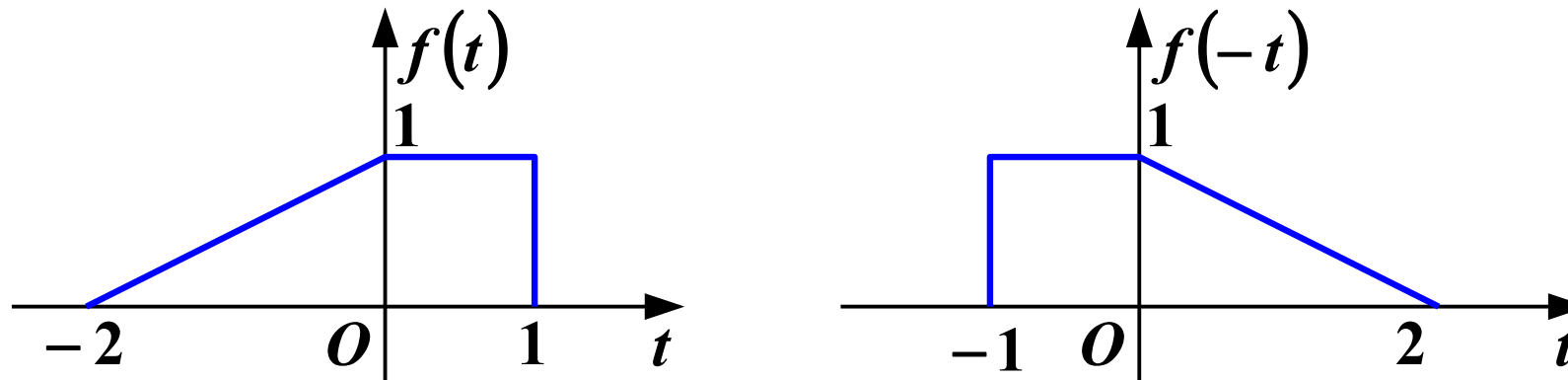
$$r(t) = a_1 e(t - \tau_1) + a_2 e(t - \tau_2) + a_3 e(t - \tau_3) + a_4 e(t - \tau_4)$$

## 3.2 Operations on continuous-time signals

### (2) Reversal

Fold the signal along the vertical axis, swapping its past and future

$$f(t) \rightarrow f(-t)$$



In reality, no physical device can achieve this operation. However, this concept can be implemented in digital signal processing—for example, using a last-in-first-out (LIFO) stack

### (3) Scaling

#### Waveform Compression and Expansion

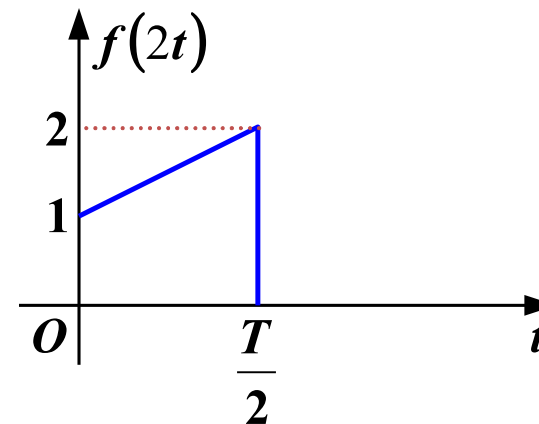
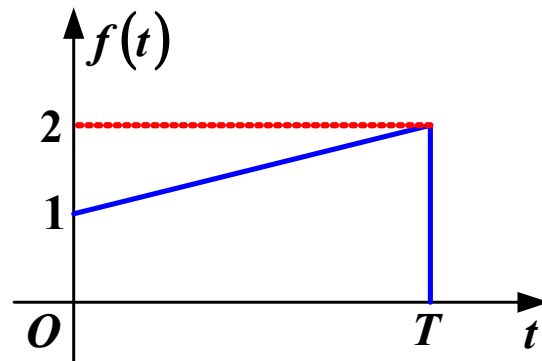
$$f(t) \rightarrow f(at)$$

$a$  is a real-valued scaling factor

## 3.2 Operations on continuous-time signals

### Example 2

Given  $f(t)$ , plot  $f(2t)$  and  $f\left(\frac{t}{2}\right)$ .



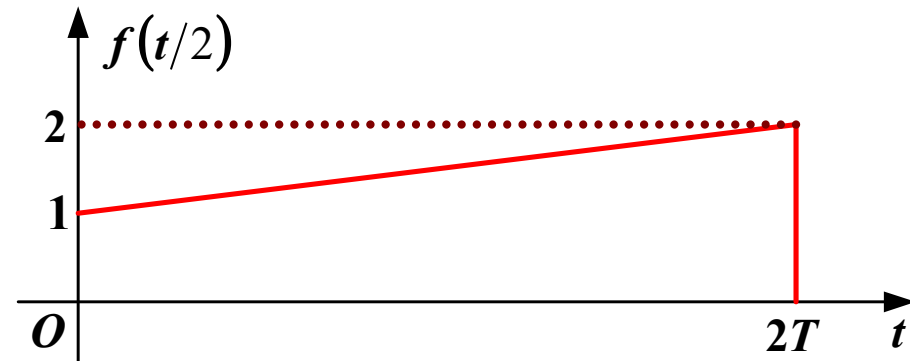
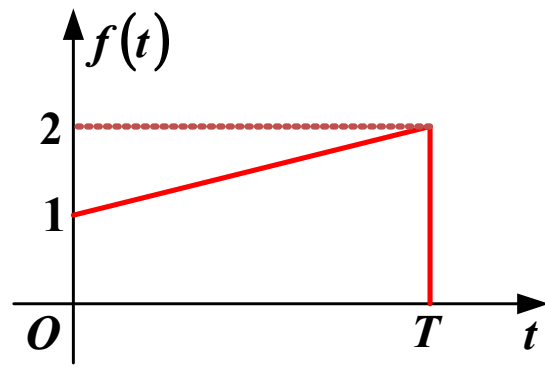
| Same argument, same function value |        |      |         | Find new coordinates |         |
|------------------------------------|--------|------|---------|----------------------|---------|
| $t$                                | $f(t)$ | $2t$ | $f(2t)$ | $t$                  | $f(2t)$ |
| 0                                  | 1      | 0    | 1       | 0                    | 1       |
| $T$                                | 2      | $T$  | 2       | $T/2$                | 2       |

$t \rightarrow 2t$ , Time scale increases, waveform compresses

## 3.2 Operations on continuous-time signals

### Example 2

$$f(t) \rightarrow f\left(\frac{t}{2}\right)$$



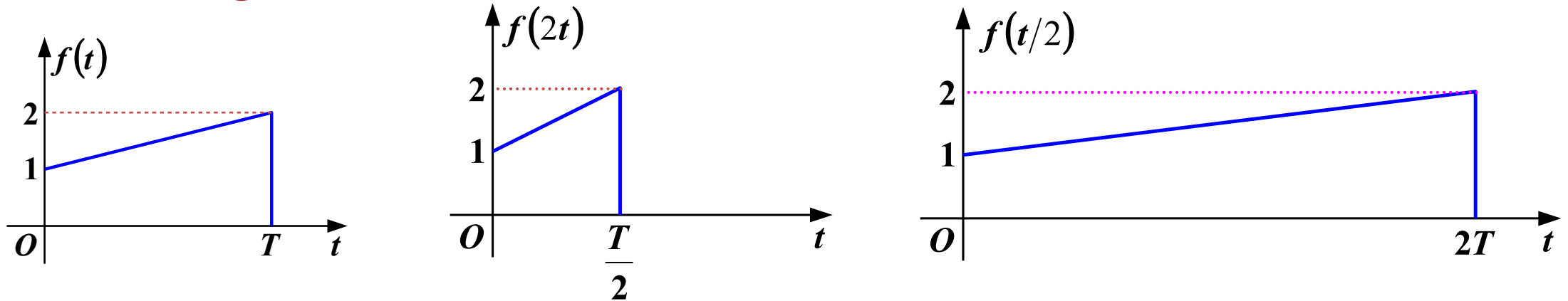
| Same argument, same function value |        |       |          |  | Find new coordinates |          |
|------------------------------------|--------|-------|----------|--|----------------------|----------|
| $t$                                | $f(t)$ | $t/2$ | $f(t/2)$ |  | $t$                  | $f(t/2)$ |
| 0                                  | 1      | 0     | 1        |  | 0                    | 1        |
| $T$                                | 2      | $T$   | 2        |  | $2T$                 | 2        |

$$t \rightarrow \frac{t}{2}$$

Time-scale compression, waveform expansion

## 3.2 Operations on continuous-time signals

### (3) Scaling



$$f(t) \rightarrow f(at) \begin{cases} a > 1 & \text{compression} \\ 0 < a < 1 & \text{expansion} \end{cases}$$

Multiplying the time variable by a coefficient is equivalent to changing the observation time scale

### (4) General form of independent variable transformation

$$f(t) \rightarrow f(at \pm b)$$



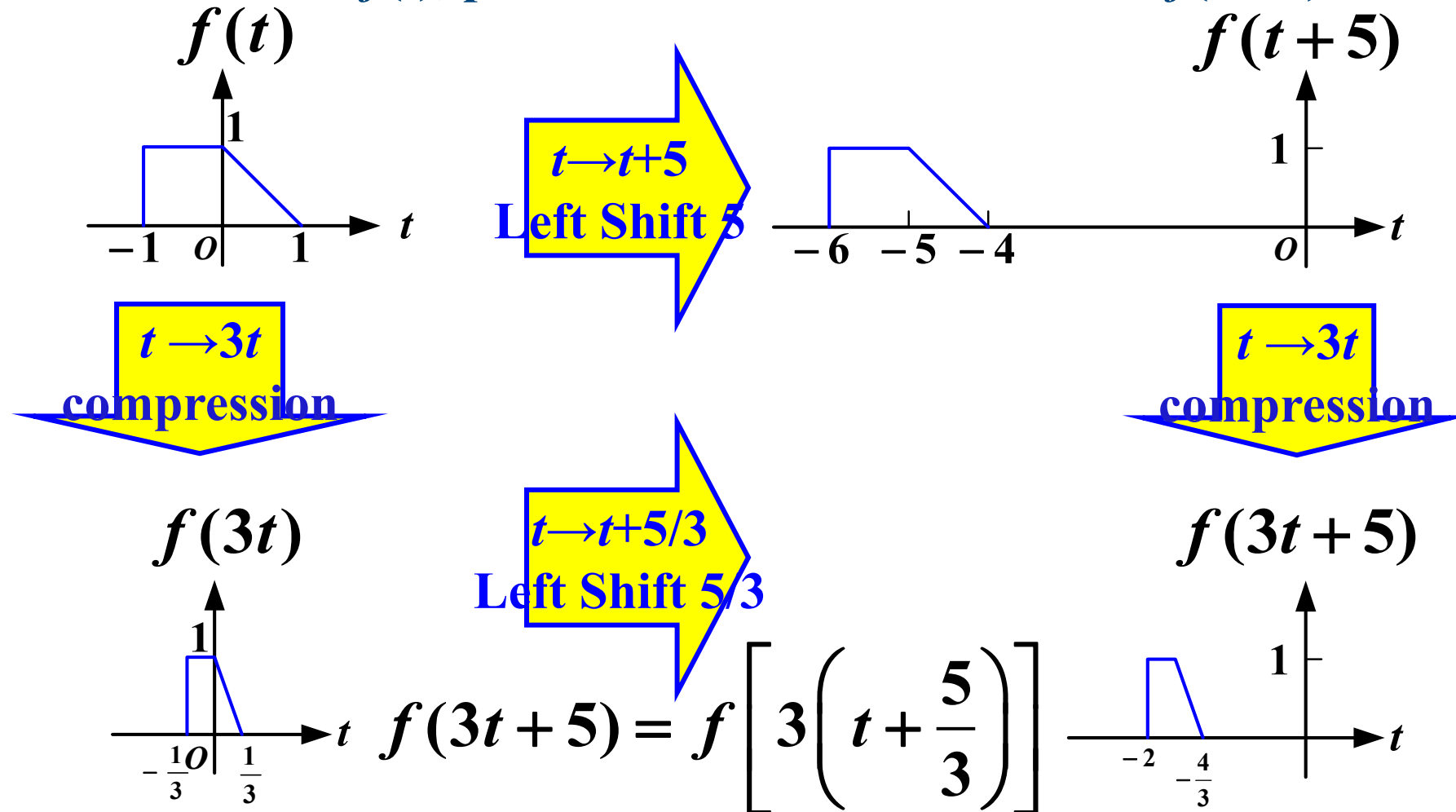
All transformations are applied to  $t$  (the independent variable)

## 3.2 Operations on continuous-time signals



### Example 3

Given the waveform of  $f(t)$ , plot the transformed waveform  $f(3t+5)$



### Summary

#### 1. Dependent Variable Operations

- Addition and Multiplication
- Differentiation and Integration

#### 2. Independent Variable Transformations

- Time Shifting (Translation)
- Time Reversal
- Time Scaling

$$f(t) \rightarrow f(at \pm b)$$

**All transformations are applied to  $t$  (the independent variable)**

# 4. Unit step signal and unit impulse signal

## Main Content

### Singularity signals

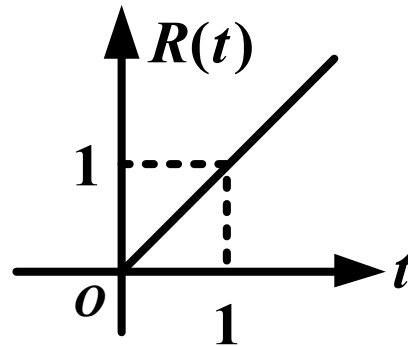
1. Unit ramp signal
2. Unit step signal
3. Unit impulse signal
4. Doublet signal

# 4. Unit step signal and unit impulse signal

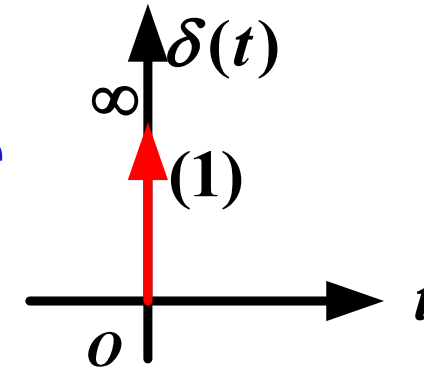
## 1. Singularity signals

Functions that themselves have discontinuity points (jump discontinuities) or whose derivatives/integrals contain discontinuities are collectively referred to as singular signals or singularity functions

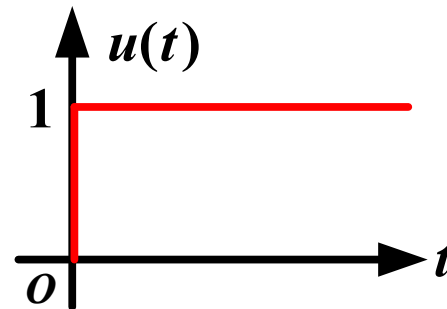
Unit ramp  
signal



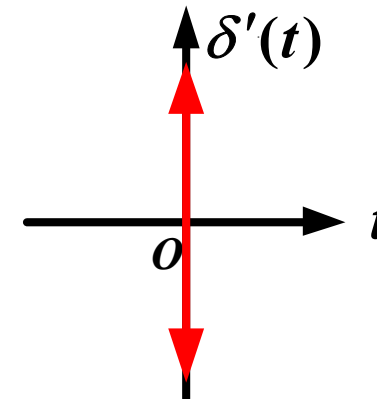
Unit impulse  
signal



Unit step  
signal



Doublet  
signal

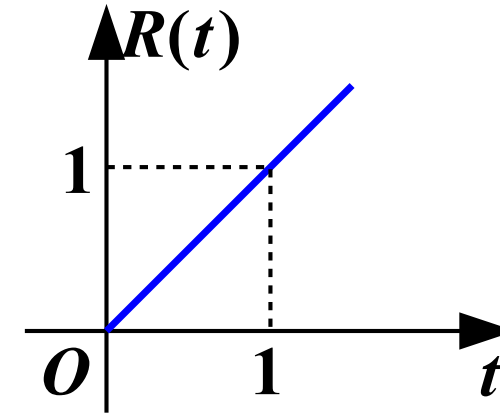


# 4. Unit step signal and unit impulse signal

## 2. Unit ramp signal

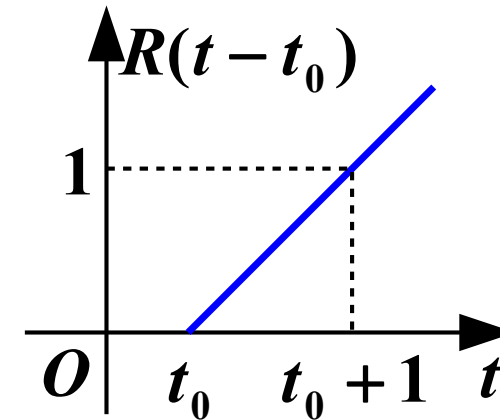
### (1) Definition

$$R(t) = \begin{cases} 0 & t < 0 \\ t & t \geq 0 \end{cases}$$



### (2) Delayed unit ramp signal

$$R(t - t_0) = \begin{cases} 0 & t < t_0 \\ t - t_0 & t \geq t_0 \end{cases}$$



From the argument  $t - t_0 = 0$ , the starting point is determined to be  $t_0$

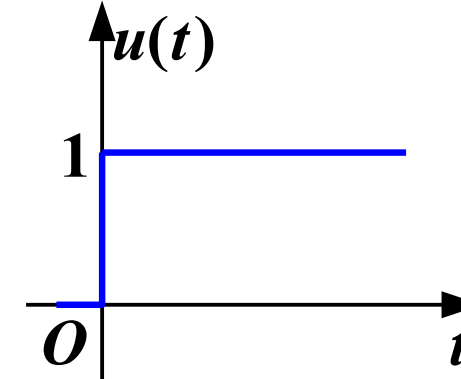
# 4. Unit step signal and unit impulse signal

## 3. Unit step signal

### (1) Definition

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases}$$

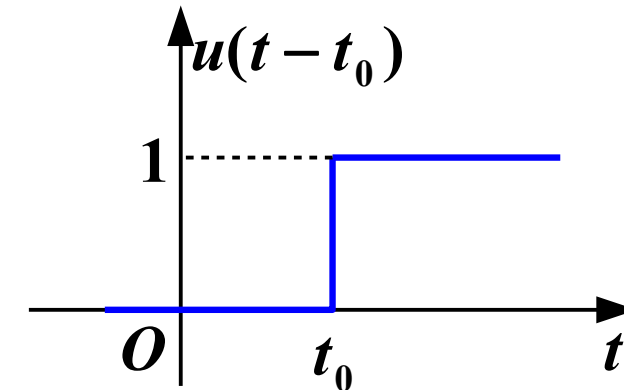
At  $t=0$ , the function is either undefined or equals  $1/2$



When the argument  $> 0$ , the function value is 1, When the argument  $< 0$ , the function value is 0

### (2) Delayed unit step signal

$$u(t - t_0) = \begin{cases} 0 & t < t_0 \\ 1 & t > t_0 \end{cases}, t_0 > 0$$



When the argument  $(t-t_0)=0$  (i.e., at  $t=t_0$ ), the function has a discontinuity (jump point)

# 4. Unit step signal and unit impulse signal

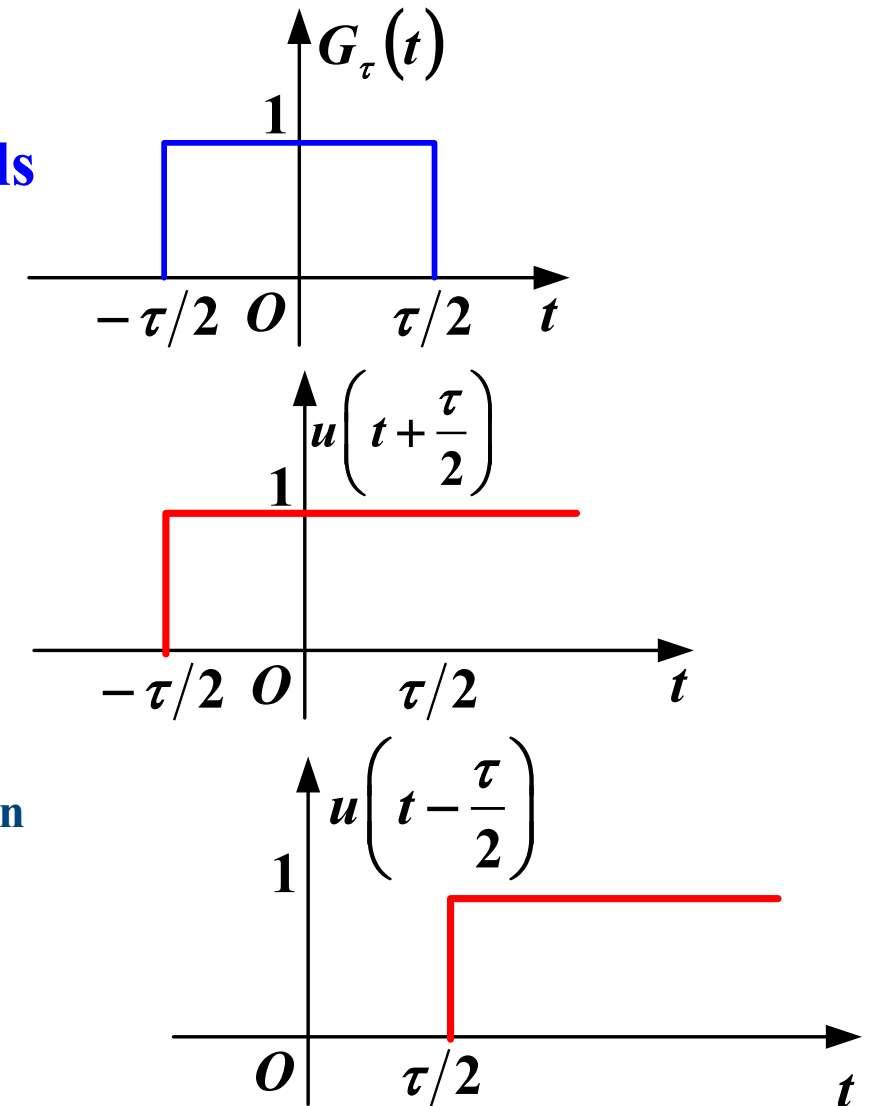
## 3. Unit step signal

### (3) Using unit step functions to represent other signals

**Gate Function** (Rectangular Window Function)

$$G_{\tau}(t) = u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right)$$

When any signal is multiplied by a gate function, only the portion within the "gate" remains



## 4. Unit step signal and unit impulse signal

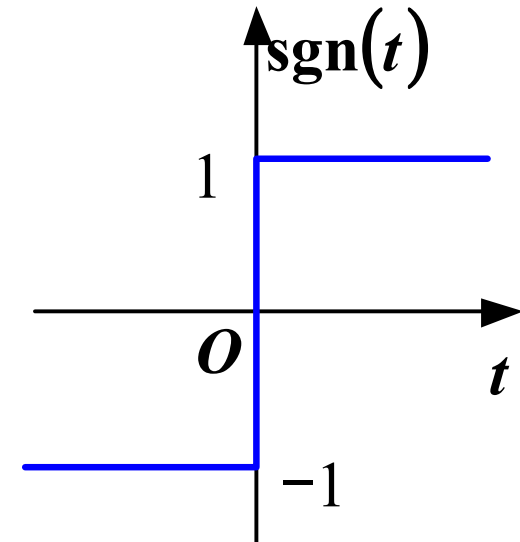
### 3. Unit step signal

#### (3) Using unit step functions to represent other signals

##### Signum Function

$$\text{sgn}(t) = \begin{cases} 1, & t > 0 \\ -1, & t < 0 \end{cases}$$

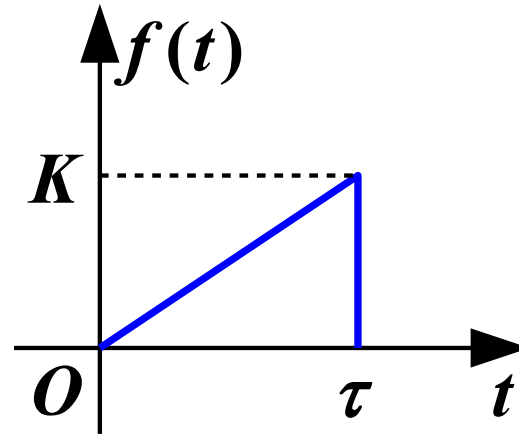
$$\begin{aligned} \text{sgn}(t) &= -u(-t) + u(t) \\ &= 2u(t) - 1 \end{aligned}$$



## 4. Unit step signal and unit impulse signal

### 3. Unit step signal

(3) Using unit step functions to represent other signals



$$f(t) = \frac{K}{\tau} t [u(t) - u(t - \tau)]$$

# 4. Unit step signal and unit impulse signal

## 4. Unit impulse signal

**Concept Introduction**

**Definition 1: Limit of Pulse Signals**

**Definition 2: Dirac Function**

**Definition 3: Generalized Function Definition**

# 4. Unit step signal and unit impulse signal

## (1) Concept Introduction

- **Impulsive Force in Mechanics**, A force that arises during collisions between objects, characterized by a sudden increase followed by rapid dissipation
- **Key Features**, Acts over an extremely short duration, can attain very large magnitudes
- **Examples**, Sports mechanics, striking a ping-pong ball, kicking a soccer ball
- **Analogous Concepts in Other Fields**, Particle dynamics, Point charge impulse, Point light source intensity spikes, Impulse signals



# 4. Unit step signal and unit impulse signal

## (2) Definition 1: Limit of Pulse Signals

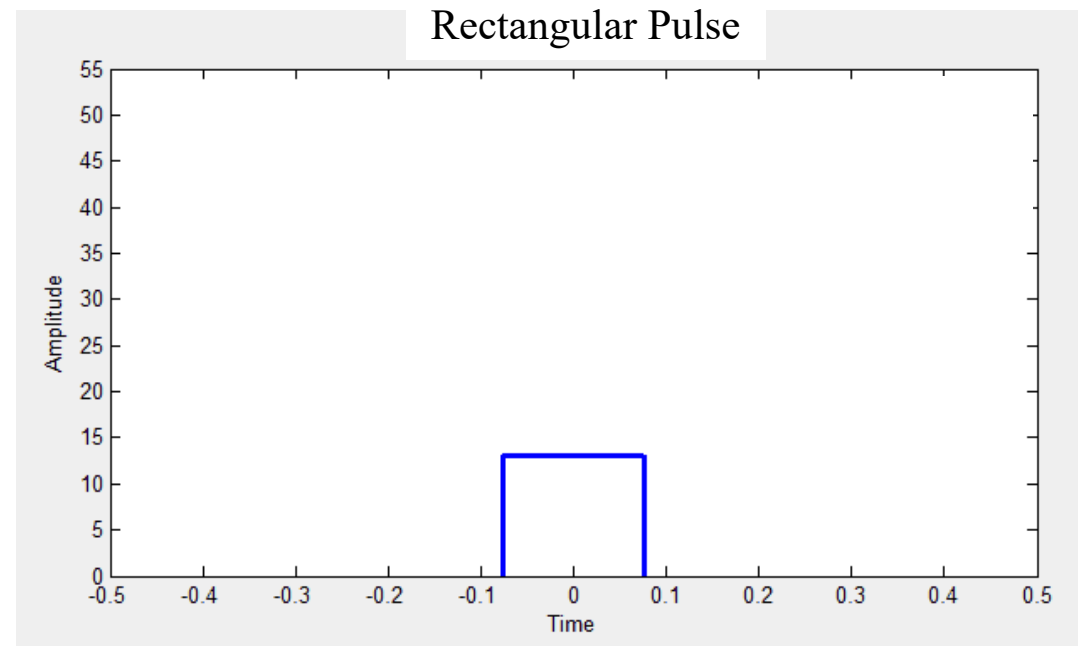
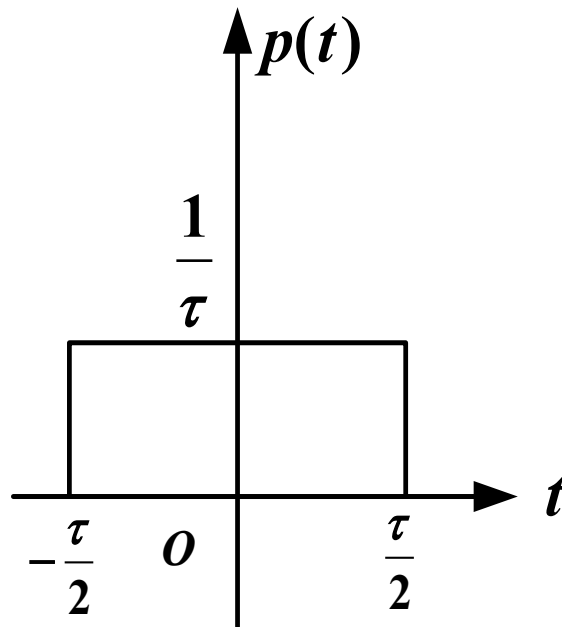
$$p(t) = \frac{1}{\tau} \left[ u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right) \right]$$

$\tau \rightarrow 0$

Area 1

Pulse Width↓

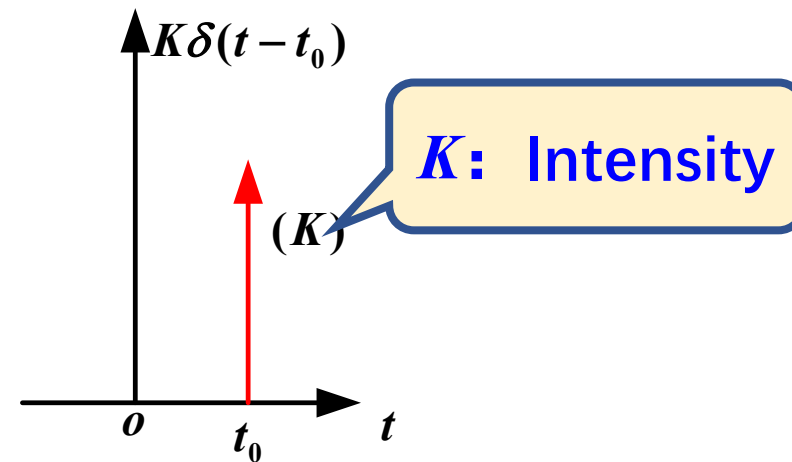
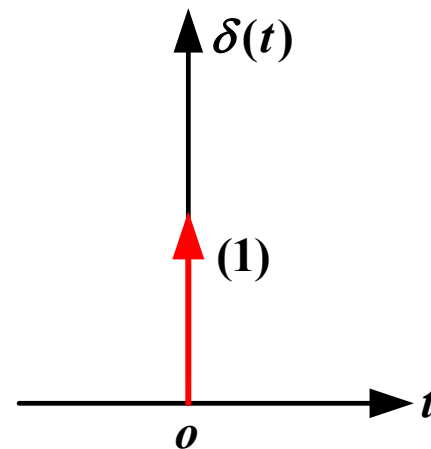
Pulse Height↑



# 4. Unit step signal and unit impulse signal

## (2) Definition 1: Limit of Pulse Signals

$$\delta(t) = \lim_{\tau \rightarrow 0} p(t) = \lim_{\tau \rightarrow 0} \frac{1}{\tau} \left[ u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right) \right]$$



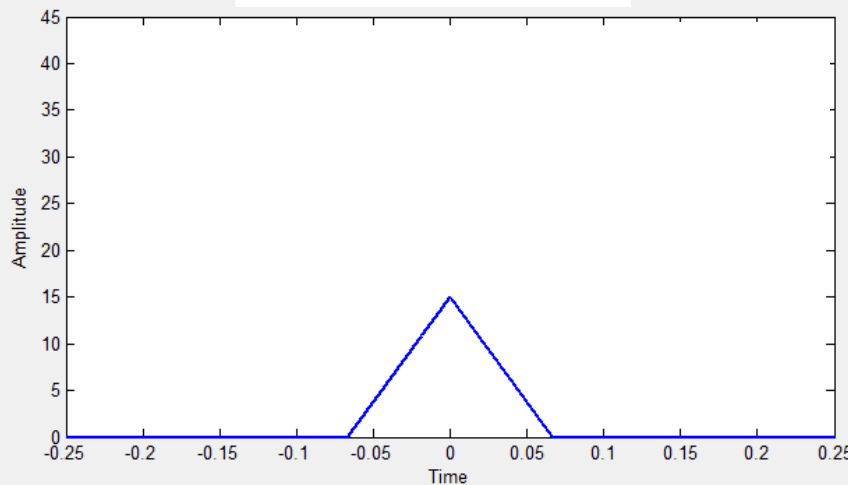
Triangular pulse, double-sided exponential pulse, Gaussian pulse, and the sinc function taking the limit as  $t \rightarrow 0$  can all be considered as impulse functions

# 4. Unit step signal and unit impulse signal

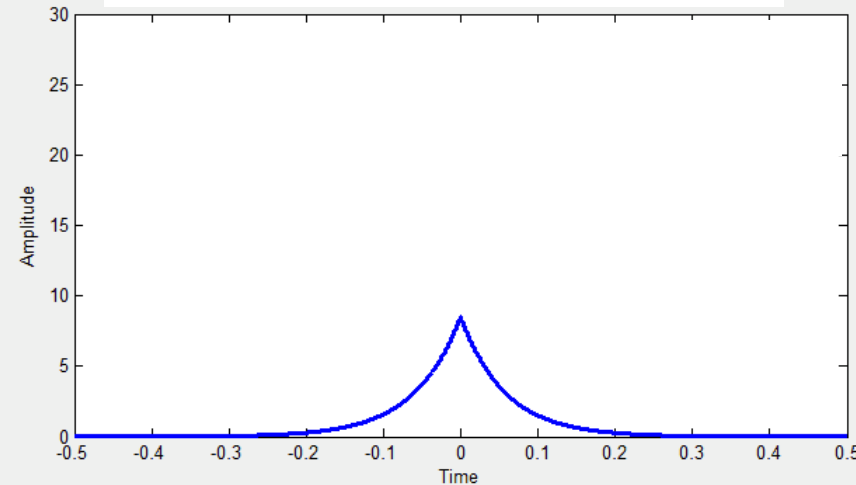
## (2) Definition 1: Limit of Pulse Signals

$$\delta(t) = \lim_{\tau \rightarrow 0} \frac{1}{\tau} \left( 1 - \frac{|t|}{\tau} \right) \quad (|t| \leq \tau) \quad \delta(t) = \lim_{\tau \rightarrow 0} \left( \frac{1}{2\tau} e^{-\frac{|t|}{\tau}} \right)$$

Triangular pulse



double-sided exponential pulse



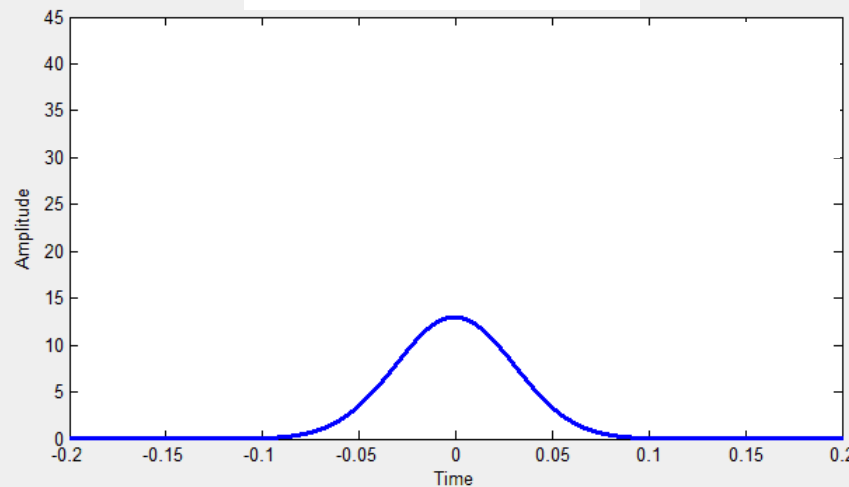
# 4. Unit step signal and unit impulse signal

## (2) Definition 1: Limit of Pulse Signals

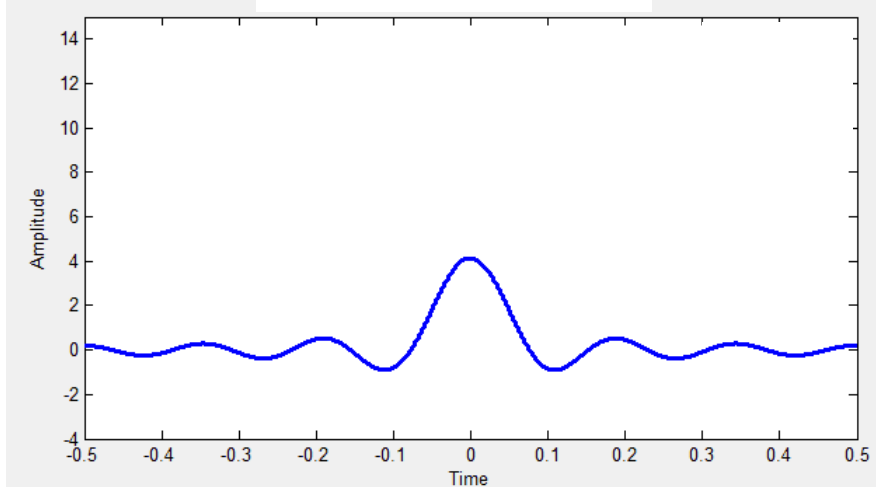
$$\delta(t) = \lim_{\tau \rightarrow 0} \left[ \frac{1}{\tau} e^{-\pi \left( \frac{t}{\tau} \right)^2} \right]$$

$$\delta(t) = \lim_{k \rightarrow \infty} \left[ \frac{k}{\pi} \text{Sa}(kt) \right]$$

Gaussian pulse



Sinc function

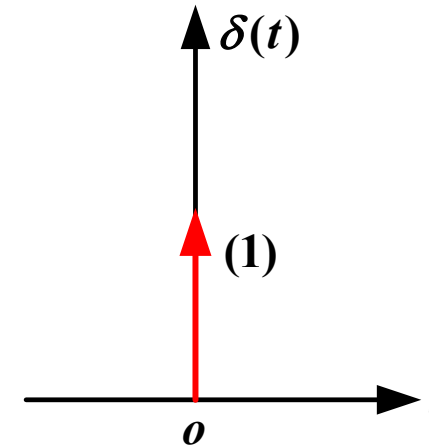


## 4. Unit step signal and unit impulse signal

### (3) Definition 2: Dirac Function

$$\begin{cases} \int_{-\infty}^{\infty} \delta(t) dt = 1 \\ \delta(t) = 0, (t \neq 0) \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(t) dt = \int_{0_-}^{0_+} \delta(t) dt = 1$$



- The function is non-zero **only** at  $t = 0$
- Its integral equals **1**
- $t = 0$ ,  $\delta(t) \rightarrow \infty$ , the function is **unbounded**

## 4. Unit step signal and unit impulse signal

### (3) Definition 2: Dirac Function



**Paul Adrien Maurice Dirac**

August 8, 1902 – October 20, 1984

British theoretical physicist, one of the founders of quantum mechanics

**Nobel Prize in Physics (1933)** for the Dirac equation, which theoretically predicted the existence of the positron (antimatter)

**Key Contributions**

**Quantum field theory:** Pioneered foundational work in quantum electrodynamics (QED)

**Gravitation:** Advanced theories in gravity and quantization of gravitational fields.

**Mathematical physics:** Introduced bra-ket notation, delta function formalism, and magnetic monopole concepts.

**Legacy:** Regarded as one of the most influential physicists of the 20th century, bridging quantum theory and relativity

## (3) Definition 3: Generalized Function Definition

$$\int_{-\infty}^{\infty} \delta(t) f(t) dt = f(0)$$

Test Functions

Distributions (Generalized Functions)

Requirements for Test Functions,  $f(t)$ :

- Must be continuous
- Must possess continuous derivatives of all orders
- And  $f(t) \neq 0, a < t < b, (-\infty < a, b < \infty)$

$\delta(t)$  is entirely characterized by its action in integrals

# 4. Unit step signal and unit impulse signal

## 4. Unit impulse signal-Properties

$$\int_{-\infty}^{\infty} \delta(t) f(t) dt = f(0)$$

To meet the requirements of signal analysis, the unit impulse function (Dirac delta function) was constructed as a generalized function (distribution). Temporally, it can be treated as a continuous-time signal in operations, as it adheres to certain rules of continuous-time signal processing. However, being a distribution, it possesses unique properties

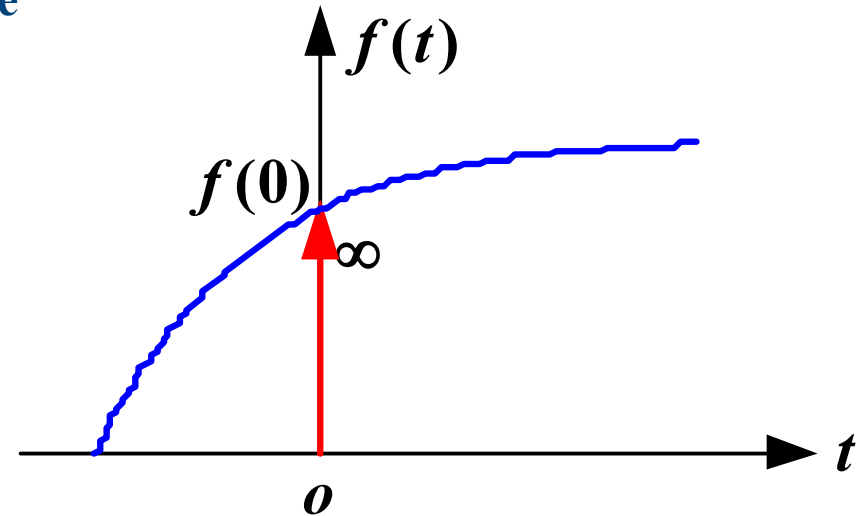
# 4. Unit step signal and unit impulse signal

## (1) Sifting Property

If  $f(t)$  is continuous at  $t=0$  and bounded everywhere, then the unit impulse signal (Dirac delta function) can be rigorously defined through its action on  $f(t)$

$$\delta(t)f(t) = f(0)\delta(t)$$

$$\int_{-\infty}^{\infty} \delta(t)f(t)dt = f(0)$$



This definition aligns with the theory of distributions (generalized functions), where  $\delta(t)$  is characterized by its integral property rather than pointwise values

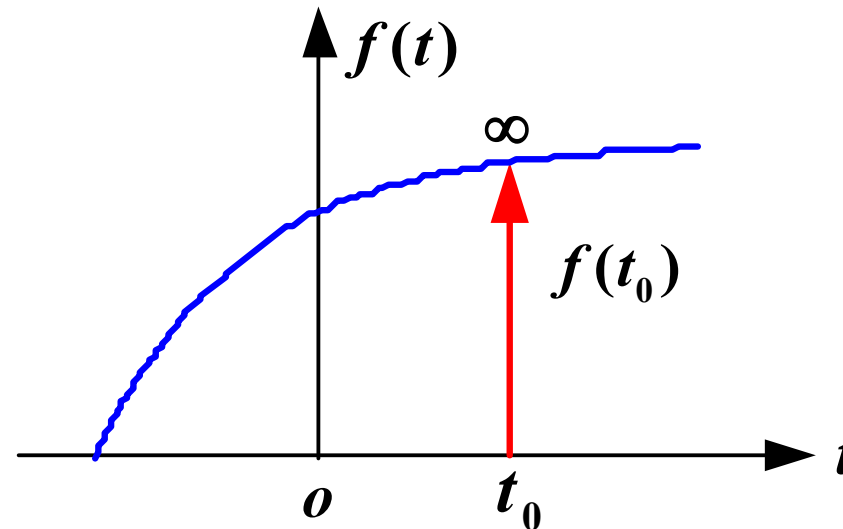
# 4. Unit step signal and unit impulse signal

## (1) Sifting Property

Shifted

$$\delta(t - t_0) f(t) = f(t_0) \delta(t - t_0)$$

$$\int_{-\infty}^{\infty} \delta(t - t_0) f(t) dt = f(t_0)$$



# 4. Unit step signal and unit impulse signal

## (2) Parity

- According to Definition 1, the rectangular pulse itself is an even function, so its limit is also an even function
- The parity property is demonstrated through the sampling property
- When proving the parity property, the focus is on the action of this function, i.e., the result of **its interaction with other function**

$$\int_{-\infty}^{\infty} \delta(t) f(t) dt = f(0)$$

$$\begin{aligned} \int_{-\infty}^{\infty} \delta(-t) f(t) dt & \stackrel{-t=\tau}{=} \int_{\infty}^{-\infty} \delta(\tau) f(-\tau) d(-\tau) \\ & = \int_{-\infty}^{\infty} \delta(\tau) f(-\tau) d\tau = f(0) \end{aligned}$$

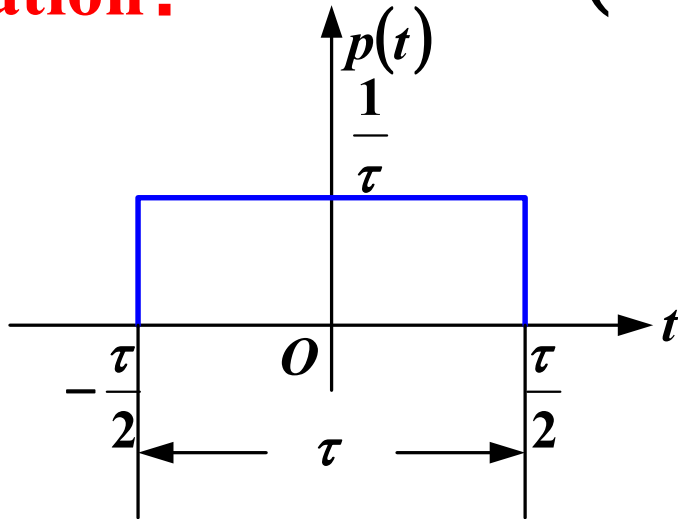
Thus  $\delta(t) = \delta(-t)$

# 4. Unit step signal and unit impulse signal

## (3) Time Scaling

**Explanation:**

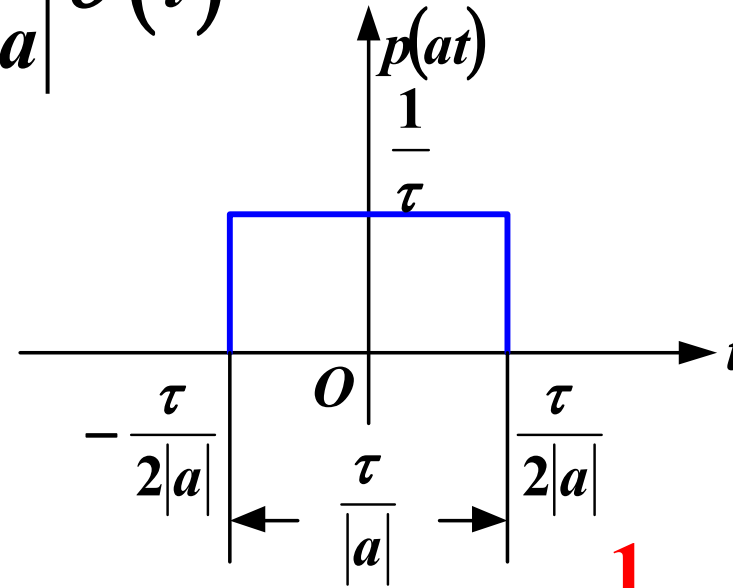
$$\delta(at) = \frac{1}{|a|} \delta(t)$$



$p(t)$  area of 1,

$\tau \rightarrow 0,$

$p(t) \rightarrow \delta(t),$



$p(at)$  area of  $\frac{1}{|a|},$

$\tau \rightarrow 0,$

$p(at) \rightarrow \frac{1}{|a|} \delta(t)$

## 4. Unit step signal and unit impulse signal

### (3) Time Scaling-Prove

**Analysis:** The proof is based on demonstrating the equality of the integral values of both sides multiplied by  $f(t)$ , considering two cases:  $a > 0$  and  $a < 0$

$$\int_{-\infty}^{\infty} \delta(at) f(t) dt \stackrel{?}{=} \int_{-\infty}^{\infty} \frac{1}{|a|} \delta(t) f(t) dt$$

(1)  $a > 0$ , let  $at = \tau$ ,  $t = \frac{\tau}{a}$

$$\int_{-\infty}^{\infty} \delta(at) f(t) dt = \int_{-\infty}^{\infty} \delta(\tau) f\left(\frac{\tau}{a}\right) d\left(\frac{\tau}{a}\right) = \frac{1}{a} f(0)$$

While  $\int_{-\infty}^{\infty} \frac{1}{|a|} \delta(t) f(t) dt = \frac{1}{a} f(0)$

$\therefore$  Both sides are equal

## 4. Unit step signal and unit impulse signal

### (3) Time Scaling-Prove

$$a < 0, \text{ let } -|a|t = \tau, t = -\frac{\tau}{|a|} \begin{cases} t: & -\infty \rightarrow +\infty \\ \tau: & +\infty \rightarrow -\infty \end{cases}$$

$$\int_{-\infty}^{+\infty} \delta(-|a|t) f(t) dt = \int_{+\infty}^{-\infty} \delta(\tau) f\left(-\frac{1}{|a|}\tau\right) d\left(-\frac{1}{|a|}\tau\right)$$

$$= \frac{1}{|a|} \int_{-\infty}^{+\infty} \delta(\tau) f\left(-\frac{1}{|a|}\tau\right) d\tau$$

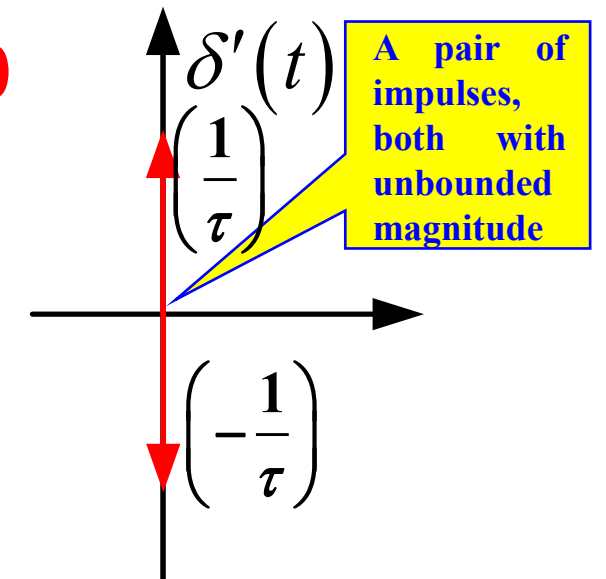
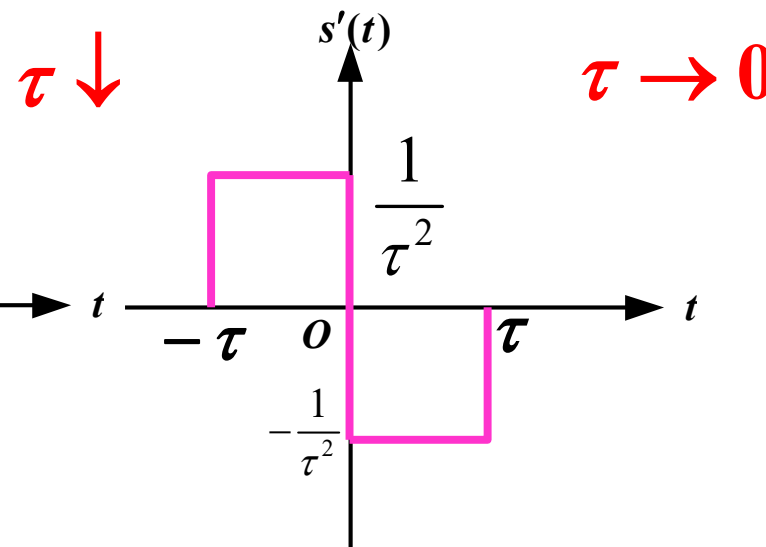
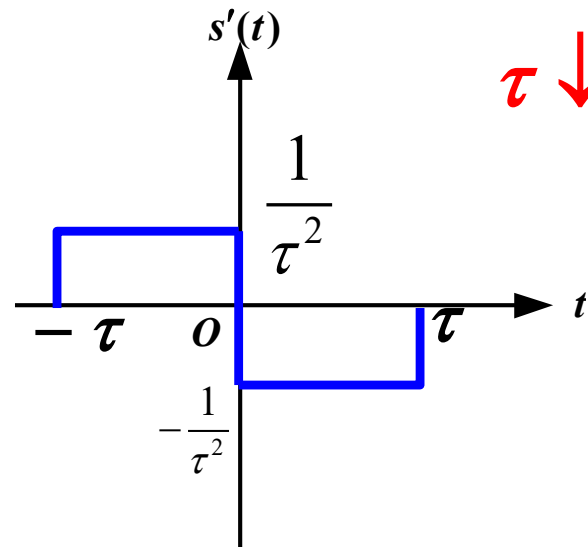
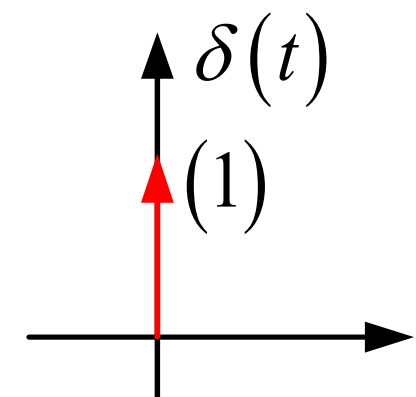
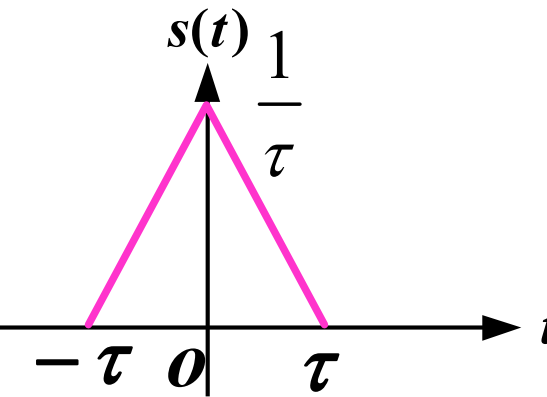
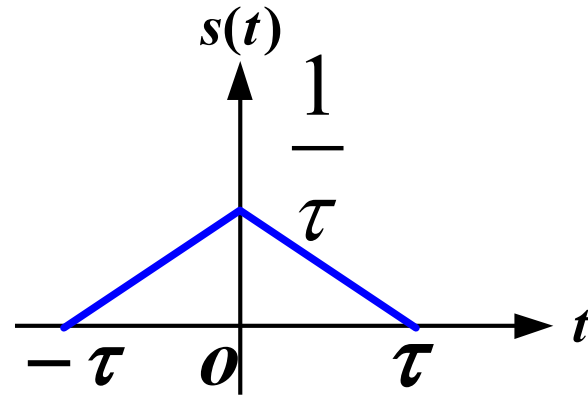
$$= \frac{1}{|a|} f(0)$$

$$\text{While } \int_{-\infty}^{+\infty} \frac{1}{|a|} \delta(t) f(t) dt = \frac{1}{|a|} f(0)$$

# 4. Unit step signal and unit impulse signal

## 5. Doublet signal

$$\delta'(t) = \frac{d\delta(t)}{dt}$$



## 5. Doublet signal

$$\textcircled{1} \int_{-\infty}^{\infty} \delta'(t) f(t) dt = -f'(0)$$

**Proof:** Using integration by parts

$$\begin{aligned} \int_{-\infty}^{\infty} \delta'(t) f(t) dt &= \int_{-\infty}^{\infty} f(t) d\delta(t) \\ &= f(t)\delta(t) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \delta(t) df(t) \\ &= - \int_{-\infty}^{\infty} f'(t) \delta(t) dt = -f'(0) \end{aligned}$$

**Shift**

$$\int_{-\infty}^{\infty} \delta'(t - t_0) f(t) dt = -f'(t_0)$$

**Generalization**

$$\int_{-\infty}^{\infty} \delta^n(t) f(t) dt = (-1)^n f^n(0)$$

## 4. Unit step signal and unit impulse signal

### 5. Doublet signal

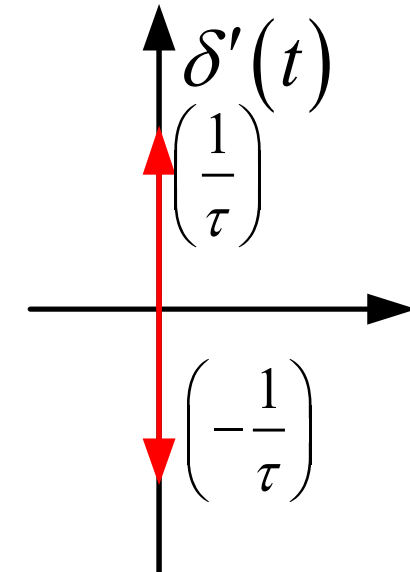
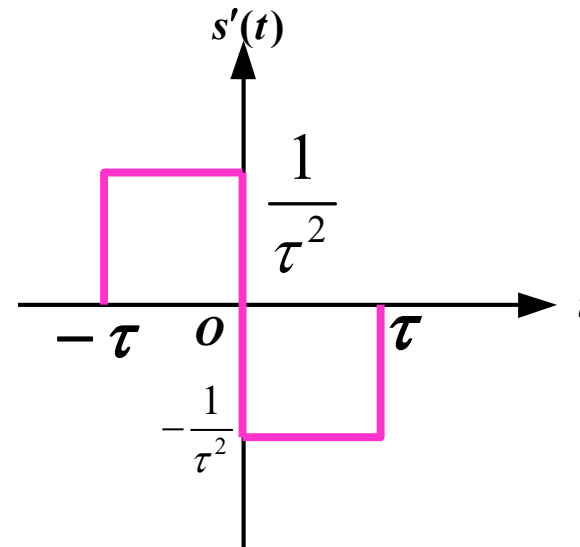
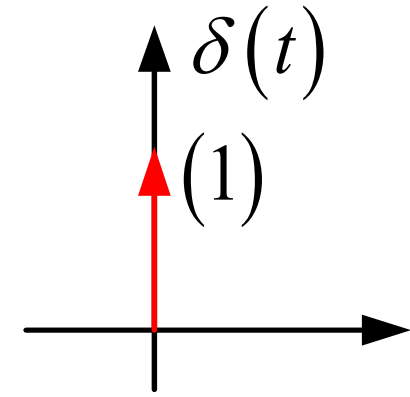
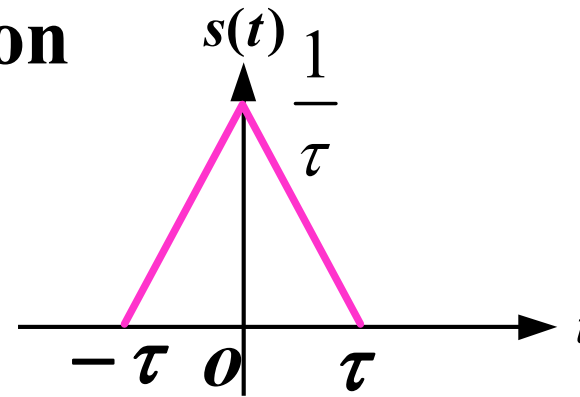
②  $\delta'(t)$  is an odd function

$$\delta'(-t) = -\delta'(t)$$

$$\delta'(t_0 - t) = -\delta'(t - t_0)$$

$$\int_{-\infty}^{\infty} \delta'(\tau) d\tau = 0$$

$$\int_{-\infty}^t \delta'(\tau) d\tau = \delta(t)$$



## 4. Unit step signal and unit impulse signal

### 5. Doublet signal

$$f(t)\delta'(t) = f(0)\delta'(t) - f'(0)\delta(t)$$

different from  $f(t)\delta(t) = f(0)\delta(t)$

**Proof:**

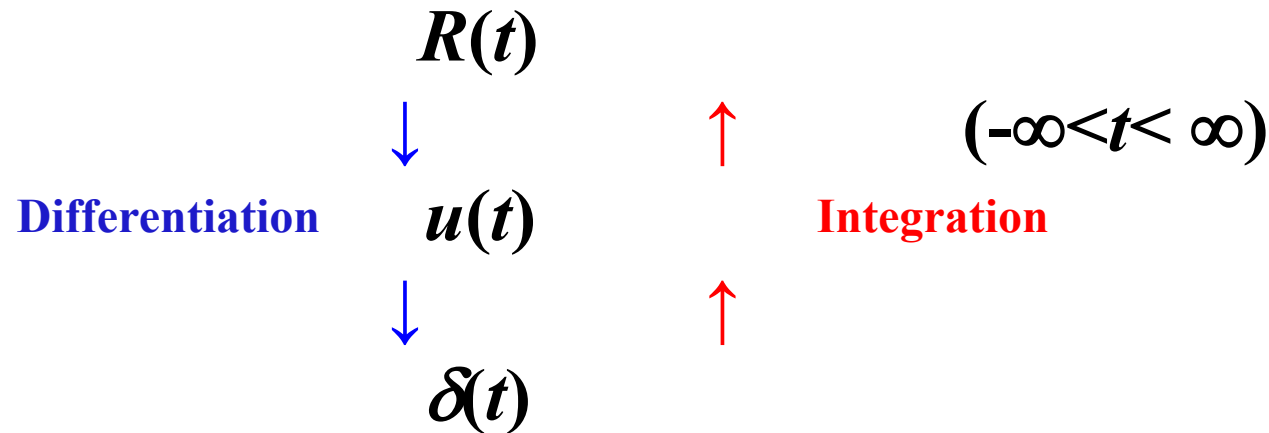
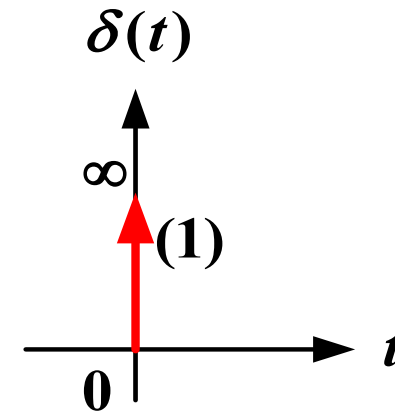
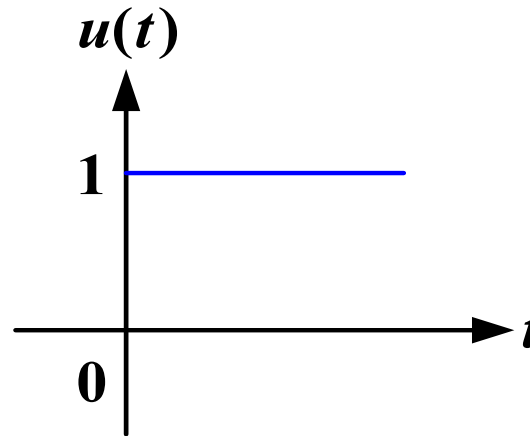
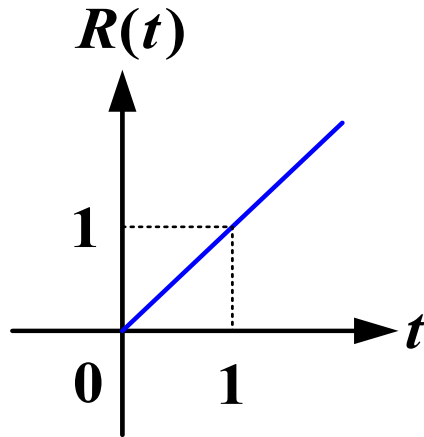
$$[f(t)\delta(t)]' = f(t)\delta'(t) + f'(t)\delta(t)$$

$$\xrightarrow[\delta(t) \text{ --- Sifting}]{}$$

$$[f(0)\delta(t)]' = f(t)\delta'(t) + f'(0)\delta(t)$$

# 4. Unit step signal and unit impulse signal

## Summary



# 4. Unit step signal and unit impulse signal

## Summary

The properties of the unit impulse signal:

(1) Sifting

$$f(t)\delta(t-t_0) = f(t_0)\delta(t-t_0)$$

$$\int_{-\infty}^{\infty} f(t)\delta(t-t_0)dt = f(t_0)$$

$$\int_{-\infty}^{\infty} \delta(t-t_0)dt = 1$$

(2) Parity

$$\delta(-t) = \delta(t)$$

(3) Scaling

$$\delta(at) = \frac{1}{|a|}\delta(t)$$

(4) Calculus Properties

$$\delta(t) = \frac{du(t)}{dt}$$

# 4. Unit step signal and unit impulse signal

## Summary

### (5) Doublet

$$\delta'(t) = \frac{d\delta(t)}{dt}$$

$$\delta'(-t) = -\delta'(t)$$

$$\int_{-\infty}^{\infty} \delta'(t) dt = 0$$

$$\int_{-\infty}^t \delta'(\tau) d\tau = \delta(t)$$

$$\int_{-\infty}^{\infty} f(t) \delta'(t) dt = -f'(0)$$

$$f(t) \delta'(t) = f(0) \delta'(t) - f'(0) \delta(t)$$

### (6) Convolution Properties

$$f(t) * \delta(t) = f(t)$$

## Main Content

- DC component and AC component
- Odd component and even component
- Real part component and imaginary part component

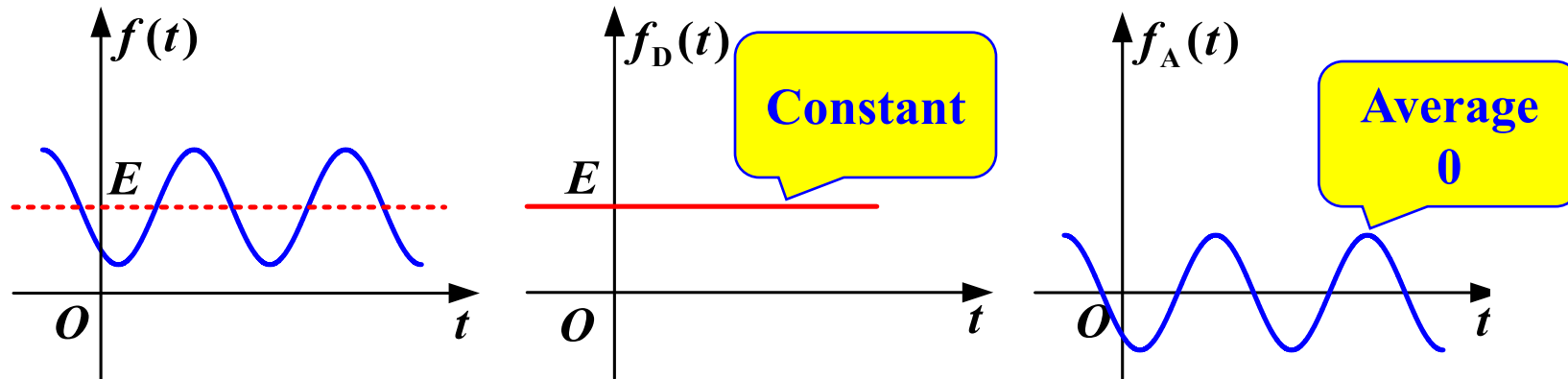
# 5. Decomposition of signals

## 1. DC component and AC component

$$f(t) = f_D(t) + f_A(t)$$

$f_D(t)$ : DC (direct current component)

$f_A(t)$ : AC (alternating current component)



$$\frac{1}{T} \int_{t_0}^{t_0+T} f(t) dt = \frac{1}{T} \int_{t_0}^{t_0+T} [f_A(t) + f_D(t)] dt = f_D(t)$$

DC component, is the average of a signal

# 5. Decomposition of signals

## 1. DC component and AC component

**Power**  $P = \frac{1}{T} \int_{t_0}^{t_0+T} f^2(t) dt$

$$= \frac{1}{T} \int_{t_0}^{t_0+T} [f_D(t) + f_A(t)]^2 dt$$

**Constant**

$$= \frac{1}{T} \int_{t_0}^{t_0+T} f_D^2(t) dt + \frac{2f_D(t)}{T} \cdot \int_{t_0}^{t_0+T} f_A(t) dt$$
$$+ \frac{1}{T} \int_{t_0}^{t_0+T} f_A^2(t) dt$$
$$= f_D^2(t) + \frac{1}{T} \int_{t_0}^{t_0+T} f_A^2(t) dt$$

**Signal's Average Power = DC Power + AC Power**

## 2. Odd component and Even component

For any real-valued signal:

$$f(t) = f_e(t) + f_o(t)$$

$$\begin{cases} f_e(t) : \text{even component} \\ f_o(t) : \text{odd component} \end{cases}$$

**Symmetry:**

$$\begin{cases} f_e(t) = f_e(-t) \\ f_o(t) = -f_o(-t) \end{cases}$$

$$f(-t) = f_e(-t) + f_o(-t)$$

$$f(-t) = f_e(t) - f_o(t)$$

**Calculation Methods:**

$$\begin{cases} f_e(t) = \frac{1}{2}[f(t) + f(-t)] \\ f_o(t) = \frac{1}{2}[f(t) - f(-t)] \end{cases}$$

# 5. Decomposition of signals

## 2. Odd component and Even component

**Signal Average Power**  $P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} [f_e(t) + f_o(t)]^2 dt$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f_e^2(t) dt + \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f_o^2(t) dt$$
$$+ \lim_{T \rightarrow \infty} \frac{2}{T} \int_{-T/2}^{T/2} f_e(t) f_o(t) dt$$

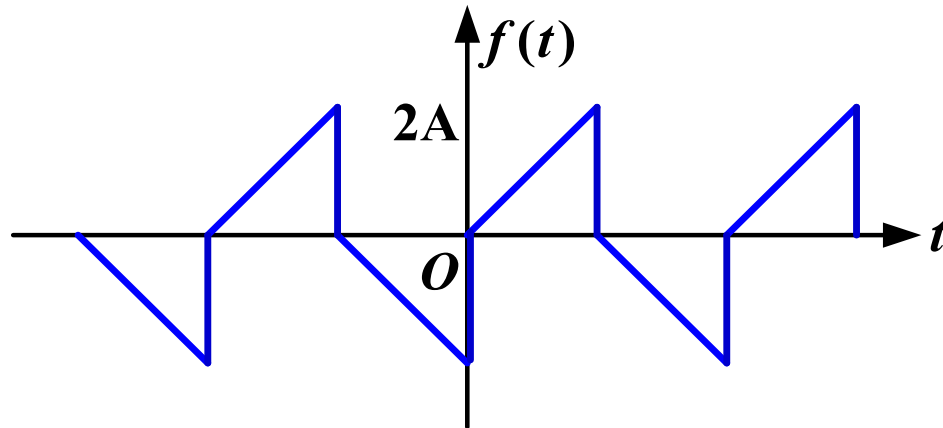
$f_e(t) f_o(t)$  odd function, the integral over a symmetric interval is zero

$$\therefore P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f_e^2(t) dt + \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f_o^2(t) dt = P_e + P_o$$

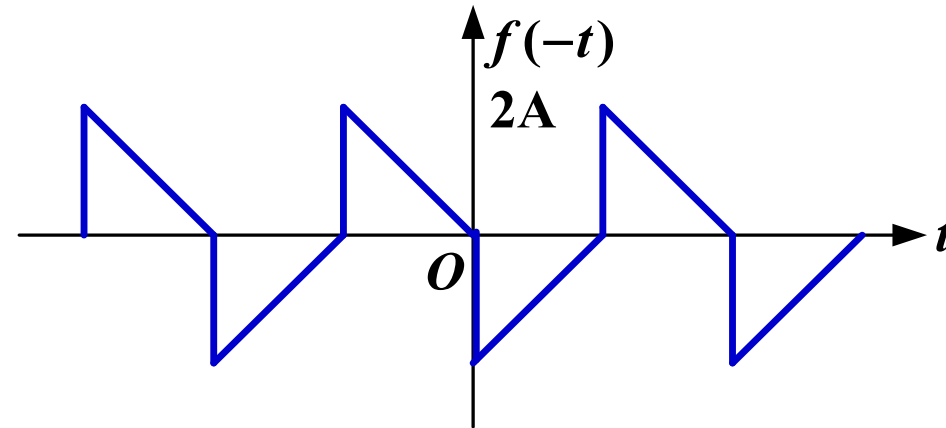
Average power of a signal = Power of **even** component + Power of **odd** component

# 5. Decomposition of signals

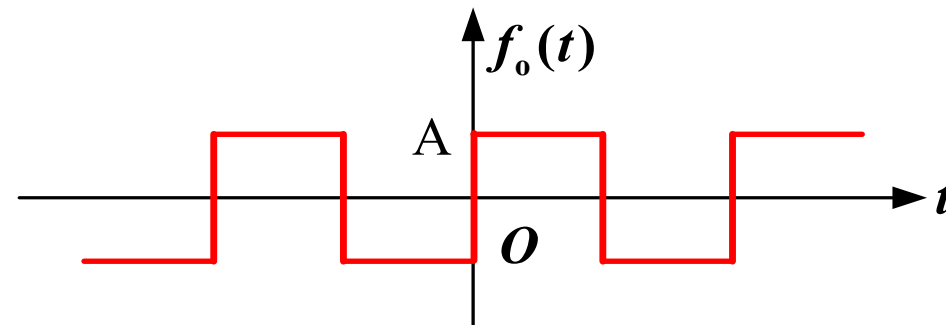
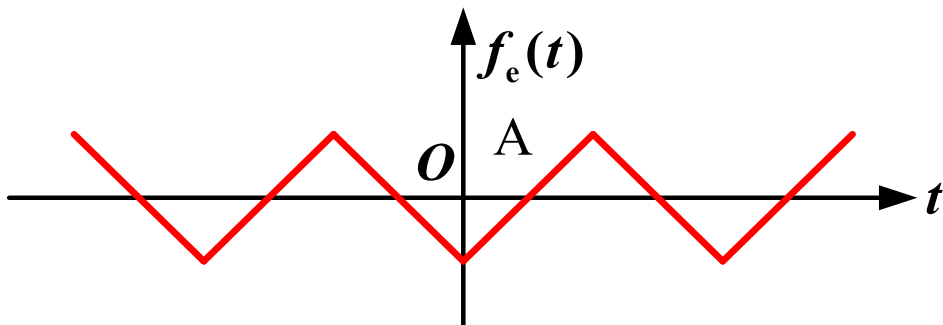
## 2. Odd component and Even component



$$f_e(t) = \frac{1}{2}[f(t) + f(-t)]$$



$$f_o(t) = \frac{1}{2}[f(t) - f(-t)]$$



# 5. Decomposition of signals

## 3. Real component and imaginary component

A signal with complex instantaneous values can be decomposed into the sum of real and imaginary parts

$$f(t) = f_r(t) + \mathbf{j}f_i(t)$$

Complex conjugate functions

$$f^*(t) = f_r(t) - \mathbf{j}f_i(t)$$

Thus

$$f_r(t) = \frac{1}{2}[f(t) + f^*(t)] \quad \mathbf{j}f_i(t) = \frac{1}{2}[f(t) - f^*(t)]$$

Real-world signals are real-valued, but complex signals can **facilitate the analysis of real signals**

## Summary

- **DC component and AC component**
- **Odd component and even component**
- **Real part component and imaginary part component**
- **Chapter 2 Impulse Components**
- **Chapter 3 Sinusoidal Components**

## Main Content

- Representation Methods of Discrete Signals
- Typical discrete signal

## 1. Representation Methods of Discrete Signals

$x(t)$  Evenly spaced  $T_s$   $x(nT_s)$ , represented  $x[n]$   $n = 0, \pm 1, \pm 2, \dots$

A **sequence** of function values, e.g.  $\left\{ \dots 9, \underset{\substack{\uparrow \\ n=0}}{8}, 4, 1 \dots \right\}$

**Waveform:** The length of each line segment represents the magnitude of the corresponding sequence value

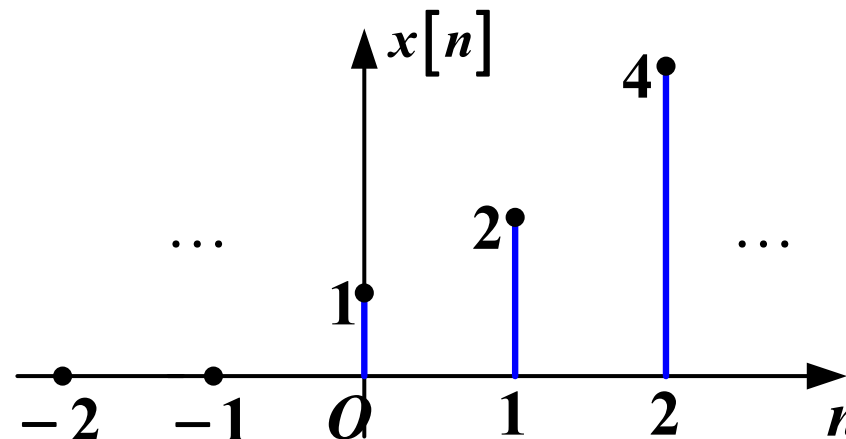
## 1. Representation Methods of Discrete Signals

$$x[n] = \begin{cases} 2^n, & n \geq 0 \\ 0, & n < 0 \end{cases}, \text{ write out its sequence form and plot the waveform}$$

Sequence form:

$$x[n] = \left\{ \dots, 0, 0, \underset{\substack{\uparrow \\ n=0}}{1}, 2, 4, 8, \dots \right\}$$

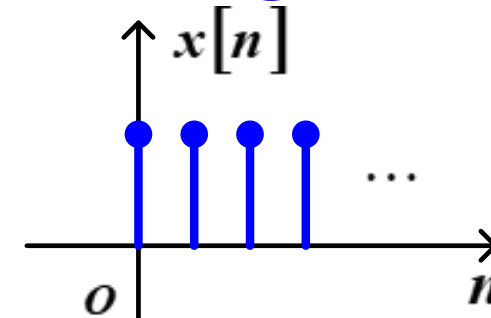
Waveform:



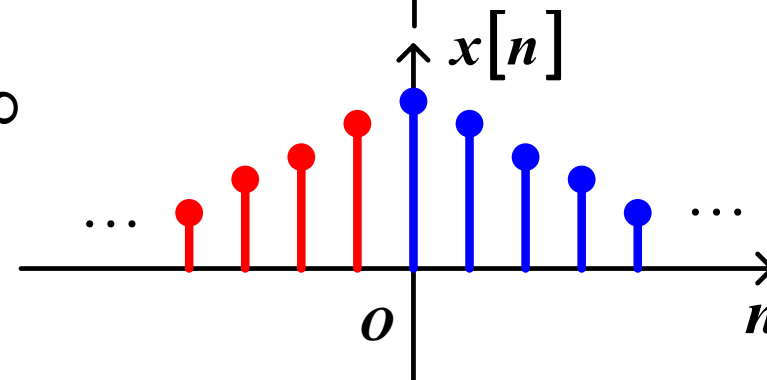
# 6.1 Typical Discrete-time signals

## 1. Representation Methods of Discrete Signals

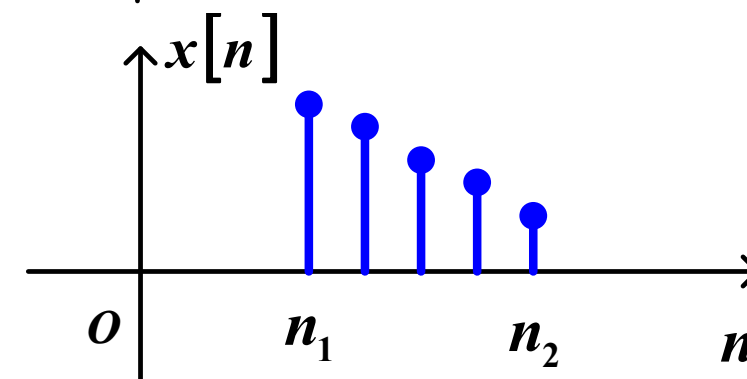
One-sided sequence  $n \geq 0$



Two-sided sequence  $-\infty \leq n \leq \infty$



Finite-length sequence  $n_1 \leq n \leq n_2$



## 2. Typical discrete signal

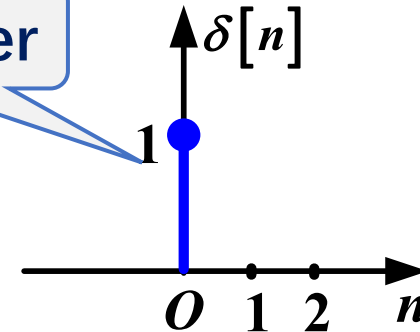
- Unit sample sequence (Discrete-time Impulse)
- Unit step sequence
- Rectangular sequence (Rectangular window)
- Ramp sequence
- One-sided Exponential Sequence
- Sinusoidal sequence
- Complex exponential sequence

# 6.1 Typical Discrete-time signals

## (1) Unit sample sequence

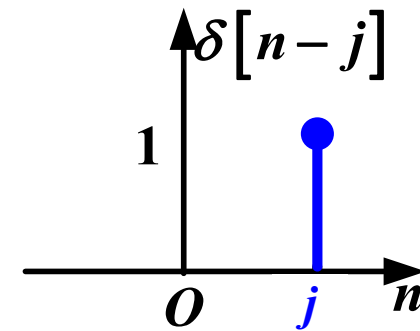
$$\delta[n] = \begin{cases} 0, & n \neq 0 \\ 1, & n = 0 \end{cases}$$

Kronecker



Time-Shifting Property

$$\delta[n-j] = \begin{cases} 0, & n \neq j \\ 1, & n = j \end{cases}$$



Scaling Property

$$c\delta[n], c\delta[n-j]$$

Sampling Property

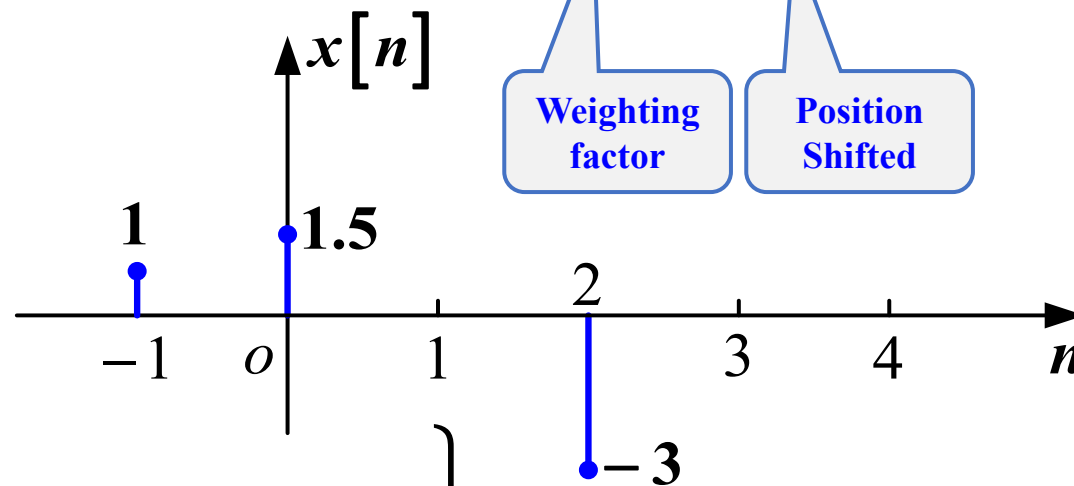
$$f[n]\delta[n] = f[0]\delta[n]$$

# 6.1 Typical Discrete-time signals

## (1) Unit sample sequence

### Representation of Arbitrary Sequences Using Unit Sample Signals

$$x[n] = \sum_{m=-\infty}^{\infty} x[m] \delta[n-m]$$

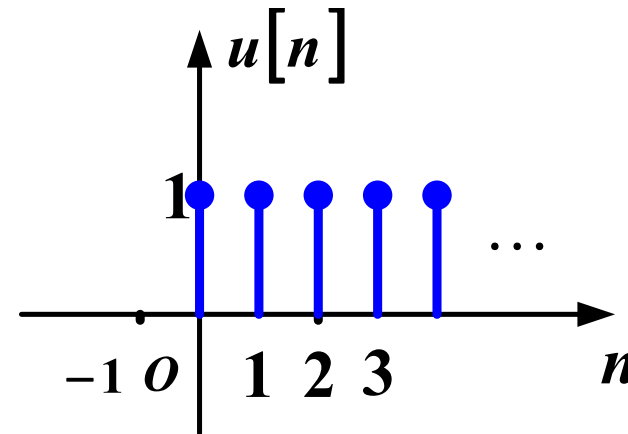


$$x[n] = \left\{ \underset{\substack{\uparrow \\ n=0}}{1}, 1.5, 0, -3, 0, 0, \right\} = \delta[n+1] + 1.5\delta[n] - 3\delta[n-2]$$

# 6.1 Typical Discrete-time signals

## (2) Unit step sequence

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



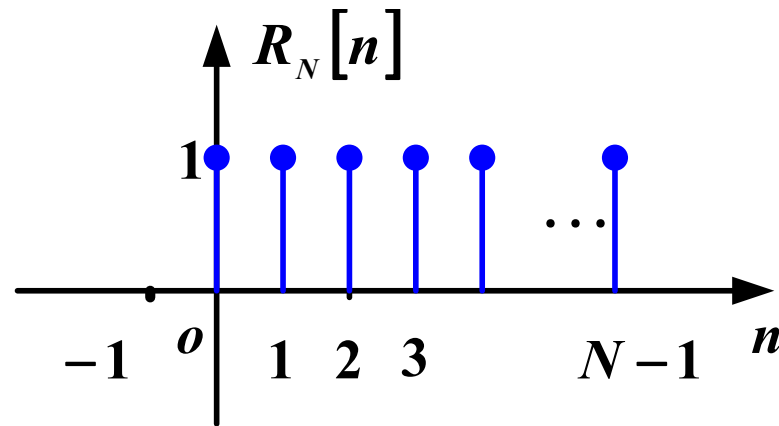
$u[n]$  can be expressed as an infinite sum of shifted unit sample signals

$$\begin{aligned} u[n] &= \delta[n] + \delta[n-1] + \delta[n-2] + \delta[n-3] + \dots \\ &= \sum_{k=0}^{\infty} \delta[n-k] \end{aligned}$$

$$\delta[n] = u[n] - u[n-1]$$

## (3) Rectangular sequence

$$R_N[n] = \begin{cases} 1 & 0 \leq n \leq N-1 \\ 0 & n < 0, n \geq N \end{cases}$$



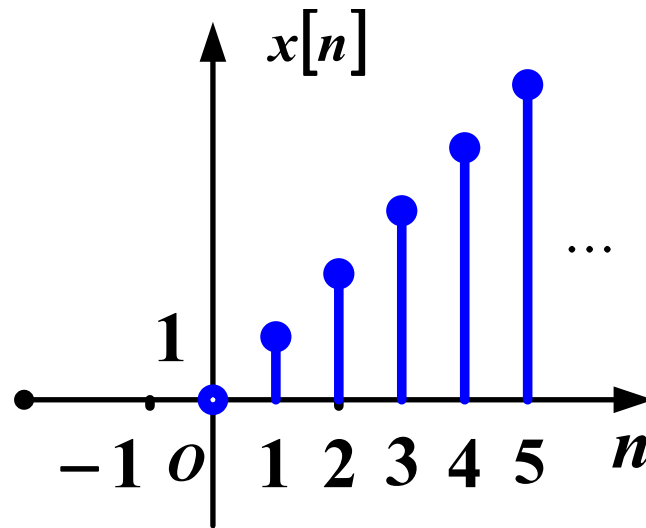
Relationship with  $u[n]$ :  $R_N[n] = u[n] - u[n - N]$

$$R_3[n] = \delta[n] + \delta[n - 1] + \delta[n - 2]$$

# 6.1 Typical Discrete-time signals

## (4) Ramp sequence

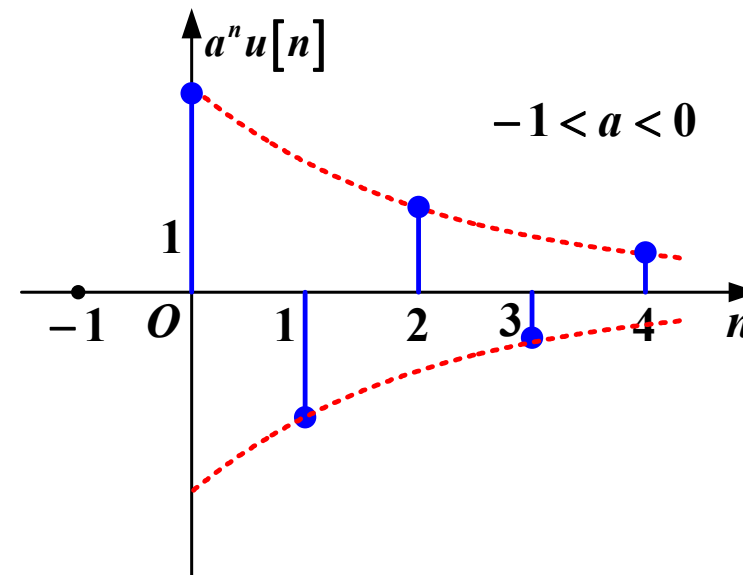
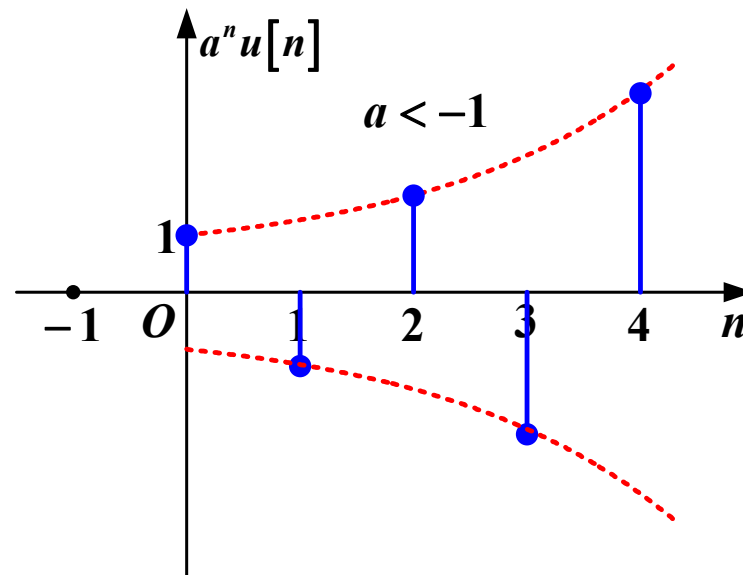
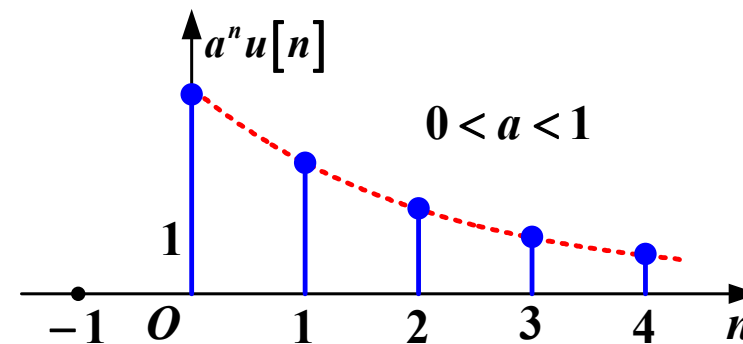
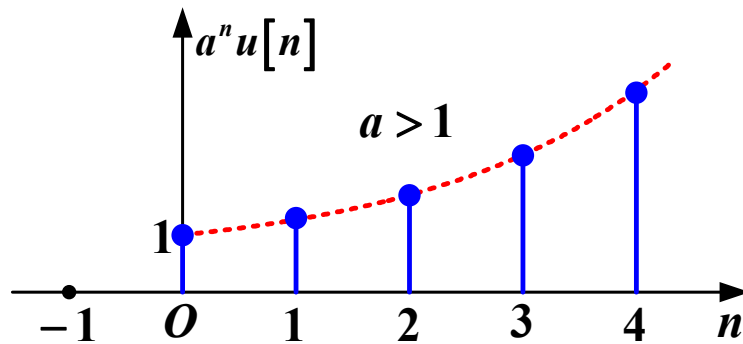
$$x[n] = nu[n]$$



# 6.1 Typical Discrete-time signals

## (5) One-sided Exponential Sequence

$$x[n] = a^n u[n]$$



# 6.1 Typical Discrete-time signals

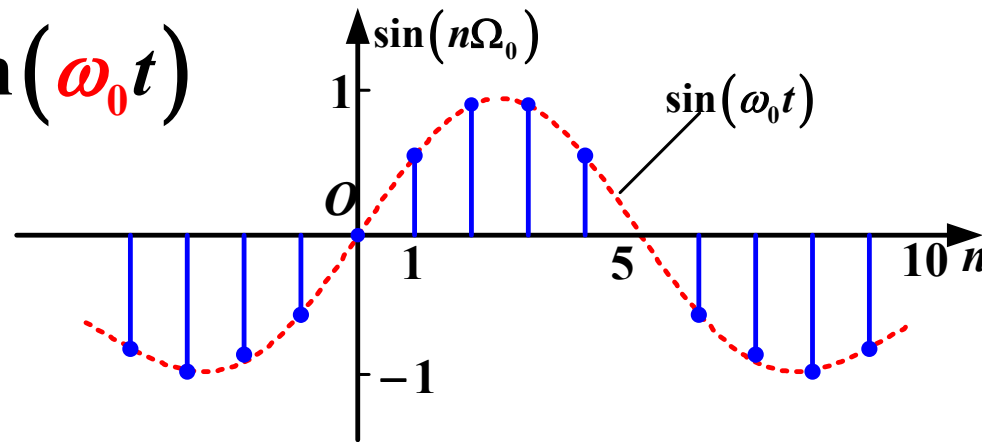
## (6) Sinusoidal sequence

$$x_a(t) = \sin(2\pi f_0 t) = \sin(\omega_0 t)$$

Sampling interval  $T_s$ , Sampling frequency  $f_s$

Sampled values at time instants  $T_s n$

$$x_a(T_s n) = \sin(\omega_0 T_s n)$$

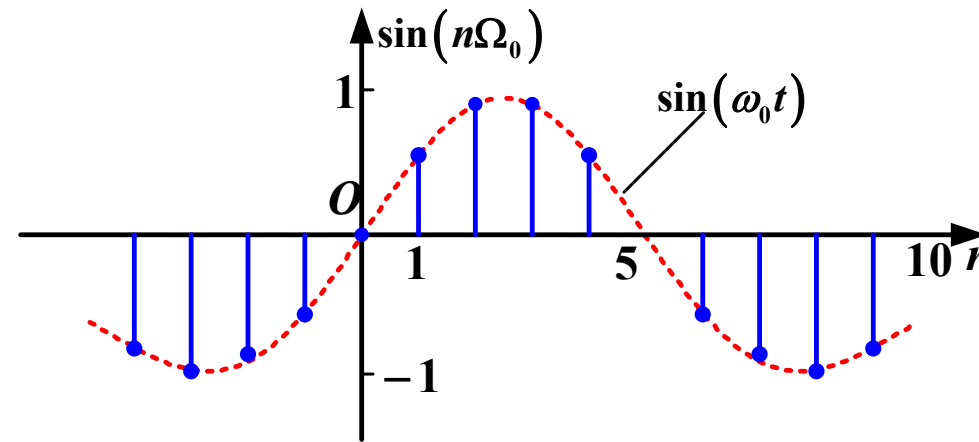


Represent the **sample values** of continuous time signals as sequences, with  $n$  as the variable

$$x[n] = \sin(\Omega_0 \cdot n)$$

Digital Angular Frequency  $\Omega_0 = \omega_0 T_s = \frac{\omega_0}{f_s} = 2\pi \frac{f_0}{f_s}$  Unit: radians/sample

## (6) Sinusoidal sequence



### Periodicity of Sinusoidal Sequences

If a discrete sinusoidal sequence is **periodic**, it must satisfy

$$x[n] = \sin(\Omega_0 n)$$

$$x[n + N] = x[n]$$

$N$  is called the **period** of the sequence and can be any **positive integer**

## (6) Sinusoidal sequence

①  $\frac{2\pi}{\Omega_0}$  is positive integer, then there is  $N = \frac{2\pi}{\Omega_0}$

$$\sin[\Omega_0(n + N)] = \sin(n\Omega_0 + N\Omega_0)$$

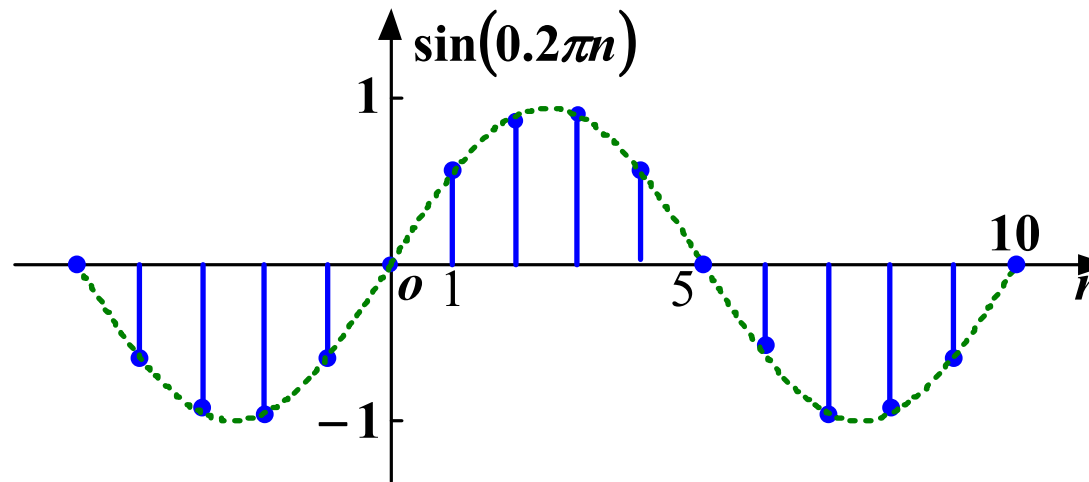
If  $N\Omega_0 = 2\pi$ , then  $\sin[\Omega_0(n + N)] = \sin(\Omega_0 n)$

$$N = \frac{2\pi}{\Omega_0}$$

# 6.1 Typical Discrete-time signals

## Example 1

$$x[n] = \sin(0.2\pi n)$$



$$\Omega_0 = 0.2\pi \quad \frac{2\pi}{\Omega_0} = 10 \quad \text{Period } N = 10$$

Repeats every **10** samples, with the sinusoidal envelope completing **1** full cycle

## (6) Sinusoidal sequence

②  $\frac{2\pi}{\Omega_0} = \frac{N}{m}$ ,  $N, m$  are positive integers, period  $N = m \frac{2\pi}{\Omega_0}$

$$\sin[\Omega_0(n + N)] = \sin(n\Omega_0 + N\Omega_0)$$

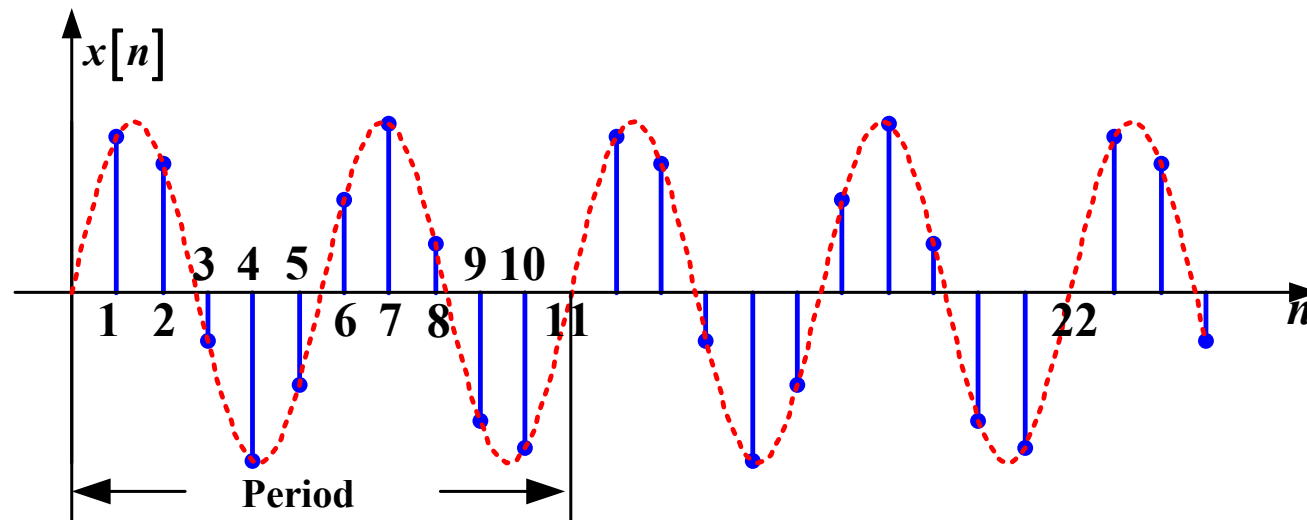
If  $N\Omega_0 = m2\pi$ , then  $\sin[\Omega_0(n + N)] = \sin(\Omega_0 n)$

Period  $N = m \frac{2\pi}{\Omega_0}$

# 6.1 Typical Discrete-time signals

## Example 2

$$x[n] = \sin\left(\frac{4\pi}{11}n\right)$$



$$N = 11$$

$$\Omega_0 = \frac{4\pi}{11}, \quad \text{Then: } \frac{2\pi}{\Omega_0} = \frac{11}{2} = \frac{N}{m}$$

Repeats every **11** samples, with the sinusoidal envelope completing **2** full cycles

## (6) Sinusoidal sequence

③  $\frac{2\pi}{\Omega_0}$  is irrational number, the sequence is aperiodic

$$\sin[\Omega_0(n+N)] = \sin(n\Omega_0 + N\Omega_0)$$

If  $N\Omega_0 \neq m2\pi$ ,  $N$ ,  $m$  are integers

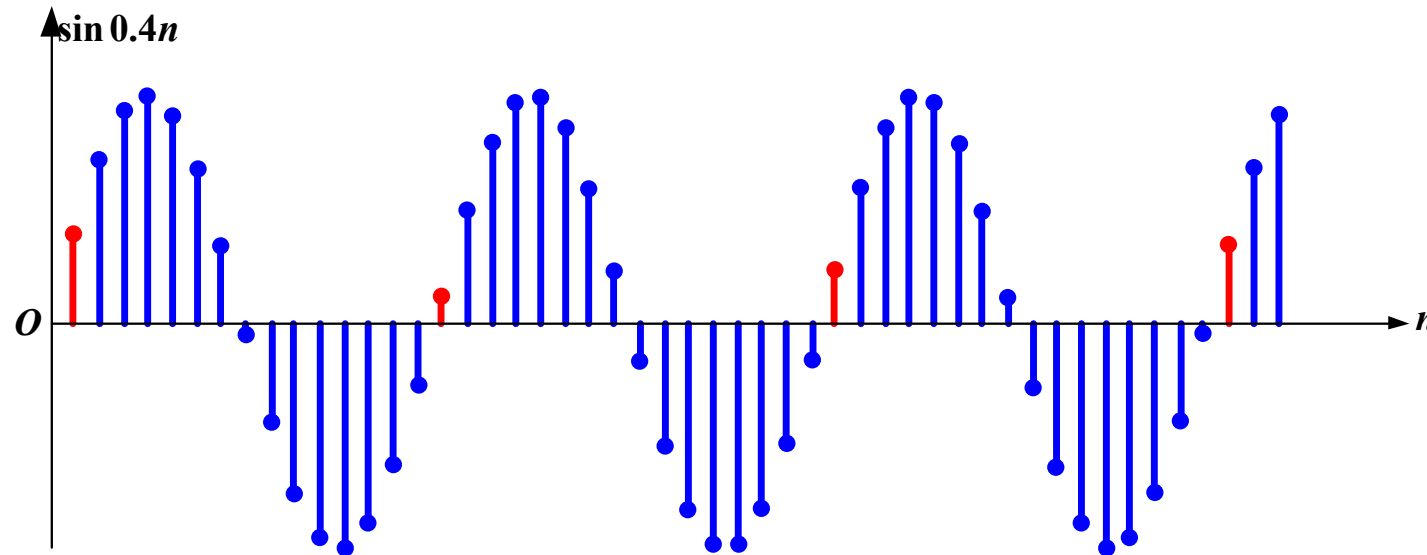
Then  $\frac{2\pi}{\Omega_0}$  is irrational, aperiodic

# 6.1 Typical Discrete-time signals

## Example 3

$$x[n] = \sin(0.4n)$$

Aperiodic  
sequence



$$\Omega_0 = 0.4 \quad \frac{2\pi}{\Omega_0} = 5\pi \text{ irrational}$$

## (7) Complex exponential sequence

$$x[n] = e^{j\Omega_0 n} = \cos \Omega_0 n + j \sin \Omega_0 n$$

Representing complex sequences in polar form:

$$x[n] = |x[n]| e^{j\arg[x[n] ]}$$

$$|x[n]| = 1$$

$$\arg[x[n]] = \Omega_0 n$$

## Summary

1. Representation Methods of Discrete Signals
2. Typical discrete signal
  - Unit sample sequence (Discrete-time Impulse)
  - Unit step sequence
  - Rectangular sequence (Rectangular window)
  - Ramp sequence
  - One-sided Exponential Sequence
  - Sinusoidal sequence
  - Complex exponential sequence

## Main Content

- **Summation**
- **Multiplication**
- **Scaling**
- **Shifting (Time Shift)**
- **Reversal (Time Reversal)**
- **Difference (Discrete Derivative)**
- **Running sum (Cumulative Sum)**
- **Reshaping: Compression (Decimation), Extension (Interpolation)**
- **Energy of a sequence**
- **Power of a sequence**

# 6.2 Operations on discrete-time signals

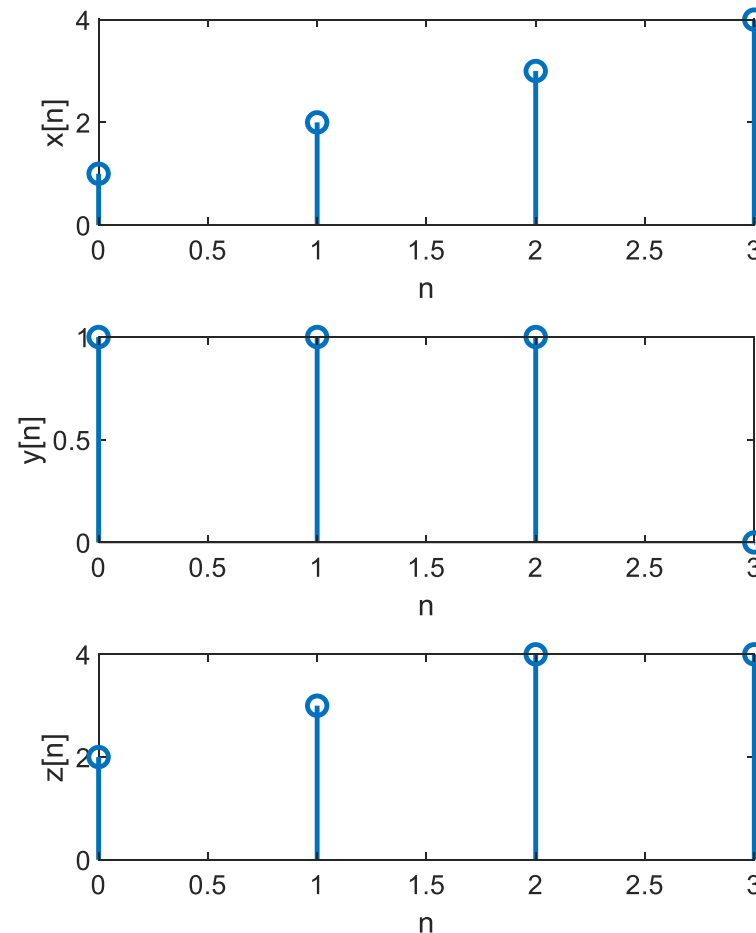
## 1. Summation

$$z[n] = x[n] + y[n]$$

**Example 1:**

$$x[n] = \left\{ \underset{\substack{\uparrow \\ n=0}}{1}, 2, 3, 4 \right\}$$

$$y[n] = \left\{ \underset{\substack{\uparrow \\ n=0}}{1}, 1, 1 \right\}$$



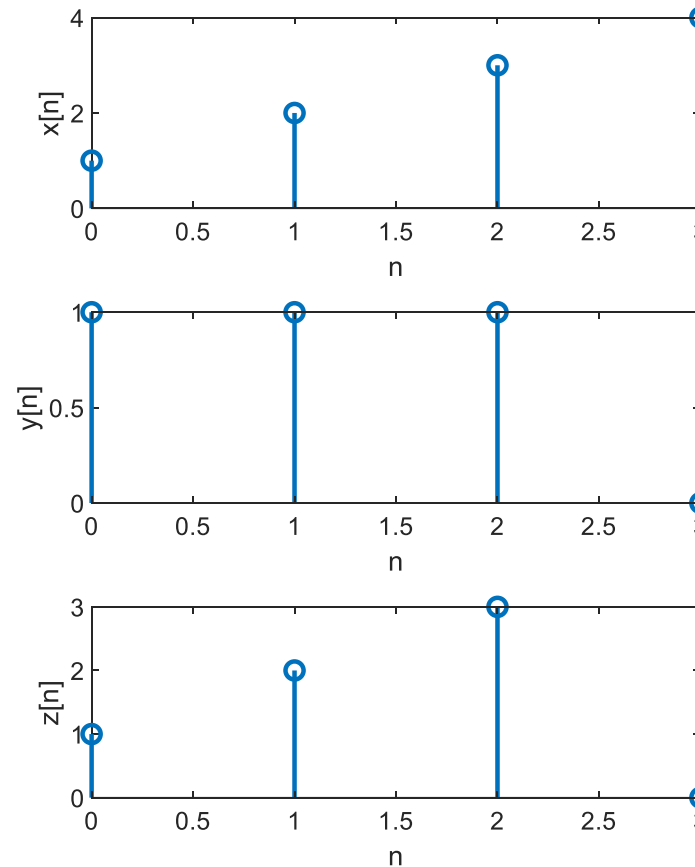
## 2. Multiplication

$$z[n] = x[n] \cdot y[n]$$

**Example 2:**

$$x[n] = \left\{ \underset{\substack{\uparrow \\ n=0}}{1}, 2, 3, 4 \right\}$$

$$y[n] = \left\{ \underset{\substack{\uparrow \\ n=0}}{1}, 1, 1 \right\}$$



## 3. Scaling

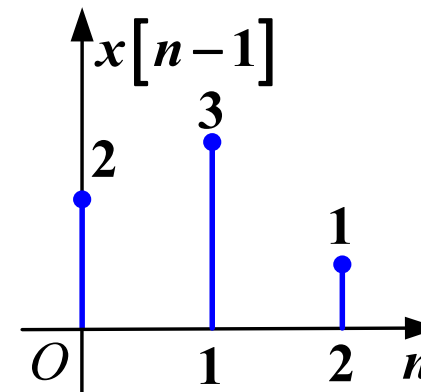
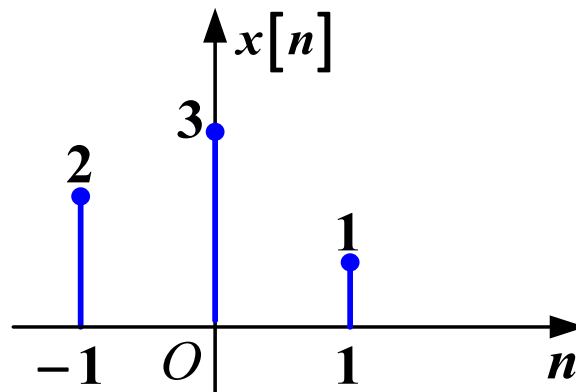
$$z[n] = ax[n]$$

### 4. Shifting

$$z[n] = x[n - m] \quad \text{Right Shift} \quad (m > 0)$$

$$z[n] = x[n + m] \quad \text{Left Shift} \quad (m > 0)$$

**Example 3:**



### 5. Reversal

$$z[n] = x[-n]$$

### 6. Difference

**Forward** difference:  $\Delta x[n] = x[n+1] - x[n]$

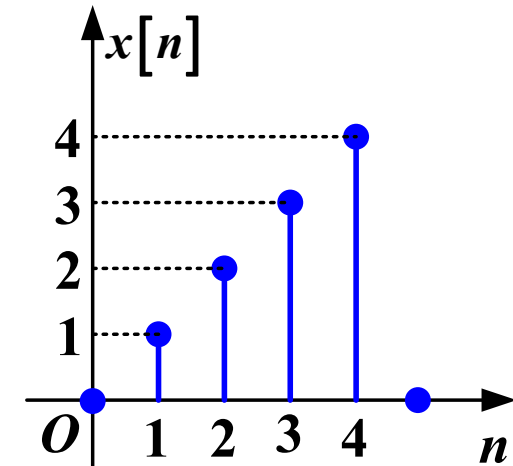
**Backward** difference:  $\nabla x[n] = x[n] - x[n-1]$

### 7. Running sum

$$z[n] = \sum_{k=-\infty}^n x[k]$$

**Example 4:** calculate the running sum  $y[n]$  of this signal

|         |                                 |
|---------|---------------------------------|
| $n = 0$ | $y[0] = 0$                      |
| $n = 1$ | $y[1] = 0 + 1 = 1$              |
| $n = 2$ | $y[2] = 0 + 1 + 2 = 3$          |
| $n = 3$ | $y[3] = 0 + 1 + 2 + 3 = 6$      |
| $n = 4$ | $y[4] = 0 + 1 + 2 + 3 + 4 = 10$ |
| $n > 4$ | $y[n] = 10$                     |



## 8. Reshaping

### Decimation

$$x[n] \rightarrow x[Nn]$$

$N$  is a positive integer

### Interpolation

$$x[n] \rightarrow x\left[\frac{n}{N}\right], N \text{ is a positive integer}$$



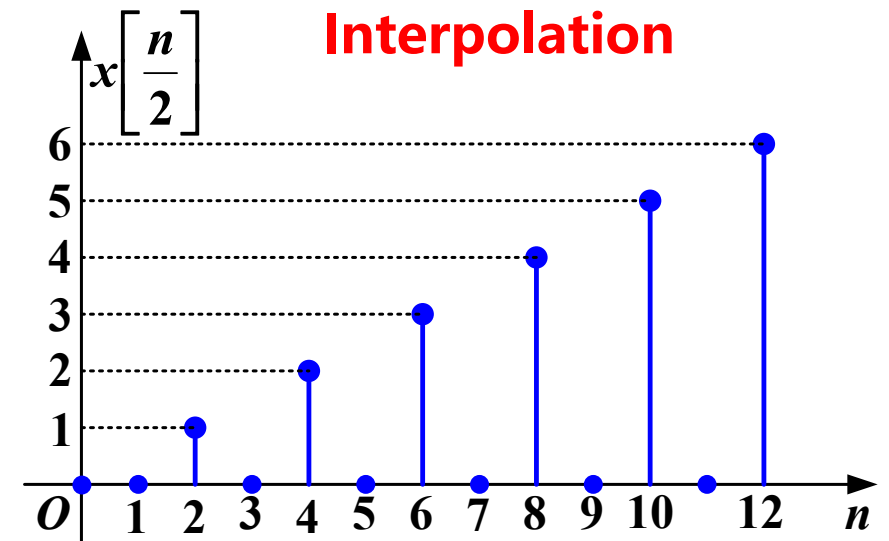
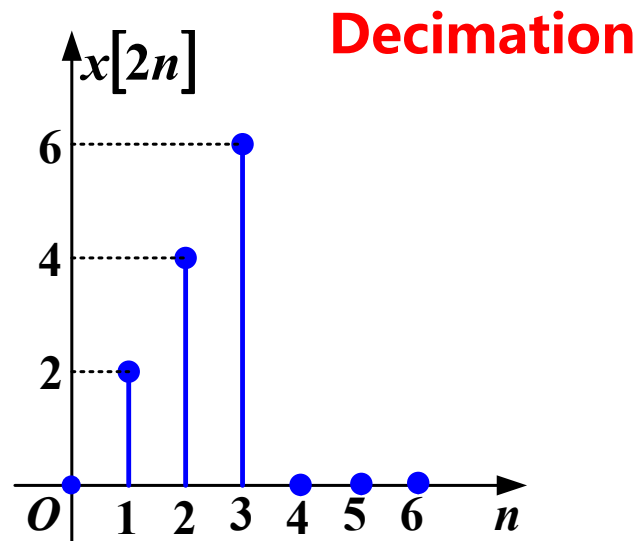
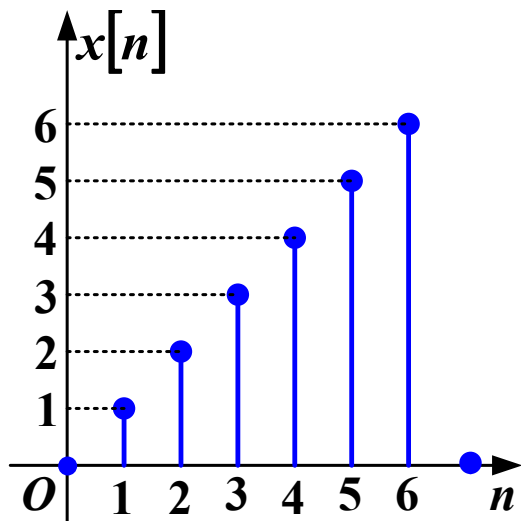
Decimation. Etching by William Hogarth in *Beaver's Roman Military Punishments* (1725)  
[https://en.wikipedia.org/wiki/Decimation\\_\(Roman\\_army\)](https://en.wikipedia.org/wiki/Decimation_(Roman_army))

**Note:** Sometimes it is necessary to remove certain points or pad with corresponding zero values

## 6.2 Operations on discrete-time signals

### Example 5

Given waveform of  $x[n]$ , please draw  $x[2n]$ ,  $x\left[\frac{n}{2}\right]$



### 9. Energy of a sequence

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

The energy of a discrete-time sequence  $x[n]$  over a finite interval  $-K \leq n \leq K$  is defined as

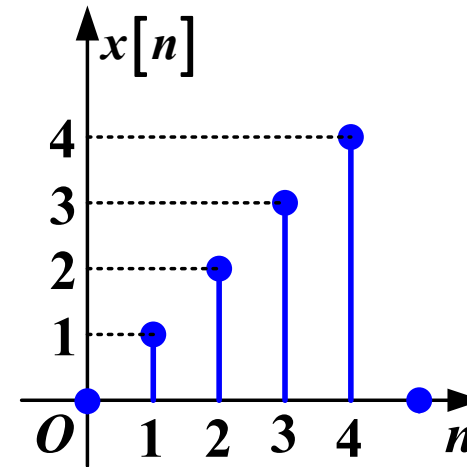
$$E_K = \sum_{n=-K}^K |x[n]|^2$$

## 6.2 Operations on discrete-time signals

### Example 6

Calculate the energy  $E$  of this signal

$$\begin{aligned} E &= \sum_{n=-\infty}^{\infty} |x[n]|^2 \\ &= 1^2 + 2^2 + 3^2 + 4^2 \\ &= 30 \end{aligned}$$



### 10. Power of a sequence

Average power of aperiodic signals

$$P = \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{n=-K}^K |x[n]|^2$$

Average power of a periodic sequence with a period of  $N$

$$P = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2$$

## Summary

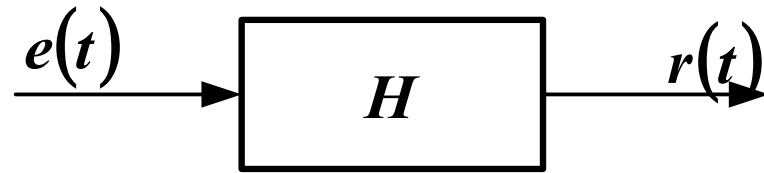
- **Summation**
- **Multiplication**
- **Scaling**
- **Shifting (Time Shift)**
- **Reversal (Time Reversal)**
- **Difference (Discrete Derivative)**
- **Running sum (Cumulative Sum)**
- **Reshaping: Compression (Decimation), Extension (Interpolation)**
- **Energy of a sequence**
- **Power of a sequence**

## Main Content

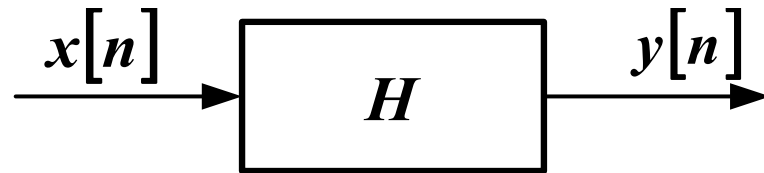
- Continuous-Time and Discrete-Time Systems
- Memoryless (Instantaneous) Systems vs. Dynamic Systems
- Lumped-Parameter Systems and Distributed-Parameter Systems
- Invertible Systems and Non-Invertible Systems
- Linear Systems
- Time-Invariant Systems
- Causal Systems
- Stable Systems

**Key Focus:** Study of lumped-parameter, linear, time-invariant (LTI) systems under deterministic signal excitation

## 1. Continuous-Time and Discrete-Time Systems



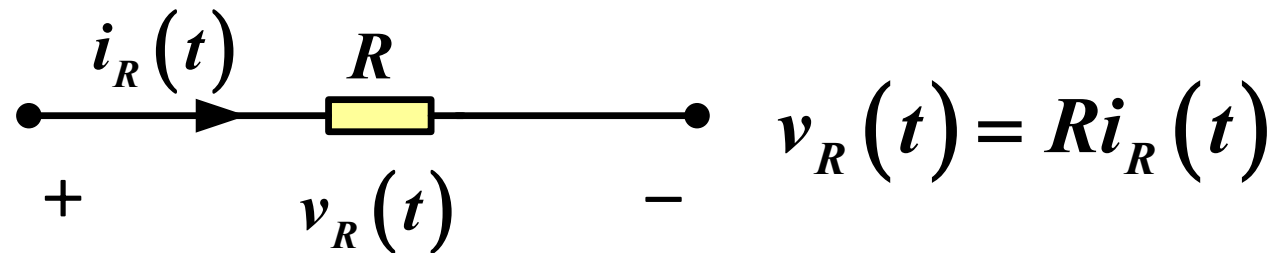
$$r(t) = H[e(t)]$$



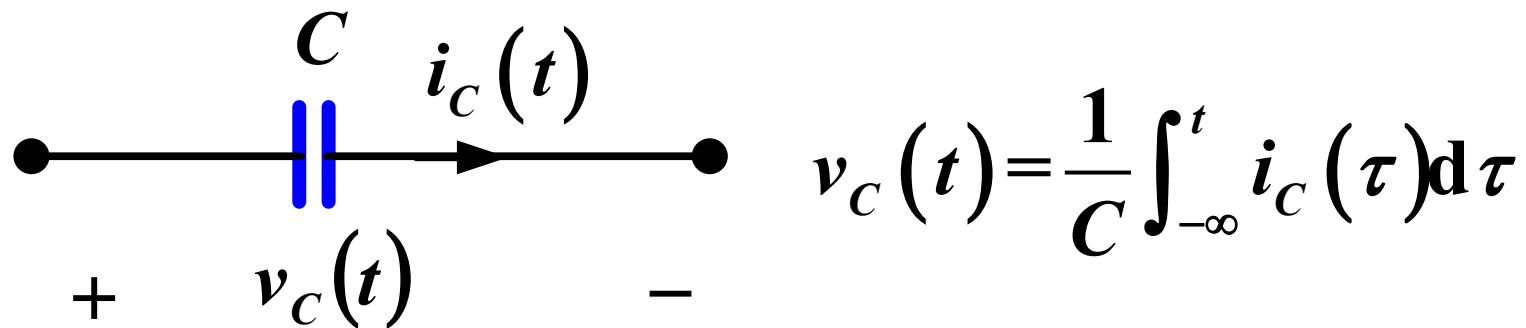
$$y[n] = H[x[n]]$$

## 2. Instantaneous Systems vs. Dynamic Systems

Instantaneous Systems (Memoryless): Algebraic Equation



Dynamic systems (systems with memory): differential or difference equations



## 3. Lumped-Parameter and Distributed-Parameter

**Lumped parameter system:** Ordinary differential equation  
(*time – dependent only:  $t$* )  
Low-frequency electrical circuits (RLC)

**Distributed Parameter Systems:** Partial Differential Equations  
(*time and space – dependent:  $t, x, y, z$* )  
Waveguides and transmission lines

## 4. Invertible Systems and Non-Invertible Systems

**Reversible system:** If the system produces different responses under different excitation signals, it is called a reversible system

**Application:** Utilizing inverse systems to achieve system compensation

$$r(t) = \eta |e(t)|^2 \quad (0 < \eta < 1)$$

**Irreversible system:** If different inputs produce identical outputs, making it impossible to determine the original input from the output alone

**Characteristics of irreversible systems:**

- The input signals  $e(t)$  and  $-e(t)$  produce the same output
- It is impossible to uniquely determine the input signal from the output
- The system is irreversible

## 5. Linear Systems

### (1) Definition

**Linear system:** A system that exhibits linear properties

**Linearity:** Characterized by homogeneity and superposition

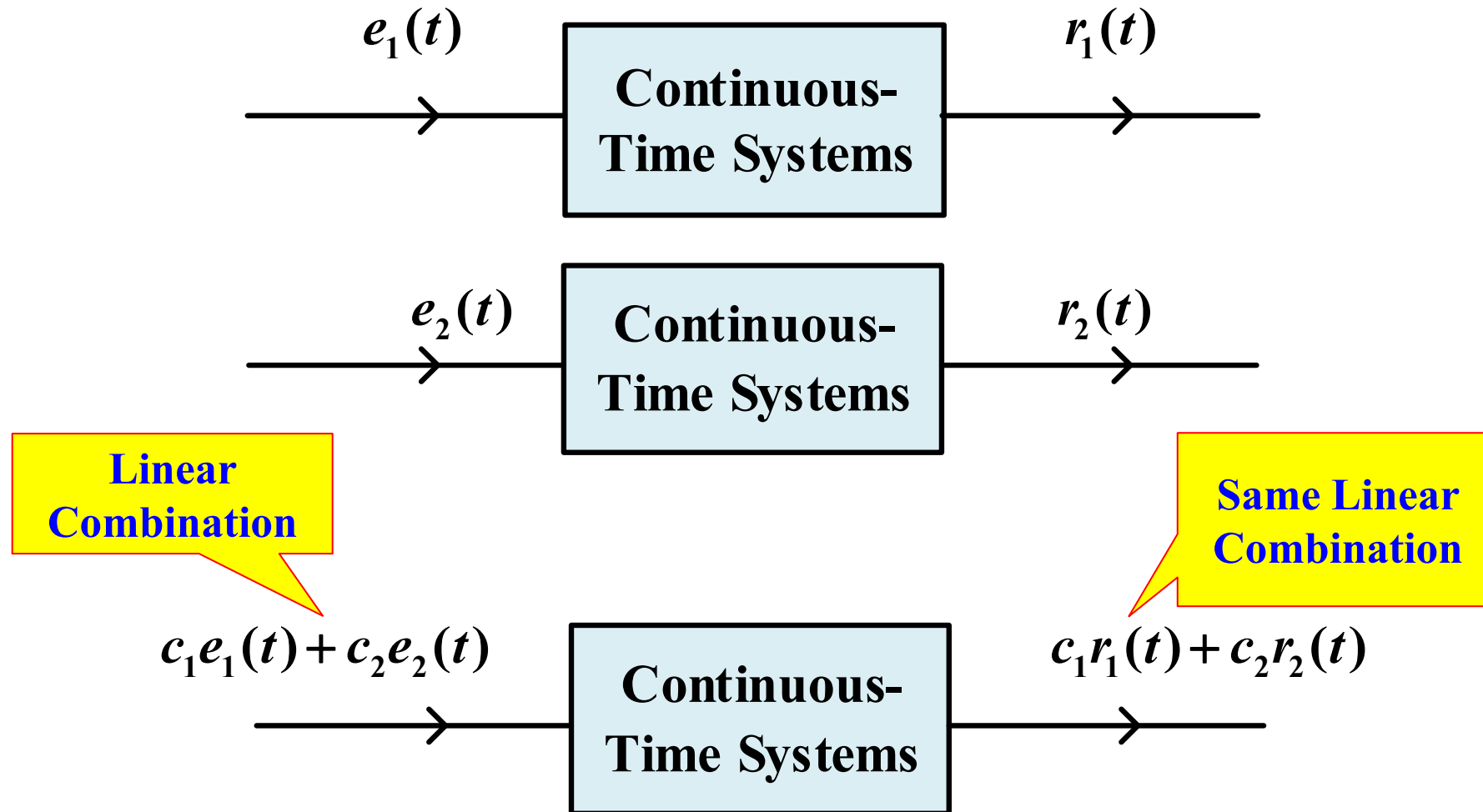
**Homogeneity:**

$$e(t) \rightarrow r(t) \Rightarrow Ke(t) \rightarrow Kr(t)$$

**Superposition:**

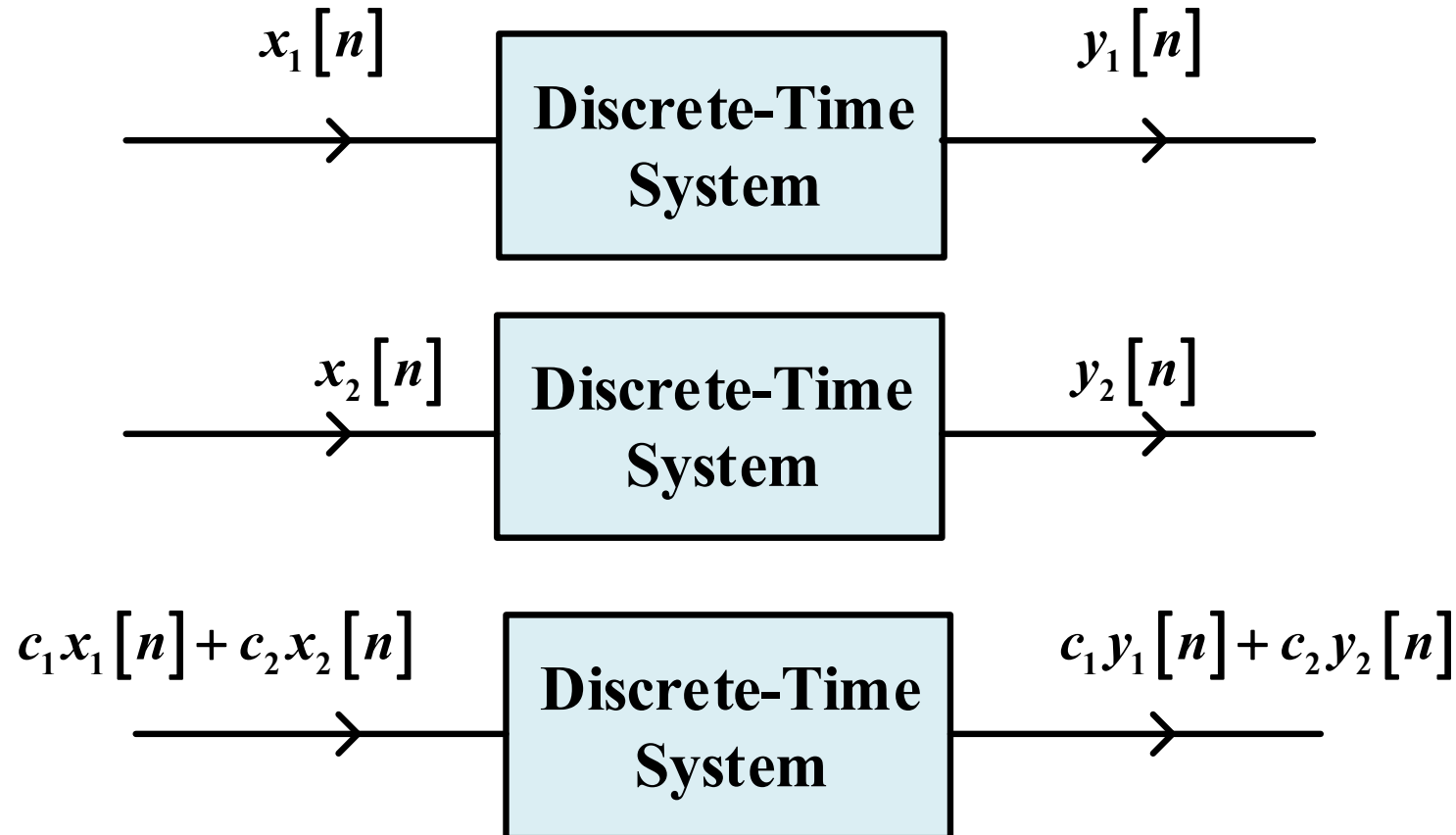
$$\left. \begin{array}{l} e_1(t) \rightarrow r_1(t) \\ e_2(t) \rightarrow r_2(t) \end{array} \right\} \Rightarrow e_1(t) + e_2(t) \rightarrow r_1(t) + r_2(t)$$

## (2) Linear Properties of Continuous Systems



## (3) Linear Properties of Discrete Systems

**Linearity: Both homogeneity and superposition hold**



## (4) Method for Judging Linear Characteristics

First perform linear combination, then pass through system = First pass through system, then perform linear combination

$$\text{If } H[C_1 e_1(t) + C_2 e_2(t)] = C_1 H[e_1(t)] + C_2 H[e_2(t)]$$

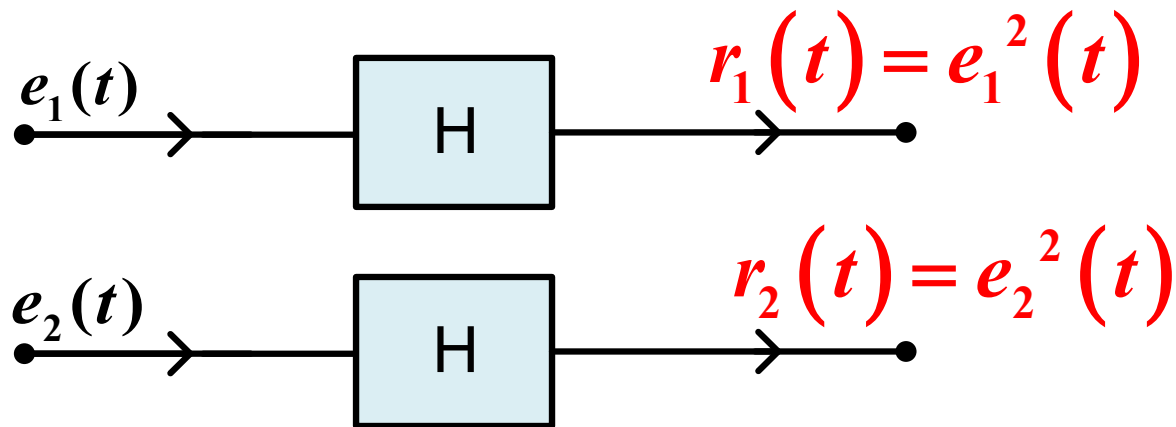
Then the system  $H[\cdot]$  is a linear system, otherwise it is a nonlinear system

**Note: The external excitation and non-zero state of the system must be handled separately**

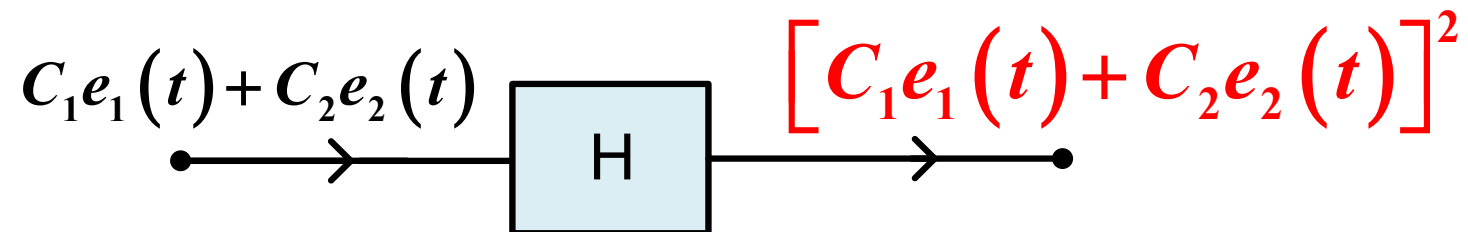
## Example 1

Is the system  $r(t) = e^2(t)$  a Linear system?

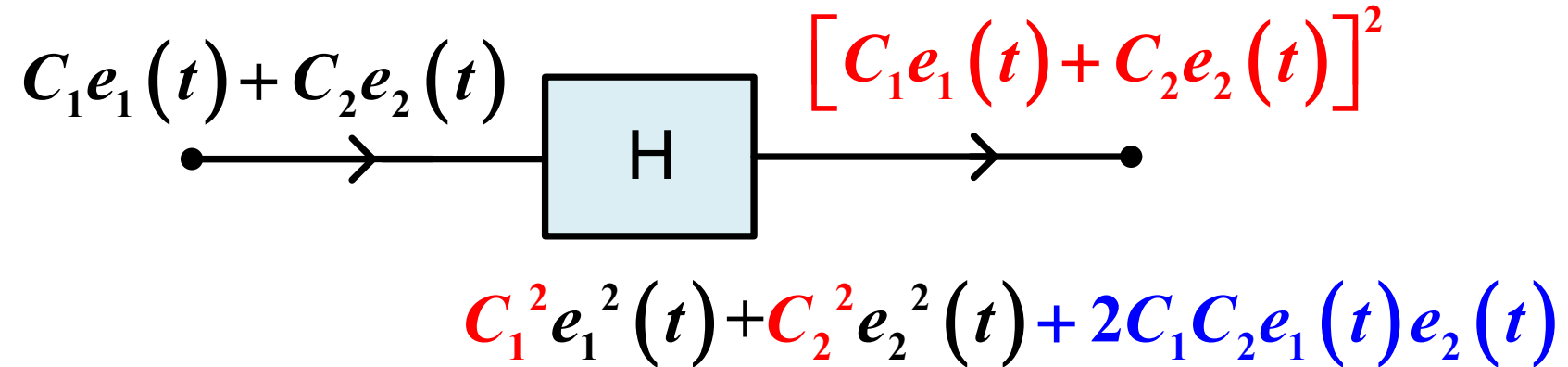
For any signals  $e_1(t), e_2(t)$



Linearly combine the two signals and input into the system



## Example 1



The linear combination of output signals  $r_1(t)$  and  $r_2(t)$  is

$$C_1 r_1(t) + C_2 r_2(t) = C_1 e_1^2(t) + C_2 e_2^2(t)$$

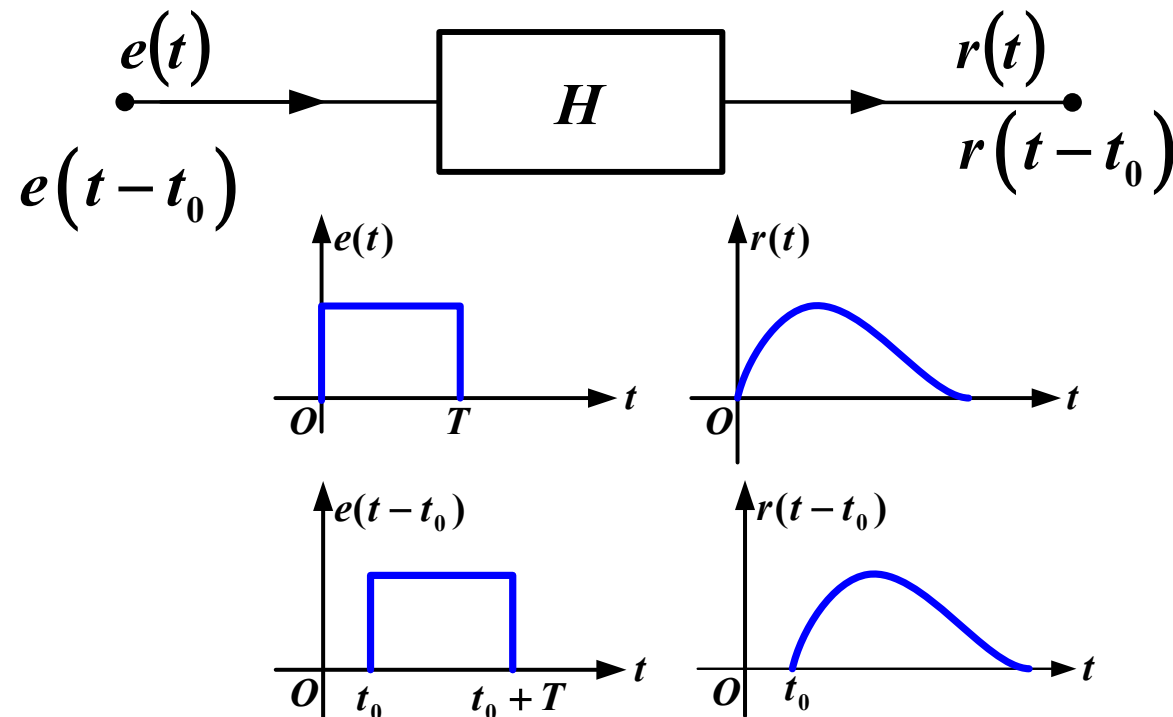
Obviously, the two results are different

Therefore, the system is **nonlinear**

## 6. Time-Invariant Systems

### (1) Definition

A system is called time-invariant if, under **zero initial conditions**, its output signal is **independent of the time** at which the input signal is applied to the system; otherwise, it is called time-varying



## (2) Time-Invariance of Continuous Systems

### Explanation:

From a circuit analysis perspective:

Whether the **component parameters vary with time**

From an equation perspective:

Whether the **coefficients of the equation vary with time**

### Judgment Method:

**First time-shift, then pass through system = First pass through system, then time-shift**

If  $H[e(t - \tau)] = r(t - \tau)$

Then the system  $H[\cdot]$  is time-invariant; otherwise, it is time-varying

## Example 2

To judge whether the system  $r(t) = \frac{de(t)}{dt}$  is a time invariant system

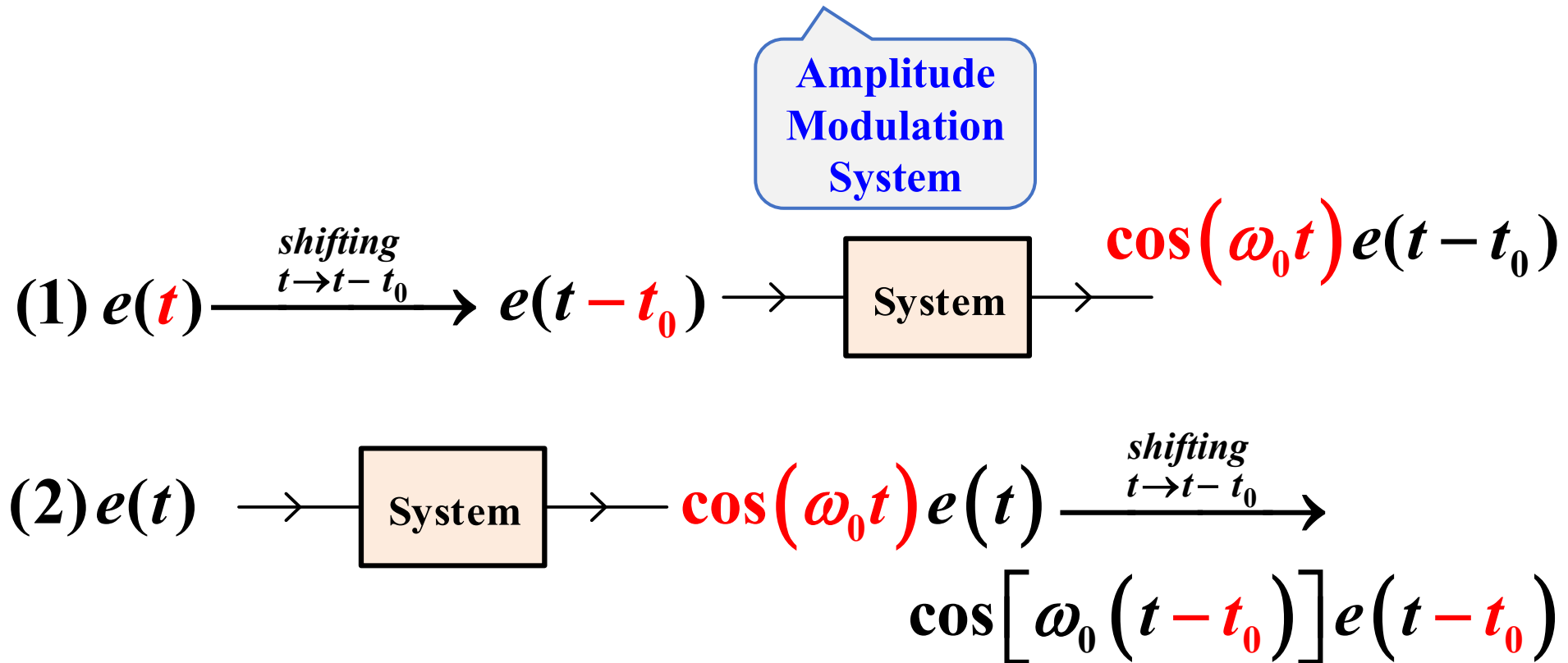
$$(1) e(t) \xrightarrow[\text{shifting } t \rightarrow t-t_0]{} e(t-t_0) \rightarrow \boxed{\text{System}} \rightarrow \frac{de(t-t_0)}{dt}$$

$$(2) e(t) \rightarrow \boxed{\text{System}} \rightarrow r(t) = \frac{de(t)}{dt} \xrightarrow[\text{shifting } t \rightarrow t-t_0]{} \frac{de(t-t_0)}{d(t-t_0)}$$

The results are identical ( $r_1(t) = r(t-t_0)$ ), therefore the system is **time-invariant**

## Example 3

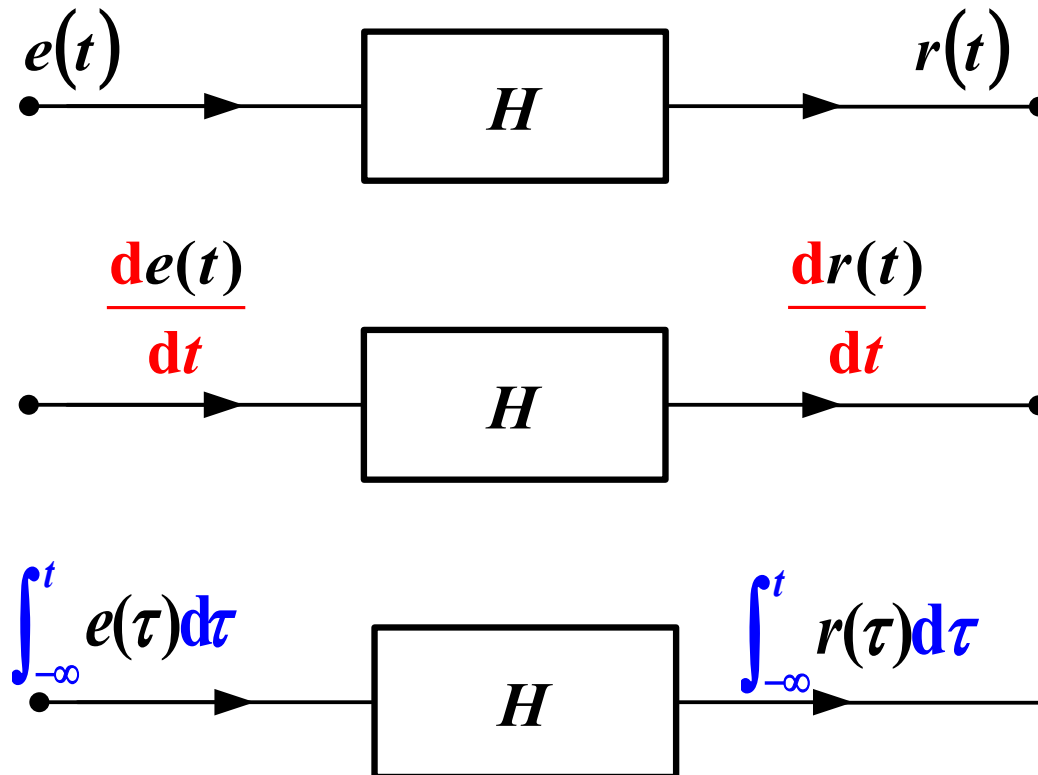
To judge whether the system  $r(t) = \cos(\omega_0 t)e(t)$  is a time invariant system



Result: Different outputs  $\rightarrow$  System is **time-varying**

## (3) LTIS, differential characteristics, integral characteristics

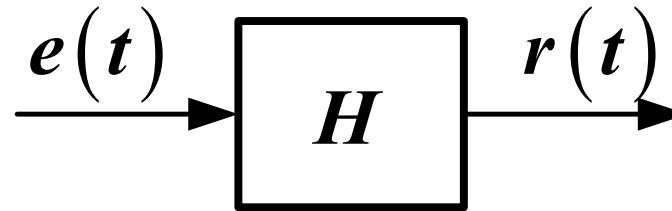
Linear and time invariant system, LTIS



# 7.1 System Characteristics

## Proof

Using Linear and Time-Invariant Properties: The methodology can be extended to higher-order systems

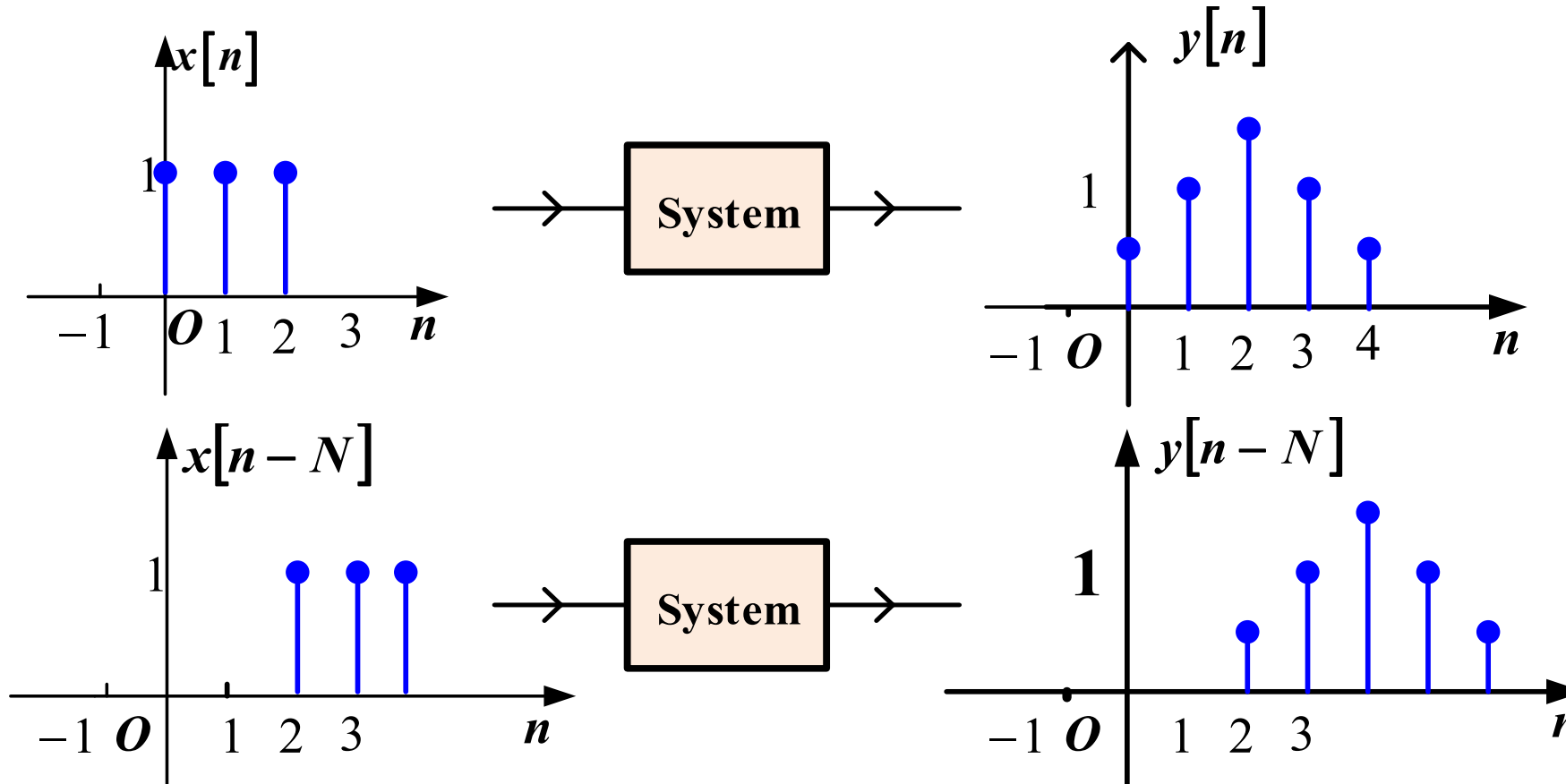


$$\frac{de(t)}{dt} = \lim_{\Delta t \rightarrow 0} \frac{e(t) - e(t - \Delta t)}{\Delta t}$$

$$\lim_{\Delta t \rightarrow 0} \frac{r(t) - r(t - \Delta t)}{\Delta t} = \frac{dr(t)}{dt}$$

## (4) Time Invariance of Discrete Systems

$$x[n] \rightarrow y[n], \quad x[n - N] \rightarrow y[n - N]$$



## 7. Causal Systems

### Causal System

A system that produces output responses only when the input signal **excites** it

This property of the system is called **causality**

### Judgment Method

Check whether the output **precedes** the input

## Example 4

### a. Analysis of System Causality

$$r(t) = e(t) + e(t - 2)$$

**Current response = current excitation + past excitation**

Therefore, the system is a Causal system

### b. For the system $r(t) = e(t) + e(t + 2)$

**Includes future excitation**

Therefore, the system is a non-causal system

## 7. Causal Systems

### Implementation Requirement

For a time system, implementation requires the system to be causal

### Non-Causal Systems

Practical Significance: Signal compression/expansion and voice signal processing utilize non-causal system characteristics

### Causal System Definition

$$e(t) = e(t)u(t)$$

Equivalent to  $t < 0, e(t) = 0$

## 8. Stable Systems

If a system only produces **bounded outputs** for any **bounded input**, it is called a stable system in the sense of bounded input bounded output (**BIBO**)

For such a system, if the input does **not diverge**, the output will **not diverge** either

From an engineering perspective, it is crucial for a system to **remain stable** under all possible conditions.

## Summary

- Continuous-Time and Discrete-Time Systems
- Memoryless (Instantaneous) Systems vs. Dynamic Systems
- Lumped-Parameter Systems and Distributed-Parameter Systems
- Invertible Systems and Non-Invertible Systems
- Linear Systems
- Time-Invariant Systems
- Causal Systems
- Stable Systems

**Key Focus:** Linear time invariant systems, from an implementation perspective, hope that the system is **causal** and **stable**

## Main Content

- **Continuous Systems:** Differential equations, block diagrams
- **Discrete Systems:** Difference equations, block diagrams

## 1. Mathematical Model of Continuous Time Systems

### Differential equations

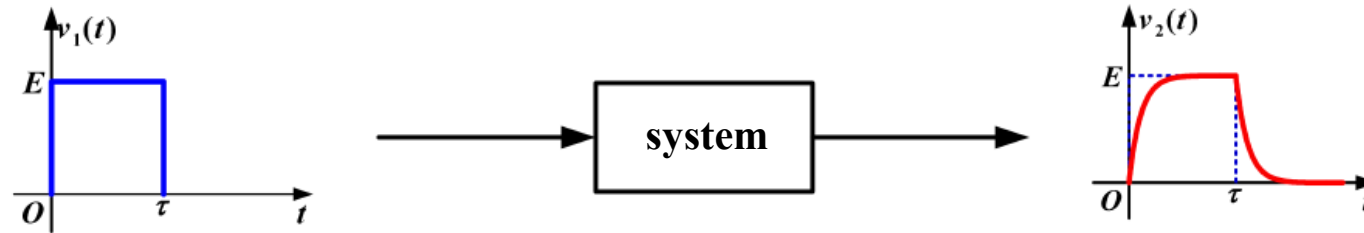
- Write the differential equations of the system based on the physical characteristics of the actual system
- Given the excitation conditions and initial state, calculate the response
- After mathematical analysis, we will return to physical reality

**Structure diagrams**, such as block diagrams and signal-flow graphs.

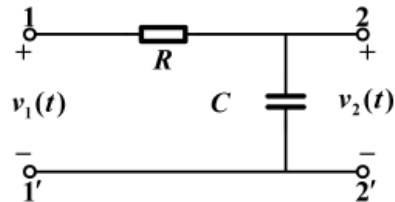
# 7.2 Representation Method for LTIS

## 1. Mathematical Model of Continuous Time Systems

How to get the response of a signal through the system?



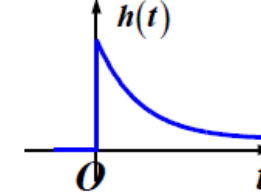
Circuit



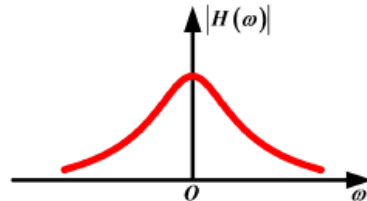
Differential equation

$$\frac{dv_2(t)}{dt} + \frac{1}{RC}v_2(t) = \frac{1}{RC}v_1(t)$$

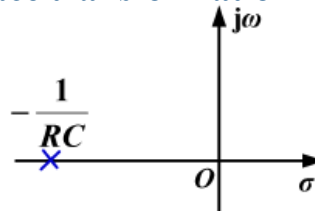
Unit impulse response



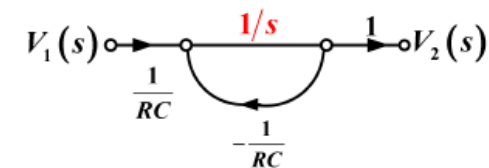
System functions



Laplace transformation



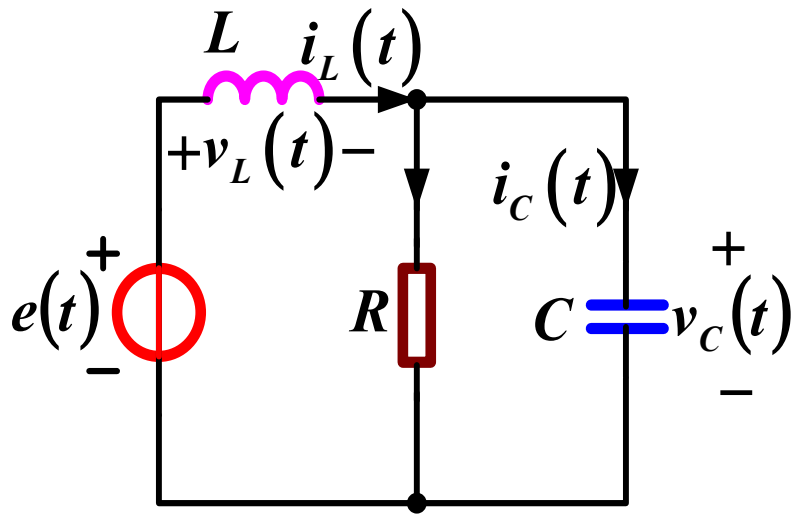
Structural diagram



## 7.2 Representation Method for LTIS

### Example 1

Please provide the differential equations that describe this system



$e(t)$ : excitation,  $v_C(t)$ : response

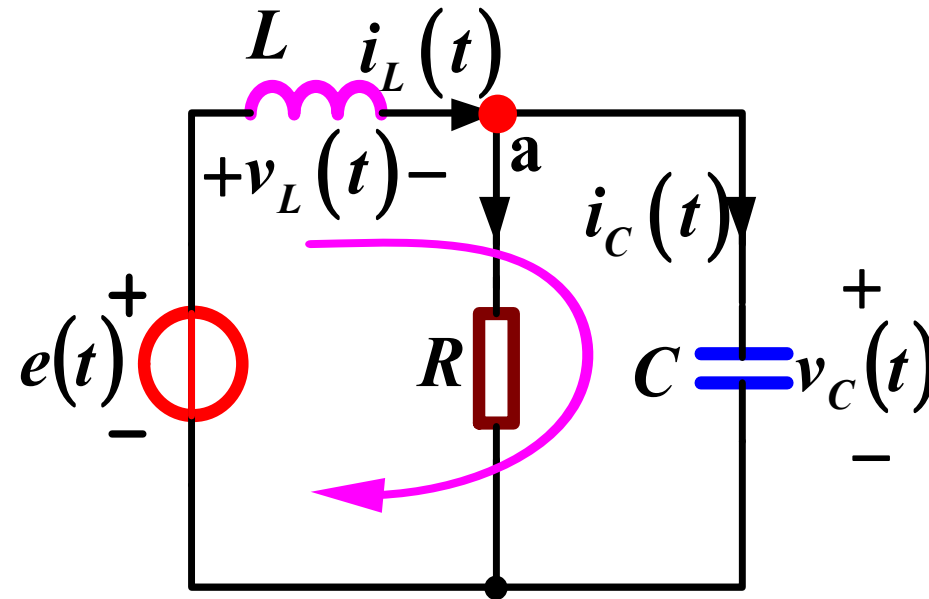
Constraint Characteristics of Components (Associated Reference Direction):

Capacitor:  $i_C(t) = C \frac{dv_C(t)}{dt}$

Inductor:  $v_L(t) = L \frac{di_L(t)}{dt}$

## 7.2 Representation Method for LTIS

### Example 1



Network topology characteristics:

**KVL:** 
$$e(t) = L \frac{d i_L(t)}{d t} + v_c(t) \quad (1)$$

**KCL(Point a):** 
$$i_L(t) = \frac{v_c(t)}{R} + C \frac{d v_c(t)}{d t} \quad (2)$$

## 7.2 Representation Method for LTIS

### Example 1

$$\begin{cases} e(t) = L \frac{d i_L(t)}{dt} + v_C(t) & (1) \\ i_L(t) = \frac{v_C(t)}{R} + C \frac{d v_C(t)}{dt} & (2) \end{cases}$$

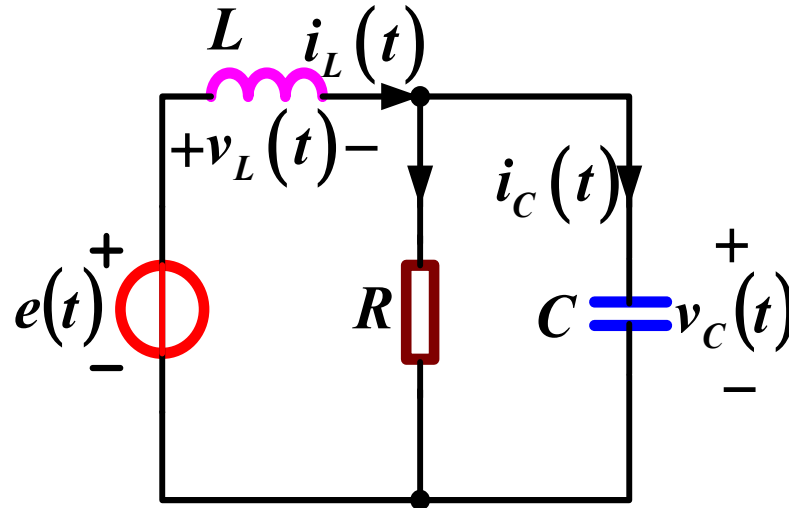
In order to **eliminate**  $i_L(t)$ , substitute (2) into (1) and organize it as follows

$$e(t) = L \frac{d \left[ \frac{v_C(t)}{R} + C \frac{d v_C(t)}{dt} \right]}{dt} + v_C(t)$$

$$\frac{d^2 v_C(t)}{dt^2} + \frac{1}{RC} \frac{d v_C(t)}{dt} + \frac{1}{LC} v_C(t) = \frac{1}{LC} e(t)$$

## 7.2 Representation Method for LTIS

### Example 1

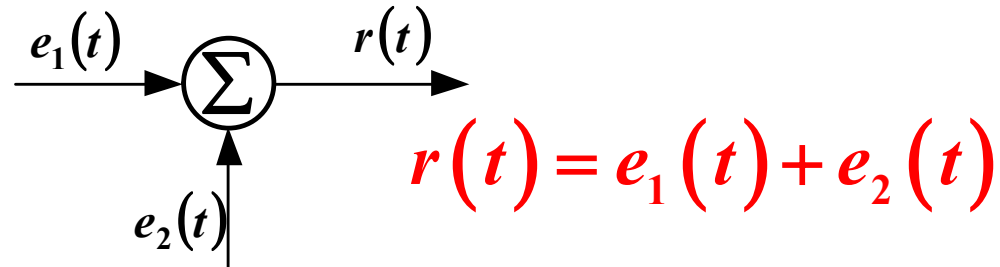


$$\frac{d^2 v_C(t)}{dt^2} + \frac{1}{RC} \frac{dv_C(t)}{dt} + \frac{1}{LC} v_C(t) = \frac{1}{LC} e(t)$$

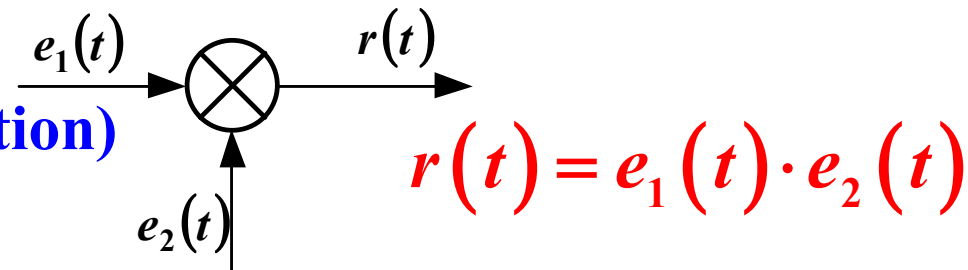
- **No** nonlinear terms (e.g., multiplication) exist in the equation  $\rightarrow$  **Linear** system
- All coefficients in the equation are constants  $\rightarrow$  **Time-invariant** system
- The **order** of the differential equation equals the order of the system

## 2. Block Diagram Representation of Continuous-Time Systems

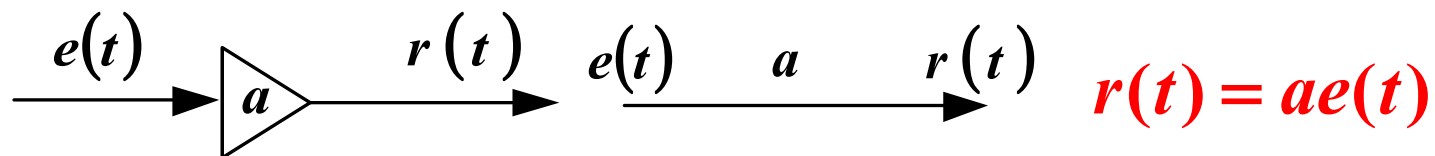
(1) Adder



(2) Multiplier  
(nonlinear operation)

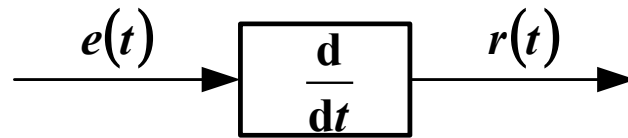


(3) Scalar multiplier (coefficient multiplier, scaler)



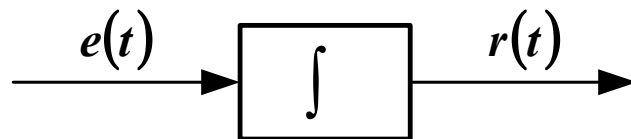
## 2. Block Diagram Representation of Continuous-Time Systems

### (4) Differentiator



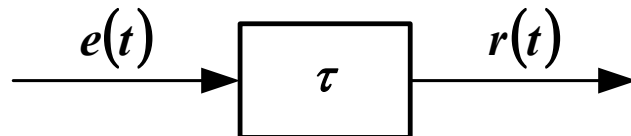
$$r(t) = \frac{de(t)}{dt}$$

### (5) Integrator



$$r(t) = \int_{-\infty}^t e(\tau) d\tau$$

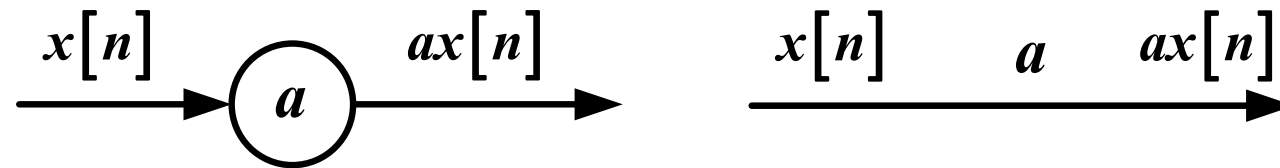
### (6) Time-delay element



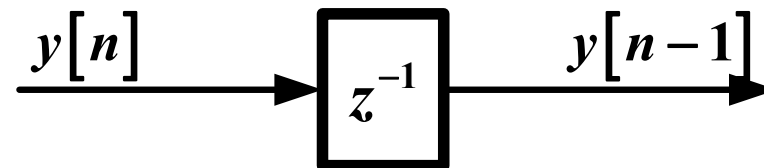
$$r(t) = e(t - \tau)$$

### 3. Representation of Discrete-Time Systems

#### Scalar Multiplier



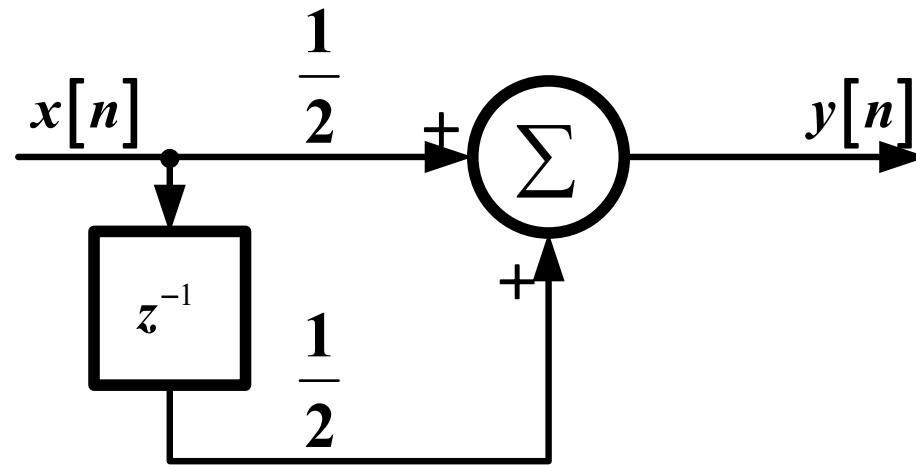
#### Unit Delay Element



A unit delay is essentially a shift register that pushes out the previous value while shifting in the new one

## 7.2 Representation Method for LTIS

### Example 2

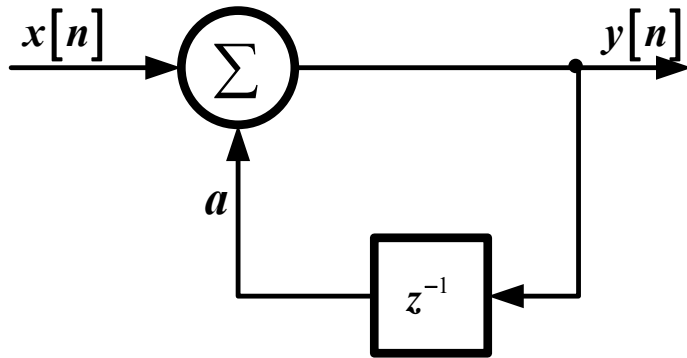


$$y[n] = \frac{1}{2}x[n] + \frac{1}{2}x[n-1]$$

**System Without Feedback Terms**

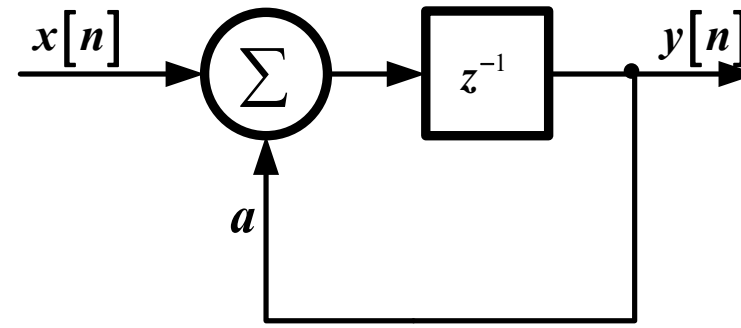
## 7.2 Representation Method for LTIS

### Example 3



$$y[n] = x[n] + ay[n-1]$$

First order backward difference equation



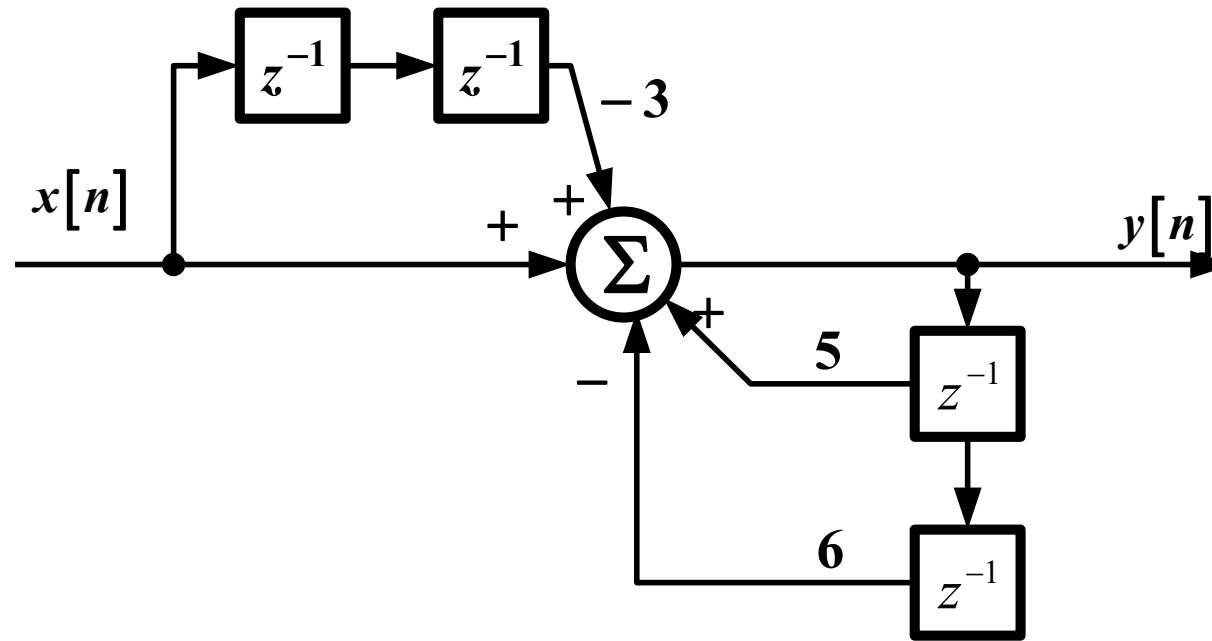
$$y[n+1] = x[n] + ay[n]$$

First-order forward difference equation

We use a backward difference equation

## 7.2 Representation Method for LTIS

### Example 4



$$y[n] = x[n] - 3x[n-2] + 5y[n-1] - 6y[n-2]$$

$$\textcolor{red}{y}[n] - 5\textcolor{red}{y}[n-1] + 6\textcolor{red}{y}[n-2] = \textcolor{blue}{x}[n] - 3\textcolor{blue}{x}[n-2]$$

A **second-order system** can be represented through multiple structural diagrams, all requiring at least **two delay elements** for implementation

### 3. Representation of Discrete-Time Systems

**General Form** 
$$\sum_{k=0}^N a_k y[n-k] = \sum_{r=0}^M b_r x[n-r]$$

**Order of a Difference Equation:** The difference between the highest and lowest argument values of the dependent variable in the equation

### Summary

- **Continuous Systems:** Differential equations, block diagrams
- **Discrete Systems:** Difference equations, block diagrams

### Main Content

- Overview of System Analysis Methods
- Two Approaches to Establish System Models
- Mathematical Model Solution Methods

## 1. Overview of System Analysis Methods

### Research Objectives:

To investigate fundamental analysis methods for **linear time-invariant (LTI)** systems, with applications to signal transmission and processing in communication systems

### System Analysis Approach:

#### (1). Mathematical Modeling

Develop equations characterizing the system dynamics

#### (2) Response Analysis

Solve for system outputs under given inputs

## 2. Two Approaches to Establish System Models

### Input-Output Description Method:

1. Focuses solely on the relationship between excitation and response, disregarding internal system variables
2. Applicable to single-input/single-output (SISO) systems
3. Formulates a single  **$n$ th-order ordinary differential equation**

### State-Variable Analysis Method:

1. Capable of determining both system response and internal variables (e.g., capacitor voltages, inductor currents)
2. Suitable for multiple-input/multiple-output (MIMO) systems
3. Establishes a set of coupled **first-order differential equations**

## 3. Mathematical Model Solution Methods

### 1. Time Domain

Classical method for solving { Continuous Systems: Differential Equations  
Discrete Systems: Difference Equations

Convolutional integral (or convolutional sum) method

### 2. Transform-Domain Analysis

- Fourier transform (FT)
- Laplace transform (LT)
- Z transform (ZT)
- Fourier Transform of Discrete Time Signals DFT

## 4. Stable Systems

**Input-Output Description  $\rightarrow$  State-Variable Description**

**Time-Domain Analysis  $\rightarrow$  Transform-Domain Analysis**

**Continuous Systems  $\rightarrow$  Discrete Systems**



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*Thank you for listening !*

